Geoelectric, Geoelectromagnetic and Combined Geophysical Methods in Groundwater Exploration in Israel



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1 Introduction

Among all geophysical methods, applied in groundwater exploration, geoelectric (GE) and geoelectromagnetic (GEM) ones are most successful for characterization of groundwater salinity (e.g., Fitterman and Stewart 1986; Goldman et al. 1988) and for delineating geometrical features of different hydrogeological targets, including the configuration of the fresh–saline groundwater interface, the thickness of alluvial fill and gravel lenses overlying the bedrock, the thickness and depth of sand and gravel lenses in till, etc. (e.g., Albouy et al. 2001; Fitterman and Labson 2005). The methods are less efficient in quantitative evaluation of hydraulic/hydrological parameters by using geoelectric ones, either measured (e.g., apparent resistivities) or interpreted in terms of true physical parameters (e.g., resistivities). In this respect, the most extensive efforts were devoted to studying the empirical relationship between the porosity and/or the pore water salinity/conductivity on one hand and the bulk electrical resistivity recovered by GE/GEM methods on the other hand (e.g., Archie 1942; Kelly and Mareš 1993; Yang et al. 1999). However, due to uncertainties in determining all the parameters used in the above relationship, such as the pore water and bulk electrical resistivities, clay content, or the type of a rock sequence, these attempts proved to be less successful than those devoted to the delineation of geometrical parameters. Somewhat better estimates can be obtained in both clastic and carbonate aquifers saturated with saline water, where the use of the time domain electromagnetic (TDEM) method allows to more accurately determine the pore water and bulk

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electrical resistivities and thus to reduce uncertainties in estimating the porosity or storativity of the aquifers (Kafri and Goldman 2005).

In the last few decades, the efforts have been also made to develop geophysical methods, which allow a more detailed characterization of aquifers by direct quantitative determination of their hydraulic properties, such as porosity/storativity and/or conductivity/transmissivity. These methods include cross-hole seismic and radar tomography, surface nuclear magnetic resonance (SNMR), seismoelectric and electroseismic methods and spectral induced polarization (SIP) method (e.g., Garambois and Dietrich 2001; Linde et al. 2008; Yaramanci 2003; Thompson and Gist 1993; Kemna et al. 2012).

In Israel, suffering not only from a shortage of freshwater, but also from severe salinization problems of both surface and groundwater resources, the main objective of geophysical surveys was aimed at detecting and accurately delineating saline ground water bodies of different origins. Taking into account the close relationship which exists between water salinity and electrical conductivity (resistivity), it is natural that the principal geophysical methods applied in Israel are the GE/GEM methods. Seismic methods, which are better suited for delineation of different rock units, are applied to discriminate aquifers and aquicludes in the subsurface (e.g., Shtivelman and Goldman 2000). The combined application of the GEM and SNMR methods in the Mediterranean and Dead Sea coastal aquifers allowed a quantitative estimation of fresh groundwater volume within the aquifers as well as the discrimination of the completely saturated brine with regard to halite from the slightly diluted Dead Sea brine (Legchenko et al. 2009). The objective of this chapter is mainly to focus on the application of GE/GEM methods for characterization of groundwater salinity in different aquifers throughout Israel.

2 Methods

The following three main GE/GEM methods, which were widely applied in groundwater exploration in Israel, are described below: DC resistivity soundings, electrical resistance tomography (ERT) and the time domain electromagnetic (TDEM) method. In addition, only one non-GE/GEM method, namely the surface nuclear magnetic resonance, SNMR (also known as proton magnetic resonance, PMR, or magnetic resonance soundings, MRS) method, is shortly described below. The description of all other geophysical methods is beyond the scope of this chapter.

2.1 The DC Resistivity Soundings

In this method, the earth is energized using a direct current. Thus, the only way to inject the current into the earth is by using a pair of current electrodes connected to a DC transmitter. Another pair of electrodes, named potential or measurement

electrodes, is used to measure the established potential difference throughout the earth surface. The surficial distribution of the potential depends on spatial resistivity distribution in the subsurface and thus the latter can be, in principle, recovered by proper measurements of the surface potential differences. This is done by moving either the current or potential or all electrodes along straight lines above a subsurface exploration target. Depending on the inter-electrode configurations, there are several methods of DC resistivity soundings, each having different advantages and drawbacks. The most popular configurations are shown in Fig. 1.

In the case of the Wenner method (Fig. 1a), all four electrodes are relocated keeping the distance (separation) between them constant. This method provides the largest signal-to-noise ratio, but is mostly time consuming and requires long cables. In the case of the Schlumberger method (Fig. 1b), only the outer (current) electrodes are relocated, keeping the potential electrodes in the center of the array. The latter are moved only occasionally, when the signal-to-noise ratio drops below some specified

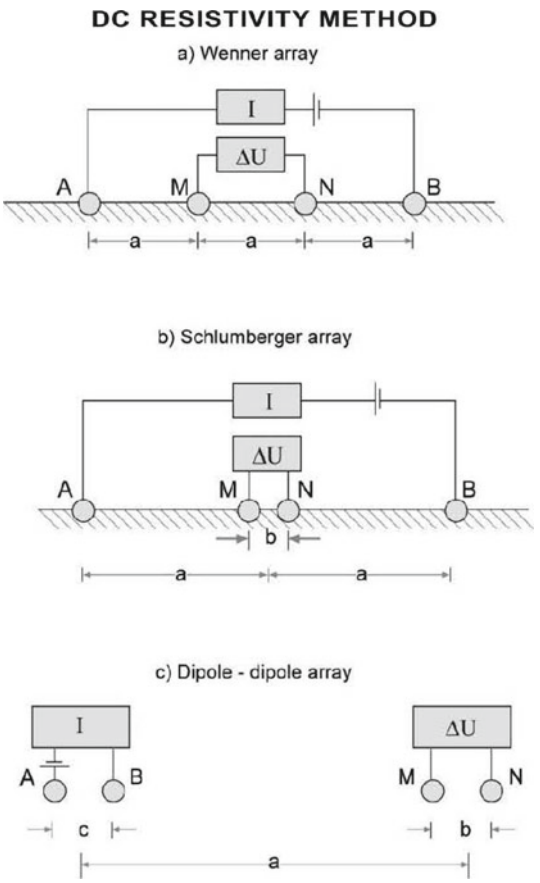


Fig. 1 DC resistivity arrays

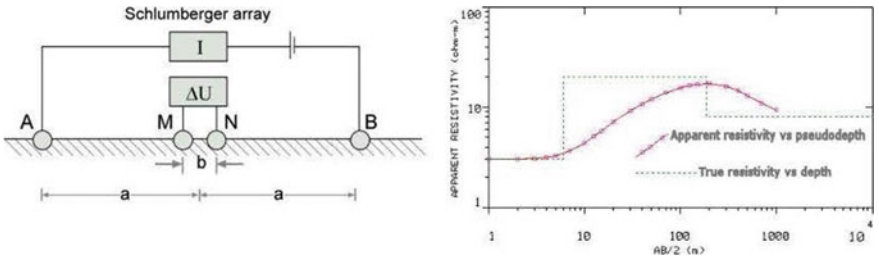


Fig. 2 Example of the apparent resistivity versus pseudo-depth curve and the true resistivity versus depth section for the Schlumberger array

level. The method provides a larger exploration depth for a given cable length and is less time consuming than the Wenner method. The dipole–dipole method (Fig. 1c) requires minimum cables lengths for a given depth, but suffers from the smallest signal-to-noise ratio.

In all DC resistivity methods, the exploration depth is achieved by increasing the separation (***a***) between the current and potential electrodes (Fig. 1). In general, the maximum separation should be at least three times greater than the required exploration depth. The voltages measured with successive separations are first transformed into apparent resistivities, namely the true resistivities of an equivalent half-space, which would produce the same signals as the actually measured ones. These apparent resistivities versus separations curves are used to build semi-qualitative resistivity pseudo-sections or are further transformed by inversion into true resistivity versus depth sections. An example of both the apparent resistivity pseudo-section and the true resistivity versus depth section is shown in Fig. 2 on the right side for the Schlumberger array inter-electrode configuration, which is shown on the left side.

Advantages

The DC resistivity methods are the most simple and intuitive GE/GM methods from both hardware and software points of view. The DC instruments are cheap and widely available. In fact, their most basic version can be easily produced in any electronic lab.

Drawbacks

The use of electrodes can be very challenging in some terrains with difficult electrical grounding conditions (e.g., dry sand dunes, hard rocks, boulders). The maximum inter-electrode separations should exceed the required exploration depth by roughly an order of magnitude. This inevitably decreases a lateral resolution of the method and distorts a quantitative estimation of the vertical resistivity distribution by 1-D inversion. In addition, the inversion suffers from particularly high non-uniqueness, as common for all potential field geophysical methods.

From the hydrological point of view, the DC methods are mainly applicable for the delineation the of the rock sequence, whether aquifers or aquicludes, in the vertical dimension. The application is restricted to a rather shallow depth, whereby

the sequence includes only a few units. The methods are not suitable in a tectonically disturbed area, where the length of a required measuring array is such that it crosses faults. In such a case, the results often represent a mixture of different tectonic blocks and 1-D inversion results might lead to erroneous hydrogeological interpretation.

2.2 The Electrical Resistance Tomography (ERT)

In order to significantly improve a resolution of the DC resistivity soundings, a multi-dimensional modification of the method was developed a few decades ago (e.g., Loke 2004). It realizes modern digital technologies, which provide automatic scanning and selection of any two pairs of electrodes among a large number of preinstalled electrodes along a profile (2D imaging) or within a selected area (3D imaging). The selected pairs are used as current and potential electrodes either in conventional arrays, as shown in Fig. 1, or in any other geometrical combination of equally spaced electrodes, required by the survey design.

An example of a field setup and the measurement procedure using the most popular Wenner inter-electrode configuration is exhibited herein (Fig. 3). In this example, 20 electrodes are preinstalled along a profile with the spacing between adjacent electrodes "a". First, all the possible measurements, with an electrode spacing of "1a", are carried out (Fig. 1). Electrode numbers 1, 2, 3 and 4 are used for the first measurement. Electrode 1 is used as the first current electrode C1, electrode 2 as the first potential electrode P1, electrode 3 as the second potential electrode P2 and electrode 4 as the second current electrode C2. Then, electrodes 2, 3, 4 and 5 are used for C1, P1, P2 and C2, respectively, for the second measurement and the measurements are repeated along the entire line of electrodes until electrodes 17, 18,

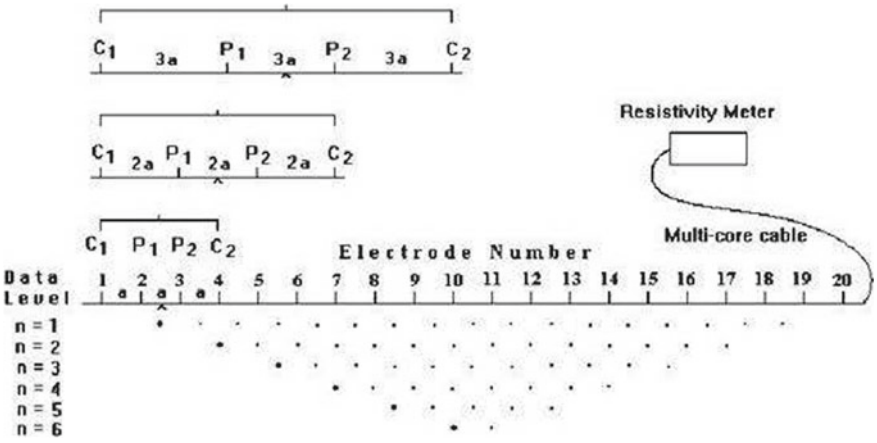


Fig. 3 Arrangement of electrodes for a 2D electrical survey and the sequence of measurements used to build up a pseudo-section (modified after Loke 2004)

19 and 20 are used for the last measurement with "1a" spacing. For a system with 20 electrodes, there are 17 (20–3) possible measurements with "1a" spacing using the Wenner array.

Upon completing the measurements with "1a" spacing, the next sequence of measurements with "2a" electrode spacing is carried out. Electrodes 1, 3, 5 and 7 are used for the first measurement. The electrodes are selected with "2a" spacing between adjacent electrodes. For the second measurement, electrodes 2, 4, 6 and 8 are used. This process is repeated along the entire line until electrodes 14, 16, 18 and 20 are used for the last measurement with "2a" spacing. For a system with 20 electrodes, there are 14 ($20 - 2 \times 3$) possible measurements with "2a" spacing.

The same procedure is repeated for measurements with "3a", "4a", "5a" and "6a" spacings between electrodes. Each set of measurements "n" corresponds to a certain pseudo-depth, whereby the greater "n" yields the greater pseudo-depth (Fig. 3). Since each measurement is related to the middle of the array, the deeper parts of the pseudo-section are shorter and the resulting pseudo-section is represented by a trapeze (Fig. 3). Depending on the array, there are different methods to calculate the pseudo-depth, but in the case of the conventional Wenner array, the pseudo-depth is normally equal to the spacing (Loke 2004).

The measured data are customarily presented in the form of the apparent resistivity pseudo-section (Fig. 4a), which provides a semi-quantitative picture of the

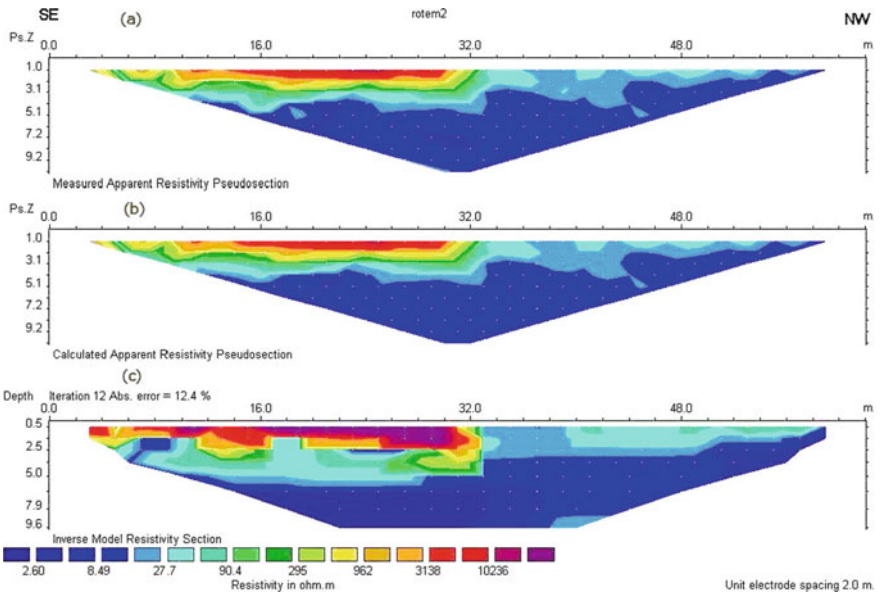


Fig. 4 Example of the ERT resistivity cross-sections. **a** Measured apparent resistivities versus pseudo-depths, **b** calculated apparent resistivities versus pseudo-depths for the best-fit model, **c** best-fit true resistivity versus depth model

true subsurface resistivity distribution. Such a presentation is feasible for the visualization of the measured apparent resistivities and as an initial model for further quantitative interpretation. In order to obtain quantitative information regarding the true resistivities and depths by using the above apparent resistivity pseudo-section, a further 2D inversion of the measured data is required (Loke 2004). An example of the inverted true resistivity versus depth model and the calculated synthetic response for this model is shown in Fig. 4b, c, respectively. A visual comparison of Fig. 4a, b shows how well the data are fitted to the measurements at different parts of the section. An overall quality of the inversion is controlled by the total RMS (root mean square) error (Fig. 4).

Advantages
In principle, the ERT method allows a quantitative determination of the resistivity distribution in all three dimensions, although at present, full 3D surveys are still rare. By increasing the number of electrodes, one can provide a high lateral resolution. The multi-dimensional forward and inverse modeling is well developed.

Drawbacks
Similarly to 1-D resistivity soundings, the use of electrodes can be most challenging in some terrains with difficult electrical grounding conditions (sand dunes, hard rocks, boulders). The inversion suffers from non-uniqueness, which is not adequately treated in the available software.

2.3 The Time Domain Electromagnetic (TDEM) Method

The TDEM sounding system consists of a transmitter (Tx) and receiver (Rx) antennae. Contrary to the DC resistivity methods, the current waveform driven through Tx is not a steady state, but consists of equal periods of time-ons and time-offs (alternative current). This current creates a primary magnetic field, which varies with time. In accordance with Faraday's law, rapid changes of the current and thus of the primary magnetic field during switch-ons and switch-offs induce an electromotive force in the subsurface. As a result, eddy currents are created everywhere in the conductive ground, the intensity of which at each point depends on the ground resistivity or conductivity. These eddy currents generate a secondary magnetic field, the time rate-of-change of which is measured on the earth surface. Thus, contrary to the DC resistivity methods, the use of electrodes as transmitters and/or receivers is only optional and not necessary in the TDEM method. Instead, an ungrounded loop of wire can be used as a transmitter and as an example a small air coil as a receiver. If the signal is only measured during time-offs, i.e., in the absence of the primary field, the receiver can be placed everywhere, both outside and inside the transmitter loop (Fig. 5), or can even be combined with the latter. Unlike the DC resistivity methods, the exploration depth in the TDEM method is not determined by the Tx-Rx separation, but rather by the time after the Tx current is switched off.

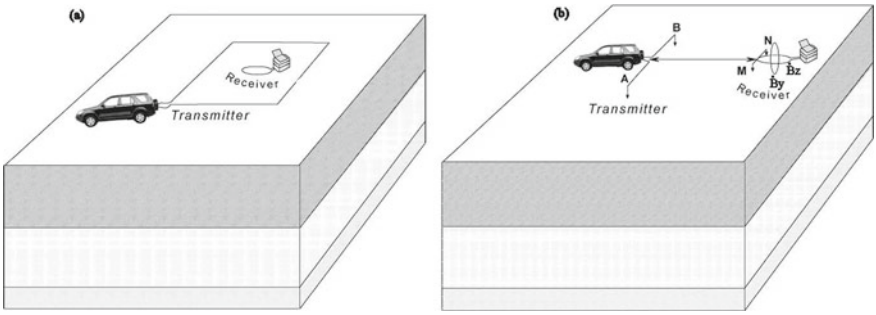


Fig. 5 Typical TDEM arrays. **a** Central loop, **b** grounded line—Ex, Bz, By

Data processing and interpretation in the TDEM method are fairly similar to those in conventional DC resistivity one. The measured signals are normally first converted to apparent resistivities and then inverted in terms of 1-D layered models. The main difference stems from a more complicated definition of the apparent resistivity, which is equivalent to a true resistivity of the half-space, not in the entire measurement time range, but only at sufficiently late times (Kaufman and Keller 1983). A typical example of the collected data in form of the late-stage apparent resistivities (left side) and the appropriate inverted model (right side) is shown in Fig. 6.

In contrast to all other GEM techniques, the time domain/transient electromagnetic (TDEM/TEM) method was developed in two different modes depending on the

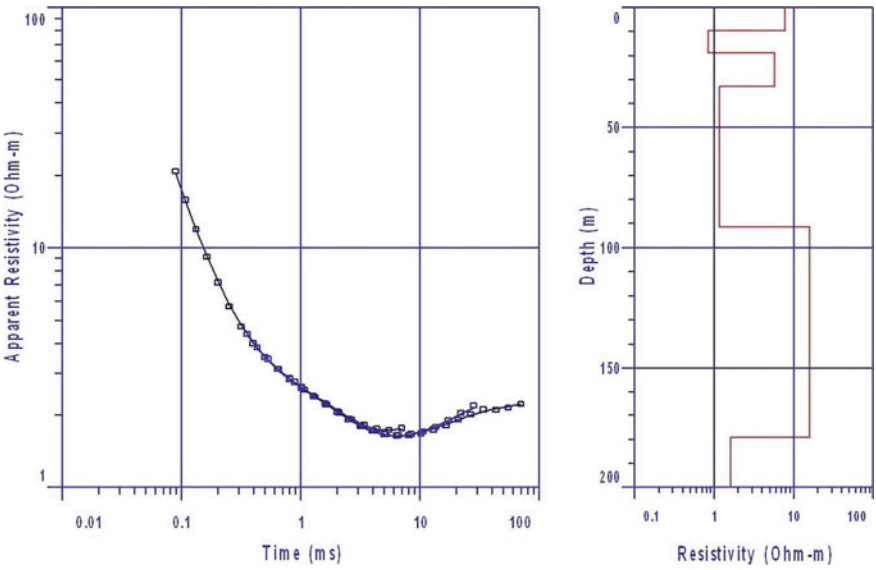


Fig. 6 An example of the late-stage apparent resistivity TDEM data (open squares) and the best-fit curve (left side) for the best-fit model (right side)

transmitter–receiver separations used. This duality was caused by the specific current waveform, including both time-on and time-off periods employed in the method. If the measurements are only carried out during the time-off periods, the response from the ground is entirely decoupled from the primary field. As a result, the receiver can be located at any distance from the transmitter, and the transmitter and receiver antennae can even be combined in a single antenna. This means that, theoretically, the method provides an exploration depth, which is independent on the Tx-Rx separation (offset). However, in reality, signal-to-noise considerations might force to place the receiver antenna far away from the transmitter (far-field) due to a very fast decay of the signal as a function of time in the close proximity to Tx (near-field). On the other hand, increasing offset causes the deterioration of the resolution and increases undesirable multi-dimensional effects caused by geologic noise. These contradictory considerations actually led to the development of two independent TDEM (TEM) methods. The first one, which uses short offset arrays, significantly shorter than the depth to the target, is named SHOTEM or near-zone TEM. The second one is named LOTEM or far-zone TEM.

SHOTEM

The array commonly used in the SHOTEM method is named central loop (CL) array (Fig. 5a). It consists of a transmitter (Tx) antenna represented by a square, sometimes rectangular, or a circular loop of insulated wire laid on the earth surface. The receiver (Rx) antenna can be either another smaller loop of wire or a multi-turn air coil placed in the center of the Tx loop or at any other convenient place inside the Tx loop ("in-loop array"). By using this array, one can detect a target at depths two to three times exceeding the Tx size. Sometimes, mainly for mineral exploration, the Rx coil is placed outside the Tx loop (slingram array) or the signal is picked-up by the same Tx loop (coincident loop or one-loop array).

Less commonly used SHOTEM arrays are those comprising electrical dipoles (Fig. 5b). This is mainly caused by current injection problems, if the dipoles are only used as transmitters. However, if the electrodes are also used to pick up the signal (electrical components), the problem is more complicated. Due to a very fast decay of the transient signal with time, most of the measured responses turn out to be severely distorted by induced polarization (IP) and/or spontaneous or self-potential (SP) effects.

The situation has been changed dramatically, when the SHOTEM method was applied in the marine environment. Both the current injection and measurement problems have been successfully overcome due to extremely low contact resistances, very homogeneous grounding conditions and, particularly, due to the slow decay of the signal in an electrically conductive environment (Goldman et al. 2011).

LOTEM

The traditional LOTEM array, which was mainly developed to explore presumably 1-D earth, consisted of a long grounded transmitter line and broadside horizontal electric dipole (Ex) and/or vertical magnetic dipole (dBz/dt) receivers (Vanyan 1967). The modern LOTEM array is also mainly broadside, but generally measures five

components of the EM field (Strack 1992), three of which are shown in Fig. 5b. The principal difference between the SHOTEM and LOTEM methods lies in the Tx-Rx separation used. Whereby the SHOTEM offset could be 2–3 times smaller than the depth to the target, the offsets used in LOTEM are normally 3–5 times larger than the depth to the target. Thus, offsets difference between the two methods for the same target depth might be in an order of magnitude. Such a huge difference leads to important geophysical consequences in lateral resolution and multi-dimensional effects.

In contrast to SHOTEM, the LOTEM signals can be measured during either turn-off or turn-on periods. In both cases, the vertical resolution of the electric field is similar to the one exhibited in the DC limit, to which the signal approaches at early times or at late times correspondingly. The lateral resolution is inherently poor due to the large offsets. The LOTEM signals decay with a time much slower than in the case of SHOTEM, thus increasing the signal-to-noise ratio in the LOTEM method as compared to the SHOTEM method for the same exploration depth. The ability to measure the electrical fields makes the LOTEM method superior to the traditional (on-land) SHOTEM one, if the target is resistive and relatively thin. However, the situation has become different in the marine environment in the presence of the discovered coastal effect, which significantly enhances the relative target response of resistive, including thin, structures in the case of the marine SHOTEM method (Goldman et al. 2011).

Advantages

The SHOTEM method provides the best lateral resolution of all 1-D GE/GEM sounding methods. With regard to conductive targets, it also demonstrates an excellent vertical resolution. By using ungrounded transmitter loops, it can be applied practically in any terrain.

The LOTEM method is very efficient in deep soundings and, despite the use of current and measurement electrodes, it can be applied practically in any terrain as it requires the grounding of just one pair of current electrodes per sounding.

From the hydrogeological point of view, the SHOTEM method has an advantage as compared to the DC and LOTEM methods. Since the dimensions of the transmitter loop are relatively small with regard to the required exploration depth, the measurements are considerably less affected by neighboring tectonic blocks and other lateral inhomogeneities.

Drawbacks

Due to extremely fast signal decay, the SHOTEM method exhibits rather poor signal-to-noise characteristics. The LOTEM method suffers from a fairly low lateral resolution.

The equipment used in both SHOTEM and LOTEM is expensive, and the interpretation software is rather complicated and is still mainly one-dimensional.

2.4 The Surface Nuclear Magnetic Resonance (SNMR) Method

In order to avoid undesirable connotations with the term "nuclear," the method is more widely named "proton magnetic resonance" (PMR) and "magnetic resonance soundings" (MRS). Although the method is based on different physical principles than GE/GEM methods, due to a great similarity in almost all aspects of applications (primary field excitation, measurements setup, data processing and interpretation), it is commonly considered among the GEM methods.

The greatest difference between the SNMR method and all GE/GEM, and in fact between all other geophysical methods, is that its inversion does not yield physical parameters, such as electrical resistivity, dielectric permittivity, acoustic velocity, but rather hydrogeological ones, such as water content and transmissivity of the aquifer. The former is directly proportional to the amplitude of the measured signal, whereby the latter is closely related to the speed of its decay with time. This unique feature of the SNMR method greatly reduces an ambiguity of the final hydrogeological interpretation, which in the case of SNMR mainly includes non-uniqueness of the geophysical inversion by itself.

The main principles of the SNMR method are shortly described below (e.g., Goldman et al. 1994a).

1. Hydrogen nuclei (protons) in water possess nonzero magnetic moments.
2. In the presence of geomagnetic field, $\mathbf{B_0}$, the protons in groundwater molecules are aligned in parallel to the field (Fig. 7a). In this equilibrium state, the protons are able to absorb electromagnetic energy that has been emitted at the Larmor frequency, f_0, where $f_0 = gB_0$ and g is the gyromagnetic ratio ($g = 4.254597 \times 10^{-2}$ radians per second per nano-Tesla).
3. If a primary electromagnetic (EM) field oscillating at the Larmor frequency is applied, the magnetic moments of the water molecules begin to precess from their initial position (Fig. 7b).

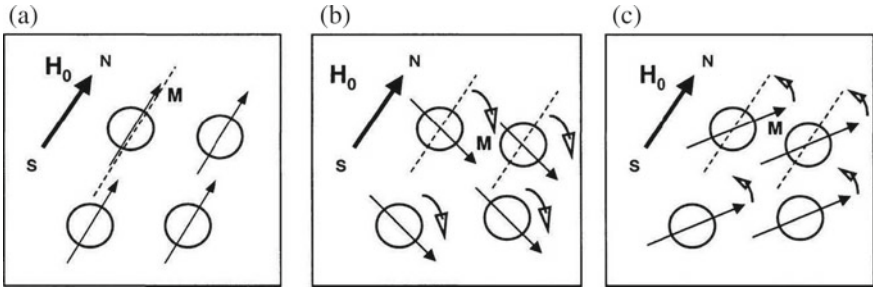


Fig. 7 General principles of the SNMR method. **a** Water protons in presence of geomagnetic field H0, **b** water protons having magnetic moments M in presence of both geomagnetic and external magnetic field emitted at the Larmor frequency, **c** water protons after the external magnetic field is switched off (modified after Legchenko et al. 2002)

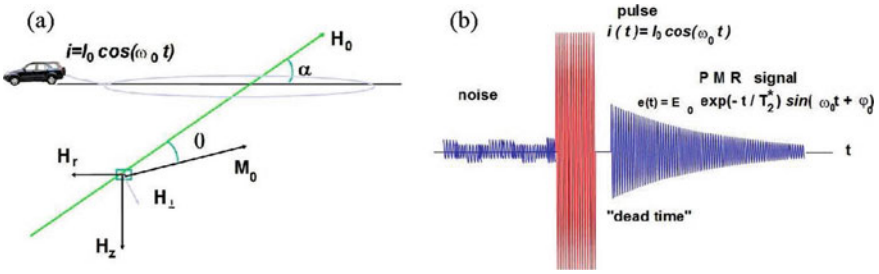


Fig. 8 **a** Typical SNMR field setup, **b** SNMR signal measured during all three stages shown in Fig. 7a–c, respectively (after Legchenko and Valla 2002)

4. When the applied EM field is stopped, the magnetic moments of the water molecules return to their original positions, generating a secondary electromagnetic field (Fig. 7c). This field also oscillates at the Larmor frequency and exponentially decays with time due to heat losses. The secondary field consists of the important information regarding groundwater and is measured at the earth surface by the geophysical NMR instrument.

A typical SNMR field set up is presented in Fig. 8a. Figure 8b shows EM signals measured by the SNMR instrument during the above-mentioned three stages (Fig. 7a–c).

The initial amplitude of the secondary field is directly proportional to the amount of groundwater, irrespective the hydraulic conductivity/transmissivity of the water-bearing rocks. Fortunately, the measured signal decays differently, depending on the transmissivity. The greater is the time constant T_2^*, the slower is the decay and the greater is the transmissivity of the aquifer. Experience shows that T_2^* for most aquicludes is less than 50 ms. The real measurements normally start after this time, named "dead time" whereby the water is detected in aquiferous rocks only.

In order to perform soundings, namely to discriminate signals from different depths, one must to use an appropriate sounding parameter. Unlike the previously described methods, the sounding parameter in the SNMR method is a product of the current pulse amplitude and the duration of the pulse, named as "pulse parameter" (i.e., the red area in Fig. 8b).

Thus, the SNMR sounding curve is represented either by the initial signal amplitude (extrapolated to the zero time from the actually measured amplitude after the "dead time") versus the "pulse parameter" or by the measured T_2^* also versus the "pulse parameter." The former curve allows to calculate the water content as a function of depth, whereas the latter curve allows to estimate the distribution of transmissivity with depth. Both estimations are carried out by means of inversion. The appropriate examples of sounding curves are shown in Fig. 9a, b.

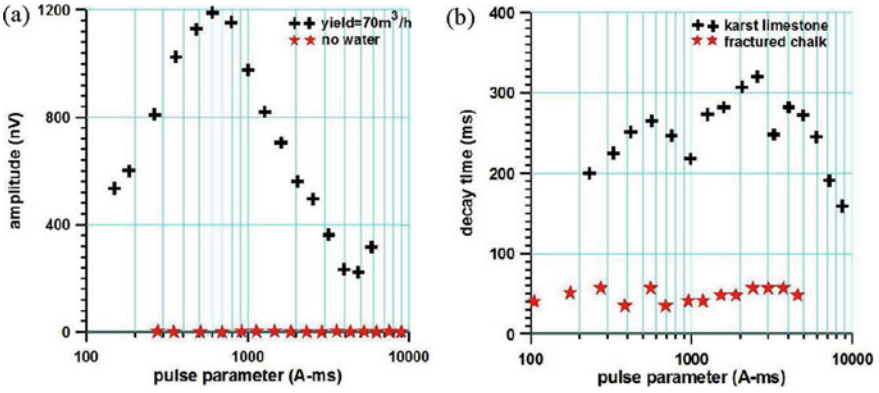


Fig. 9 Examples of the SNMR data. **a** Initial amplitude versus pulse parameter, **b** time constant (decay time) T_2^* versus pulse parameter (after Legchenko and Valla 2002)

Advantages

The SNMR method is the only geophysical tool, which allows a direct detection of groundwater and quantitative estimation of the water content within aquifers and, under favorable conditions, the transmissivity of the aquifer.

Drawbacks

Due to the necessity to use a specific and relatively high frequency (roughly 2 kHz), the method has principally a limited exploration depth (roughly 100 m). The instrument is expensive, and measurements are very slow.

3 Geoelectric Characterization of Groundwater Salinity in Clastic Coastal Aquifers

3.1 The Mediterranean Coastal Aquifer

The regional coastal aquifer is one of the major water resources of the country. As such, due to the heavy water demand, the aquifer is heavily over-exploited. This in turn leads to water table decline below sea level, resulting in the encroachment of seawater into the aquifer up to approximately 2 km inland. Due to its importance, the hydrogeology of the aquifer was dealt with numerous studies. The following is a concise summary of the configuration of the aquifer, based on Kafri and Goldman (2006) (see also Chapter "General Information and Hydrogeology of the Mediterranean and Dead Sea Coastal Aquifers and Their Relation with Their Base Level").

The aquifer, approximately 110 km long, extends from the Gaza Strip in the south to the Mount Carmel in the north. It is divided into strips, 2 km wide, perpendicular

to the shoreline (Fig. 10). The west–east (hereafter W-E) extension of the aquifer changes from some 5 km in the north to about 30 km in the south (Fig. 10).

A generalized hydrogeological cross-section of the aquifer which includes also the lithological units, the water table and the seawater intrusions into the different sub-aquifers is shown in Fig. 11. The thickness of the aquifer is approximately 150 m close to the shoreline, and it decreases to the east and to the north. The aquifer consists of a Quaternary sequence of alternating marine and continental deposits, composed of calcareous sandstones, sands, clay and loam interlayers. The entire sequence overlies impermeable shales and clays of the Pliocene Saqiye group which form the base of the aquifer. In the western portion of the aquifer, up to some 4 km from the

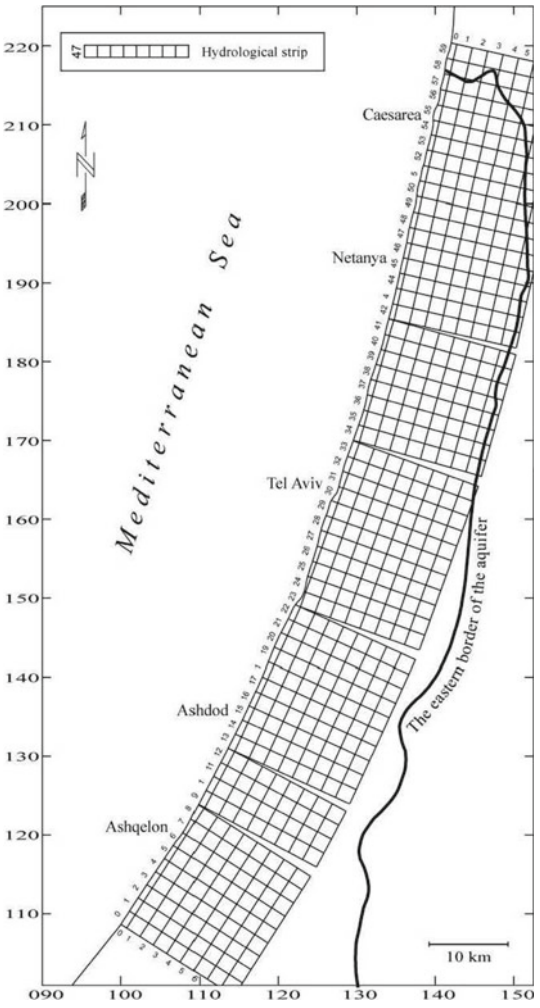


Fig. 10 Schematic map of the Mediterranean coastal aquifer of Israel

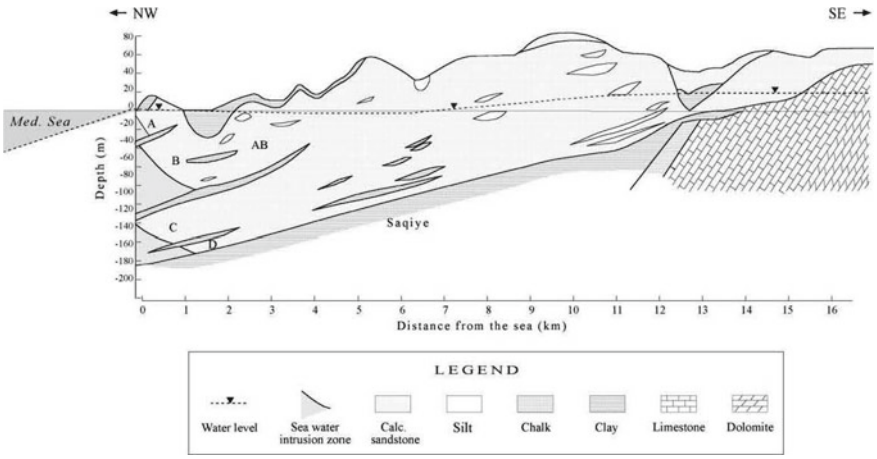


Fig. 11 Typical hydrogeological cross-section of the Mediterranean coastal aquifer

shoreline, the above-mentioned impermeable, mostly clay, horizons subdivide the aquifer into generally four sub-aquifers from top to bottom (Fig. 11), denoted herein as A (phreatic), B (phreatic or confined), C and D (confined) sub-aquifers.

As of today, the exploitation of the aquifer is being carried out mostly from the upper (A and B) sub-aquifers. However, fresh or brackish water was encountered in the lower C and D, sub-aquifers (hereafter CD), below a seawater intrusion in only a few existing deep observation wells. These sub-aquifers were seldom exploited despite the growing water demand, and the fact that these sub-aquifers are protected from surficial pollution. As a result, very little was known as to this rather deep freshwater body and whether it is subjected to seawater encroachment.

Both problems are crucial for rational exploitation of the lower sub-aquifers, namely the potential of freshwater as well as to its protection from salinization in case that the CD are blocked to the sea, as suggested by Kolton (1988).

The DC resistivity method

Since the 50s of the last century, an intensive drilling program for groundwater exploitation was carried out in Israel. The drilling was carried out in both the shallow coastal clastic aquifers and the rather deep mountain carbonate aquifers. In many cases, in order to obtain a drilling forecast of the lithological sequence, DC resistivity (Schlumberger) soundings were carried out at the chosen site prior to the drilling. This was done even in the case of a multi-layered subsurface deep sequence. In general, the success in accurately defining and delineating the different lithological units was found to be poor due to the limitations of the DC method as described above.

The first geoelectric measurements targeted to detect and monitor the fresh–seawater interface in the Mediterranean coastal aquifer were carried out in the 60s and 70s of the last century using the DC resistivity (Schlumberger) soundings (Ginzburg

and Levanon 1976). The measurements, carried out near existing observation wells, showed the feasibility of the method to detect and roughly evaluate the depth to the interface and the resistivity of seawater intrusion. However, the accuracy of the measurements and interpretation was rather low due to the above-mentioned drawbacks of the DC resistivity soundings. For example, Fig. 12 shows the apparent resistivities measured in 1960 and 1975 at the same location near an observation well drilled into the Mediterranean coastal aquifer (Ginzburg and Levanon 1976). It can be seen that the obtained resistivities for the seawater intrusion are greatly overestimated (roughly 5 ohm-m instead of the real 1.5 ohm-m). Unfortunately, no quantitative borehole information regarding the depth to the interface has been provided in the article, but the general conclusion was that the vertical resolution of the DC resistivity soundings is insufficient for monitoring the seawater intrusion (Ginzburg and Levanon 1976).

On-land TDEM measurements

In order to increase the accuracy and both lateral and vertical resolution of the geophysical measurements, the TDEM method was applied for the first time in Israel in the mid-80s of the last century (Goldman et al. 1988). The first feasibility study in the Mediterranean coastal aquifer included just four central loop TDEM stations located in the Nitzanim area, roughly 1 km to the south of observation wells 12-0, 12-B and 12-A (Fig. 13). One of the stations (N4) was located directly on observation well T_1. The interpretation of all collected data revealed a very low-resistivity body, gradually dipping from west to east, which was undoubtedly identified with a seawater intrusion. The resistivity of that unit was extremely consistent varying within a narrow range between 1.4 and 1.8 ohm-m. The most exciting result was obtained at station N4. The interpretation of the data collected at this station detected

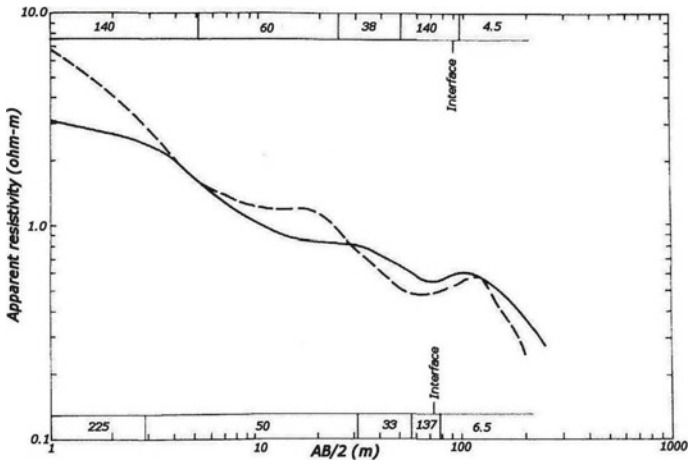


Fig. 12 DC apparent resistivities measured by the Schlumberger array in the Mediterranean coast of Israel. Solid line, in 1960; dashed line, in 1975 (modified after Ginzburg and Levanon 1976)

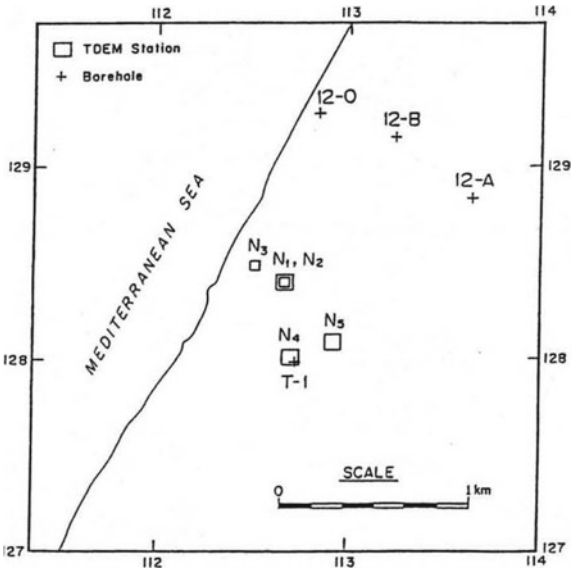


Fig. 13 Location map of the TDEM feasibility study survey at the Nitzanim area

the top of the very low-resistivity body at a depth of 70 m. According to the measured borehole salinity (conductivity), the fresh–seawater interface on top of a half sea water salinity (17,500 mg/l TDS) was encountered at the depth of 67 m, whereas a seawater salinity of 35,000 mg/l TDS was encountered at a depth of 70 m (Goldman et al. 1988).

Such an exceptionally high accuracy, however, needed further statistically significant verification, especially taking into account that the use of TDEM in ground water exploration was pioneering at that time, not only in Israel but all around the world. Therefore, the next feasibility study, carried out shortly after the first survey, included 100 TDEM stations distributed throughout the entire coastal plain, predominantly in proximity of observation wells (Goldman et al. 1991). The main results of the survey are shown in Fig. 14. It is evident that the resistivities of the seawater-intruded coastal aquifer vary in a narrow range between 1.1 and 2.9 ohm-m, with the mean value, of 1.7 ohm-m. Out of 31 measurements, which detected the resistivities in the above range, 21 showed the discrepancy between the geophysical and the borehole measurements of less than ±5 m, while greater errors were observed in case of a wide transition zone or due to remoteness from an observation well (Goldman et al. 1991). It should be noted that such differences can also stem from the fact that due to temporal groundwater-level fluctuations, the actual depth to the interface at the time of the TDEM measurements has already changed since the borehole salinity was measured.

Based on the above results, it was decided to perform regular TDEM measurements at locations of abandoned observation wells to monitor the sea water intrusion and to map the interface in areas of scarce hydrogeological information.

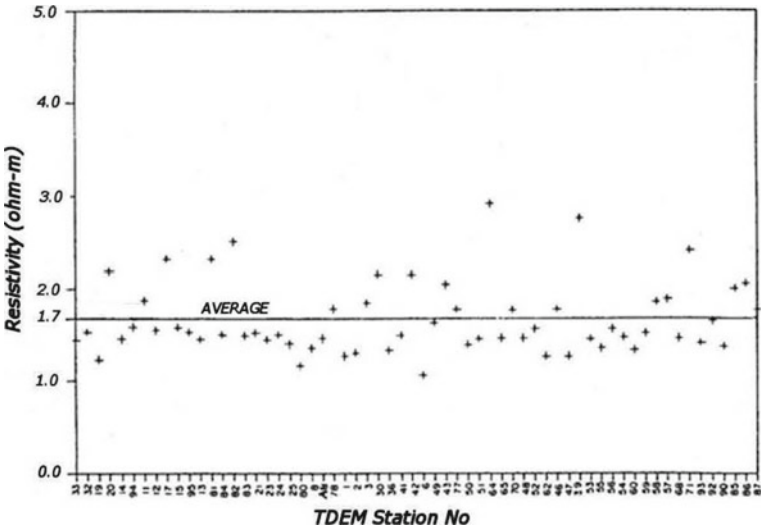


Fig. 14 TDEM resistivities of the sea water intrusion into the Mediterranean coastal aquifer (modified after Goldman et al. 1991)

During the last decade of the past century, roughly 700 TDEM measurements were carried out throughout the entire coastal aquifer. In the course of these surveys, resistivity "reversals" (higher resistivities below lower ones) were detected in the CD of the coastal aquifer, indicating that they might be saturated with fresher water underneath the seawater-intruded upper sub-aquifers. This assumption has been successfully verified near one of the very few observation wells, which at that time managed to measure the water salinity within the CD. Figure 15 shows the comparison of borehole water salinities (conductivities) measured in Ashdod 112 borehole with the resistivities obtained by the nearby surface TDEM measurements.

One can see that higher TDEM resistivities opposite to CD correspond to lower than seawater salinities measured in the well at appropriate depths. Similar resistivity behavior has been found in many TDEM measurements both isolated and along traverses (Fig. 16).

Based on the above, a TDEM survey aimed to reveal the resistivities of the entire sequence of the aquifer, including those of the underlying Saqiye Group aquiclude, was executed along the entire coastal plain (Kafri and Goldman 2006). A hundred TDEM measurements were done, followed by drilling of ca 200-m-deep observation wells, in which water conductivity logging was performed in all four sub-aquifers. To enable correlation, 14 TDEM measurements have been carried out adjacent to these observation wells. In all cases, the interface that was detected in the wells within the upper sub-aquifers was also accurately detected by the TDEM measurements (Fig. 17). The bulk resistivity of the seawater-intruded portion of the aquifer varied

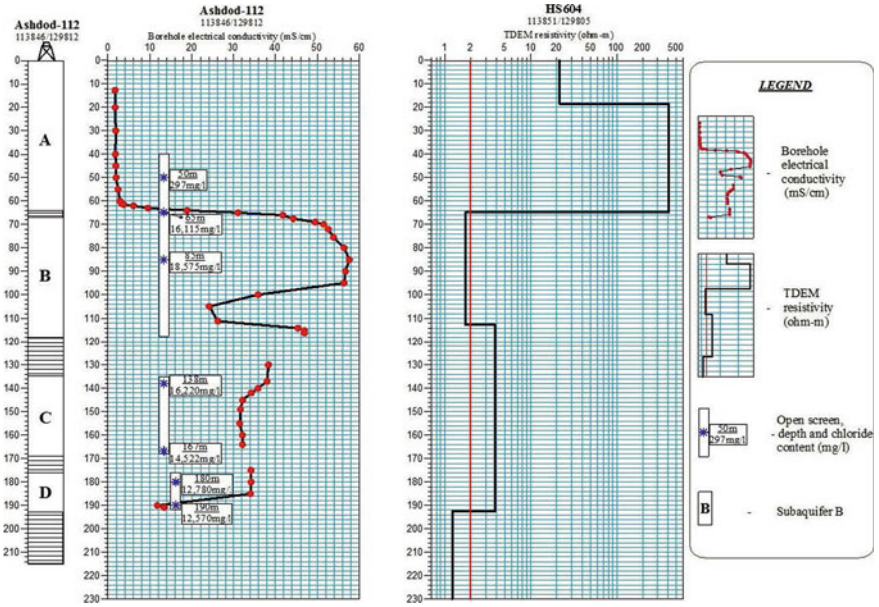


Fig. 15 Comparison between the EC profile measured in the observation well Ashdod-112 and TDEM resistivity cross-section obtained in the nearby TDEM station HS 604

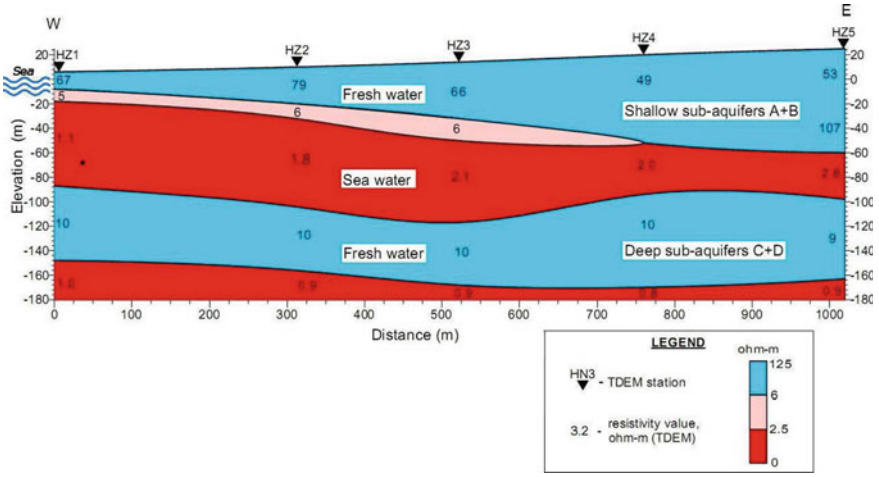


Fig. 16 Quasi 2D resistivity cross-sections along the TDEM traverse perpendicular to shoreline in the Ziqim area

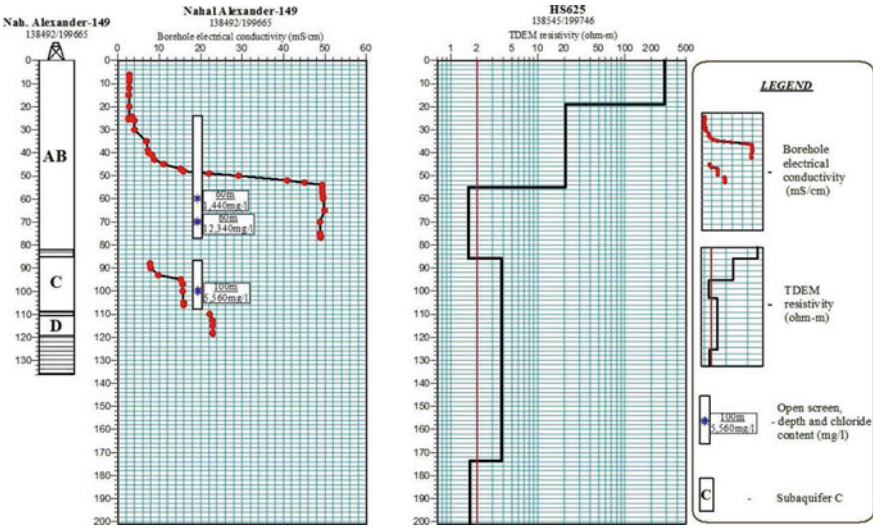


Fig. 17 Correlation of TDEM resistivities and borehole conductivity measurements in the observation well Nahal Alexander-149. Dashed line roughly represents the threshold resistivity value for seawater

between 1.5 and 1.9 ohm-m, averaging a value of 1.7 ohm-m as found in the previous TDEM surveys that were correlated with mostly shallow observation wells (Goldman et al. 1991).

An increase of the resistivity was observed below the depth of 86 m within the CD sub-aquifer as shown in Fig. 17. This is in accordance with the adjacent borehole, significantly lower water conductivity, than that of the overlying portion identified as seawater intrusion.

In contrast to the above, Fig. 18 shows an example whereby the borehole water conductivity profile does not fit the relevant lateral TDEM resistivities profile within the deep sub-aquifers, despite the good fit in the upper sub-aquifers. High TDEM resistivities and low borehole conductivities were found down to the approximate depth of 70 m, as expected. Below this depth, a consequent drop of resistivity and a parallel rise of conductivity are observed. A further drop of bulk resistivity to around 2 ohm-m (equivalent to seawater salinity) is observed at a depth of 90 m, whereby the borehole water conductivities below this depth were low.

This same phenomenon was found in approximately 20% of the correlation measurements. It should be noticed that in all these cases, the TDEM resistivities within the CD consistently reflect higher salinities than those actually analyzed in the observation wells water samples. It implies that the disagreement is caused by a real hydrogeological setup rather than by measurement errors or an ambiguous geophysical interpretation (Goldman et al. 2003). The most reasonable proposed explanation to the above disagreement is as follows;

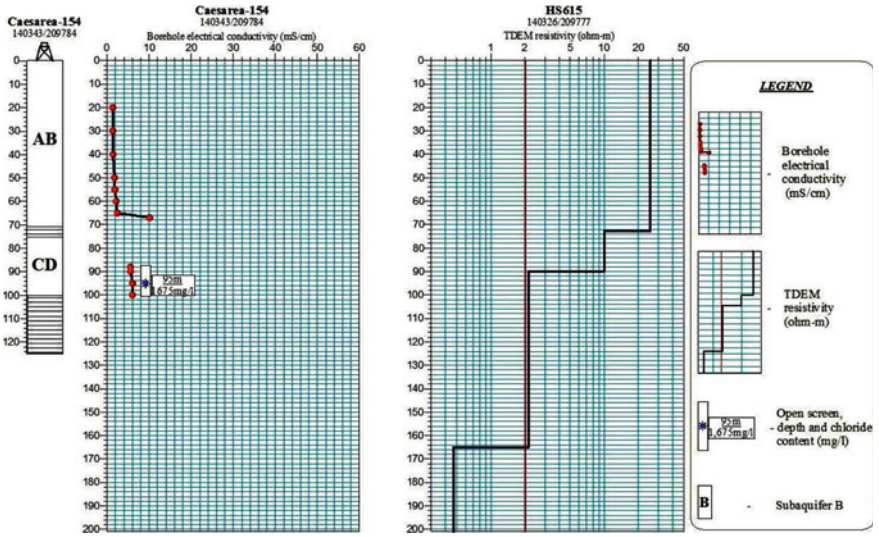


Fig. 18 Correlation of TDEM resistivities and borehole conductivity measurements in the observation well Caesarea-154. Dashed line roughly represents the threshold resistivity value for seawater

Water salinities that were obtained through analysis of borehole samples reflect only the aquiferous portion of the sequence which contains freshwater. At the same time, the TDEM bulk resistivity measurements of a certain sequence include and integrate also the clay interlayers which seldom contain trapped or interstitial saline water of low resistivity. The latter are more abundant, proportion wise, in the lower sub-aquifers as compared to the upper ones. The end result is that the upper sub-aquifers exhibit higher integrated bulk resistivities, whereas the lower ones exhibit lower bulk resistivities due the low resistivity contribution of the impermeable saline interlayers.

Therefore, the ability of TDEM to accurately characterize the water salinity within the lower sub-aquifers is limited since it strongly depends on the sparse available information regarding aquifer/aquiclude thickness ratio, the depth to the lower sub-aquifers, as well as the salinity distribution within the upper sub-aquifers. However, taking into account the small probability of misinterpretation (three out of fourteen calibration measurements) and the fact that TDEM consistently overestimates the water salinity in the CD, one might expect that the method can be successfully applied to the regional mapping of the entire coastal plain or to its significant parts. This is particularly true regarding those parts, where the resistivities detected in the CD are fairly high and continuous in space.

The distribution of the TDEM resistivities of the CD and those of the underlying Saqiye beds (designated S) is exhibited on a longitudinal profile parallel and close to the seashore (Fig. 19). One can see continuous high resistivities within the CD along the segment between 27 and 47 km of the profile (North Ashdod-Bat Yam).

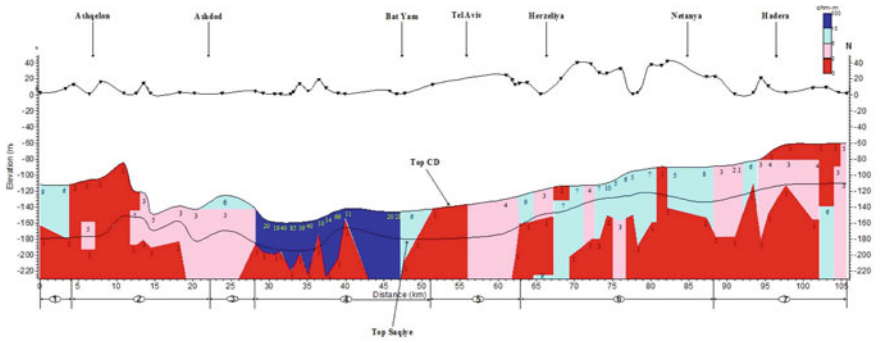


Fig. 19 Quasi-2D resistivity cross-section along the entire Mediterranean coastal line of Israel (after Kafri and Goldman 2006)

Regarding this part of the coastal aquifer, it is evident, with confidence, that the CD is occupied by freshwater. Along the segment between 70 and 87 km of the profile, the CD is apparently saturated with brackish water. In the rest of the coastal aquifer, the water in the CD is either saline or, if fresh, the thickness of the CD is most likely relatively small.

Since only one measurement is available in the entire metropolitan Tel Aviv area (roughly between kilometers 50 and 60 of the profile), the detected low resistivity of CD could be caused by the above-mentioned artifact and the real extent of the freshwater body within the CD might be more than 30 km. In order to estimate the real south–north and east–west dimensions of the discovered freshwater body within the CD, an offshore continuation of the on-land TDEM measurements was required. These measurements were necessary to estimate the amount of freshwater within the CD, and also to decide whether the CD is laterally blocked to the sea by the Saqiye clays, or alternatively whether it is hydraulically connected to the sea.

Offshore TDEM measurements

In the case of the offshore TDEM measurements, the required target is considerably more resistive than the background medium and thus the LOTEM method provides a better detectability than the SHOTEM method, as noted above. On the other hand, accurate detection of the target boundaries requires a better lateral resolution of the SHOTEM method. Therefore, the offshore survey included both methods.

The marine SHOTEM method

The use of the above-described conventional SHOTEM central loop array is not feasible in a wavy sea, where laying down a large transmitter loop is a real challenge. The use of the theoretically equivalent so-called AB-loop array (also known as Ex-Bz) consisted of a grounded transmitter line and an equatorial small receiver coil is technically simpler in the sea. Contrary to on-land applications, grounding current electrodes in water is very simple. Calculations show that in a shallow sea (up to 50–100 m deep) the sensitivity of the marine SHOTEM method with regards to a sub-seafloor resistive target (e.g., freshwater) practically does not depend on the location

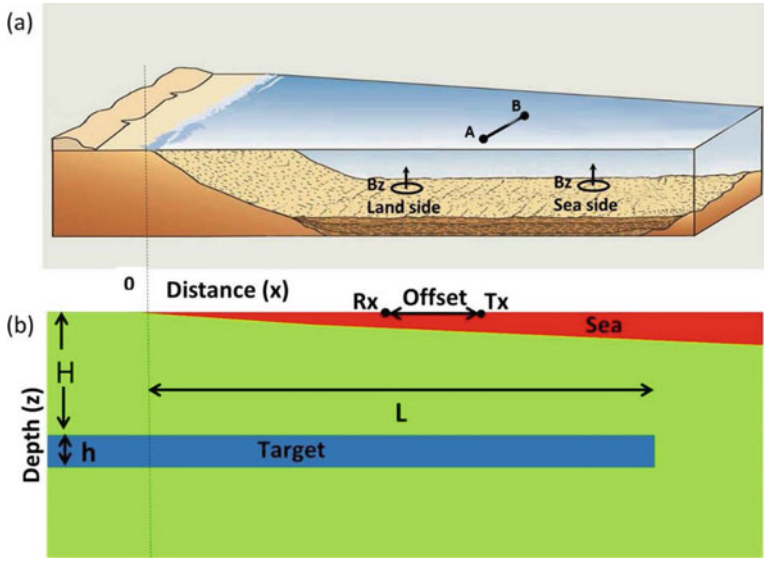


Fig. 20 **a** Marine TDEM system; **b** conceptual geoelectric model (modified after Levi and Goldman 2017)

of either the transmitter or the receiver within the sea. Therefore, due to technical considerations, the selected array consisted of a floating surface transmitter line (AB) and a bottom located receiver coil (Bz) (Fig. 20a).

In a 1-D Earth, the measured signal is invariant with respect to which side of the transmitter line the receiver coil is located. However, this is not the case in the presence of a strong 2D resistivity contrast, which exists between the very conductive sea water and the relatively resistive coast. Both feasibility study measurements and the follow-up 2D/3D calculations showed that the land-side AB-Bz array (Fig. 20a) is considerably more sensitive to the presence of the sub-seafloor resistive target than either the sea-side AB-Bz (Fig. 20a) or the signal in the 1-D case, where no sea–land resistivity contrast exists. This coastal effect was investigated and discovered during the marine TDEM survey offshore the Palmahim beach in Israel. It provided a new and highly efficient tool for exploration of sub-seafloor resistive targets (Goldman et al. 2011). Indeed, it exhibits a high lateral resolution, as all conventional SHOTEM methods do, but contrary to them, it was found to be also very sensitive to the presence of high-resistivity sub-seafloor targets. The use of the marine LOTEM method remains however important, since the discovered coastal effect only takes place at relatively short distances from the shoreline, up to 3 km under specific bathymetric and the sub-bottom geoelectric conditions of the study area.

The offshore SHOTEM measurements included four marine campaigns, during which a total of 35 measurements, using 20 different transmitter locations, were carried out. The first two campaigns were used as a feasibility study that tested different arrays and acquisition systems. The results of the feasibility study led to the

development of a novel highly efficient land-side marine TDEM system, described in details by Goldman et al. (2011). The next two campaigns (Levi et al. 2018) included a 15-km south–north profile (P-1) between Bat Yam and Tel Aviv parallel to the Mediterranean shoreline at an approximate distance of 600 m offshore (Fig. 21). The profile consisted of eight measurements, including those carried out during the previous feasibility study campaigns. In addition, two east–west profiles were run perpendicular to the shoreline up to approximately 3000 m offshore (Fig. 21), up to a water depth of 33 m. The southern profile (P-2) aligned south of Bat Yam included four measurements, and the northern profile (P-3) included three measurements.

All the measurements along the south–north (hereafter S-N) profile were interpreted using a standard 1-D inversion software (TEMIX-XL and IX1D by Interpex Ltd.) and were verified afterward by the appropriate 2D forward modeling, which included the sea–land resistivity contrast. Typical examples of the acquired data and the interpreted 1-D models are shown in Fig. 22 in case of an existing target (a) and in the absence of a target (b).

The interpretation of the E-W measurements was complicated. In order to determine the western boundary of the target as accurate as possible, the initial quasi-2D resistivity cross-section obtained by 1-D inversions was not simply verified, but was rather used as an initial model for a further full 2D trial-and-error interpretation (Levi et al. 2018). The resulting real 2D model, obtained from P-2, is shown in Fig. 23.

Based on all 35 offshore measurements, the interpreted high-resistivity (freshwater) body is roughly 30 km long extending continuously from North Ashdod to North Tel Aviv at the depth range between 100 and 250 meters below sea level. Its

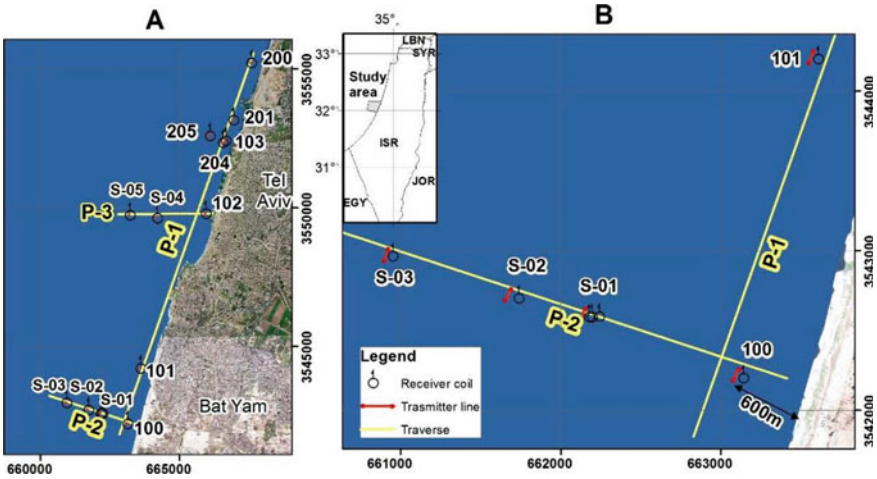


Fig. 21 Location map of the marine TDEM surveys (modified after Levi et al. 2018)

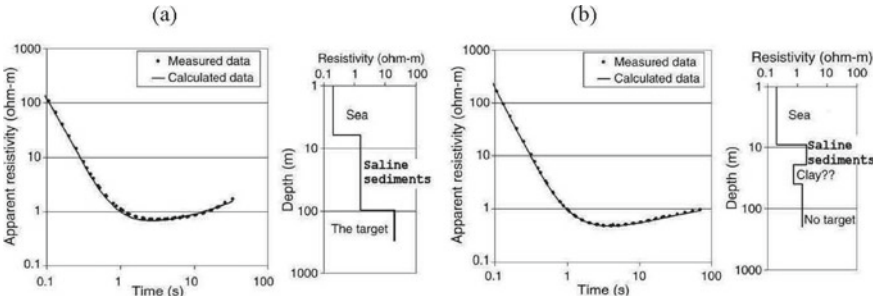


Fig. 22 Examples of measured and interpreted offshore TDEM data (apparent resistivities), **a** with a target; **b** without a target (modified after Goldman et al. 2011)

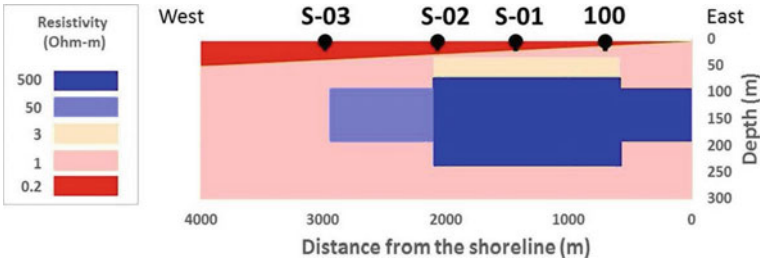


Fig. 23 The interpreted E-W 2D resistivity cross-section. Freshwater in CD extends to 3 km west of the shoreline (after Levi et al. 2018)

lateral western boundary is determined less accurately and estimated as no less than 3 km west of the sea shoreline.

The marine LOTEM method

Due to the more complicated acquisition procedure, the marine LOTEM survey was carried out along only one E-W profile in the vicinity of the city of Bat Yam. It included eight transmitter and fourteen receiver locations using both in-line and mainly broadside transmitter-receiver (T/R) configurations (Fig. 24). The T/R offsets varied between 400 and 1200 m with the first receiver located on-land near the shoreline and the last receiver, located approximately 5 km west of it (Lippert 2015).

Out of 14 measurements, nine were of good quality, which allowed a reliable interpretation, the results of which are shown as a columnar resistivity cross-section in Fig. 25 (Lippert 2015). Both Figs. 23 and 25 show a general similarity between the SHOTEM and LOTEM results with regard to the resistivity, the depth and the thickness of the target. The main difference manifests itself in the location of the western boundary of the target. According to the SHOTEM model, the boundary is located at a distance of approximately 3 km offshore, whereas according to the LOTEM model, it is located some 3700 m offshore. The LOTEM results seem to be more reliable, especially taking into account the recent follow up measurements,

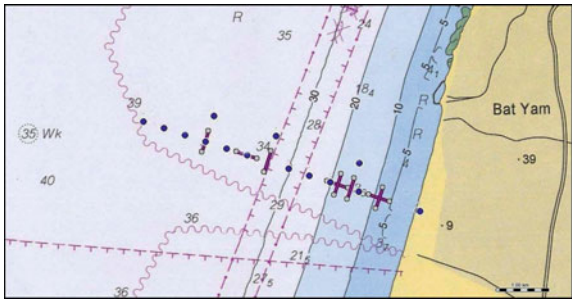


Fig. 24 Location map of the marine LOTEM survey (modified after Lippert 2015)

using a novel marine differential electric dipole (DED), that led to approximately the same results as did LOTEM (Haroon et al. 2018).

Taking into account all the above considerations, the general hydrogeological interpretation of the marine geophysical measurements can be briefly described as follows:

The top of the freshwater body within the CD is located at approximately 100 m below sea level, and its thickness varies roughly between 100 and 150 m. It extends continuously from north Ashdod to north Tel Aviv and westward offshore to approximately between 3 and 3.7 km from the shoreline, whereby the latter distance seems to be more reliable.

Combined ERT-TDEM measurements aimed to delineate diluted seawater intrusions from the Alexander stream estuary to the coastal aquifer

The water salinity at the lower reaches of the Alexander stream, as well as at some other coastal streams along the Mediterranean coast of Israel (e.g., Yarkon, Qishon), may attain seawater salinity. Since the riverbeds are below sea level, they are subjected to seawater intrusion along them up to about several kilometers upstream. As a result, a stratified water column is developed with a saline water mass at the bottom. Similar to seawater intrusion into coastal aquifers, the saline water within the streams may

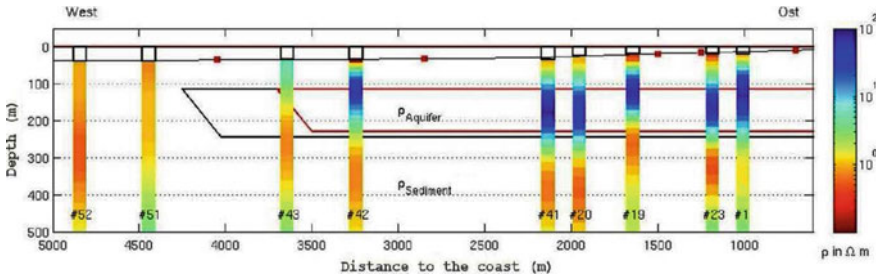


Fig. 25 Interpreted columnar resistivity cross-section along the profile shown in Fig. 24. Resistivity of the aquifer is 50 ohm-m; resistivity of the sediment is 1 ohm-m (modified after Lippert 2015)

contribute to salinization of the groundwater. Although the process of seawater intrusion into coastal aquifers is well known and properly treated, the local-scale potential salinization from the stream estuaries has never been studied in Israel until recently. A study carried out by Shalem et al. (2014) investigated the salinization mechanism of groundwater along seawater-intruded streams and estuaries. The study, among other methods, included also a combined high-resolution electrical resistance tomography (ERT) and time domain (transient) electromagnetic (TDEM/TEM) methods, which were applied at several measurement sites along the lower Alexander stream within an approximately 1.5 km distance from the shoreline (Fig. 26).

Contrary to the expectations, the geophysical measurements clearly showed a saline water intrusion (resistivity of 1–2 ohm-m) from the river into the aquifer only at the most eastern, and remote from the sea sites located approximately 1.5 km east of the shoreline (lines 03 and 04). The resistivities measured at the western sites, close to the coast, were significantly higher, besides just a few meters from the stream. Thus, contrary to traditional seawater intrusion into coastal aquifers, the saline intrusion from coastal estuaries is site dependent due to hydraulic heterogeneity of the shallow river bed along the stream. Indeed, the eastern sites, that are more remote from the sea, consist of a sandy sequence, whereas the western ones are silty. Therefore, salinization in the east extends laterally to more than 80 m off the stream and only up to 20 m in the west as shown in Fig. 27.

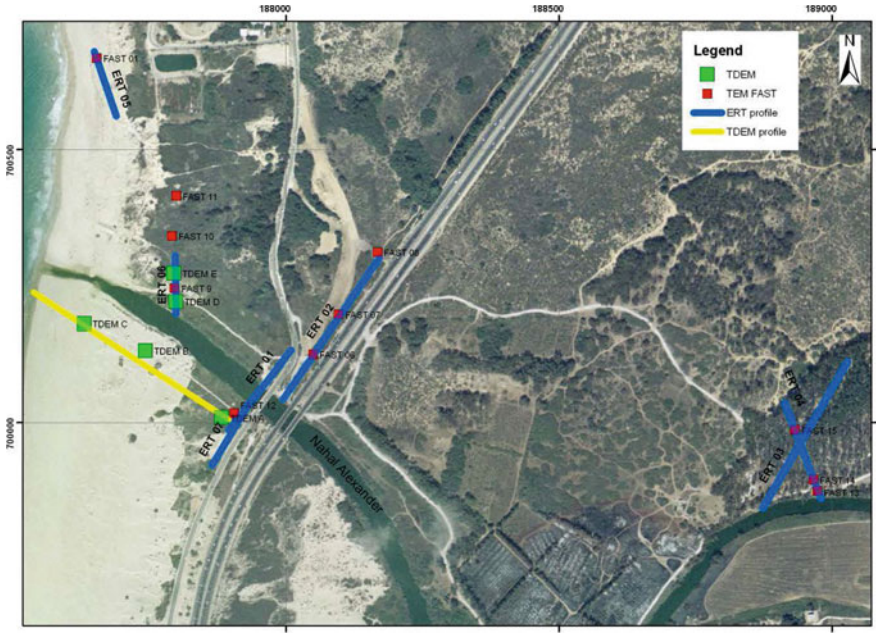


Fig. 26 Location map of the ERT and TDEM (TEM) measurements at the Alexander stream

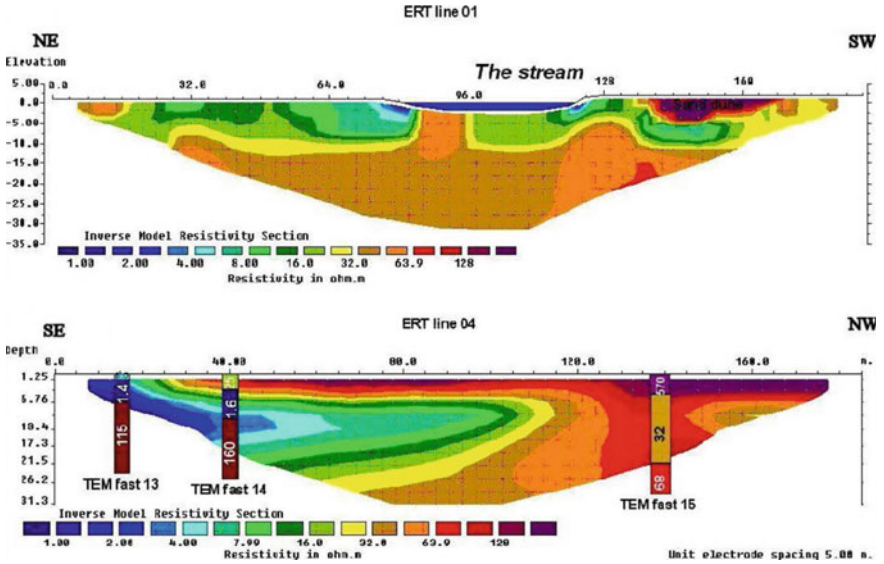


Fig. 27 ERT resistivity cross-sections along the western line 01 and along the eastern line 04 (see location map in Fig. 26). Line 01 crosses the Alexander stream in the N-S direction. Line 04 runs perpendicular to the stream on its northern bank and is accompanied by TEM soundings (modified after Shalem et al. 2014)

Geoelectric (ERT) and TDEM measurements aimed to delineate submarine groundwater discharge (SGD)

The SGD phenomenon is known, from all around the world (see also Chapter "Submarine Groundwater Discharge Along the Israeli Eastern Mediterranean Coast and in Inland Basins"). It was studied by geochemical, hydrological as well as of geophysical methods, as were employed in Israel, in the Dor Beach site, which became a major test site for studying SGD since 2006. The first geophysical survey included high-resolution time-series ERT measurements, using a 112-m-long cable (Fig. 28). It provided information regarding the coastal hydrogeology, up to the maximum exploration depth of approximately 20 m (Swarzenski et al. 2006).

In summer 2008, the deeper parts of the site were explored, using roughly two times longer shore diagonal and perpendicular ERT lines combined with onshore TDEM measurements (Fig. 29). These yielded information regarding the distribution of the electrical resistivity up to a maximum depth of roughly 100 m whereby a high-resistivity body was detected at a depth of nearly 63 m, which most likely represents the underlying Judea Group aquifer, apparently saturated with groundwater, with salinity below that of intruding seawater. Two saline/freshwater interfaces were detected at depths of approximately 17 and 49 m (Fig. 30). The upper interface between fresh and brackish water was detected by both the ERT and TDEM methods, whereas the lower fresh and seawater interface was detected only by the TDEM method.

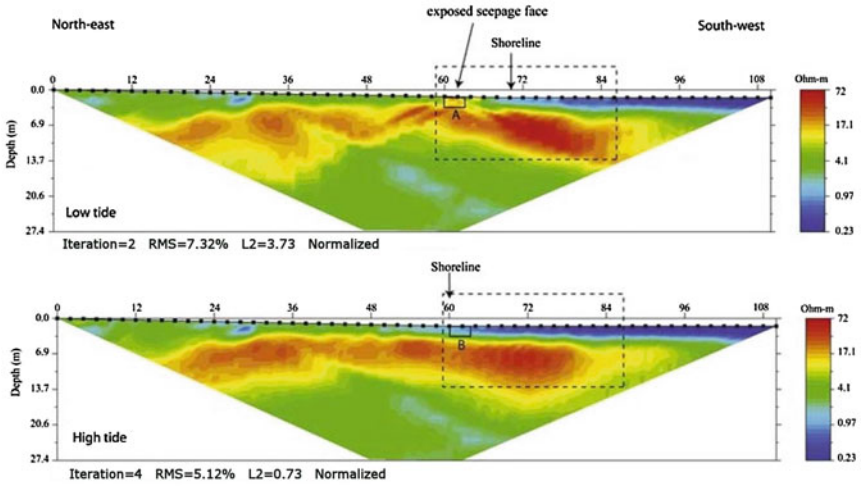


Fig. 28 ERT resistivity cross-sections at the Dor Beach site during low tide (upper) and high tide (lower) periods (modified after Swarzenski et al. 2006)

3.2 The Gulf of Eilat (Aqaba) Coastal Aquifer

The Gulf of Eilat (Aqaba) is a base level to the convergent groundwater flow from western and eastern mountain aquifers into the young Rift Valley fill aquifer. The latter consists of intercalations of aquiferous detrital coarse sediments and thick impermeable marls and clays horizons that divide the aquifer into a number of sub-aquifers.

Several TDEM soundings as well as traverses have been carried out in the western and northern coastal strips of the Gulf of Eilat (Aqaba). The results of the northern ones are discussed below.

TDEM measurements in the northern coastal strip

The setup of the northern coastal strip is characterized by the following elements that influence groundwater salinity: (1) A gulf consisting of two artificial "fjords" that penetrate several hundred meters inland to the north of the shoreline (Fig. 31).

(2) A deep saline end member, (3) The surficial artificial evaporation ponds of the salt industry, which host concentrated brines and halite deposits (Fig. 31).

The TDEM measurements have been carried out along two lines, perpendicular to each other, namely the F-F′ and G-G′ lines, The F-F′ line runs close to the "fjords" and the evaporation ponds. An additional shorter E-E′ line runs from the seashore inland to the north, adjacent to the state border and crossing the evaporation ponds (Fig. 31).

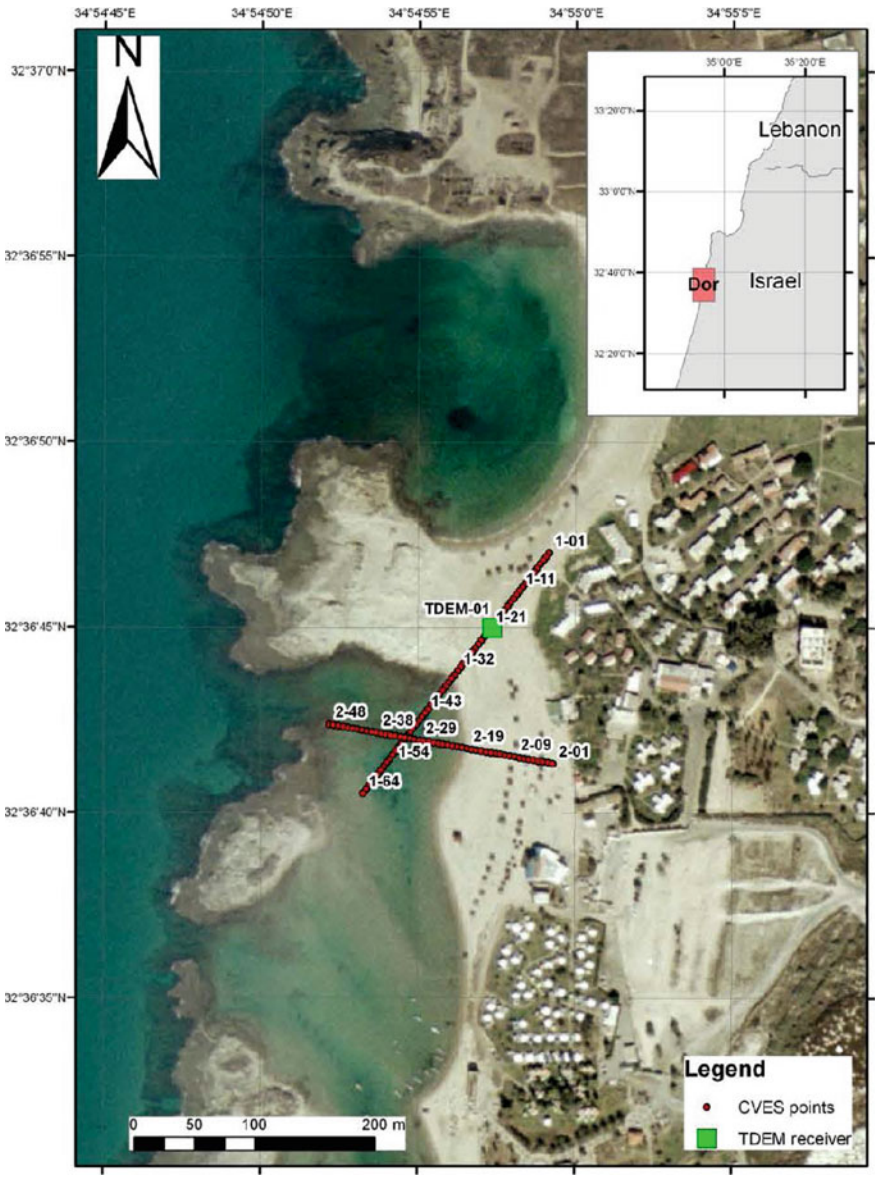


Fig. 29 Location map of the combined ERT-TDEM measurements at the Dor Beach

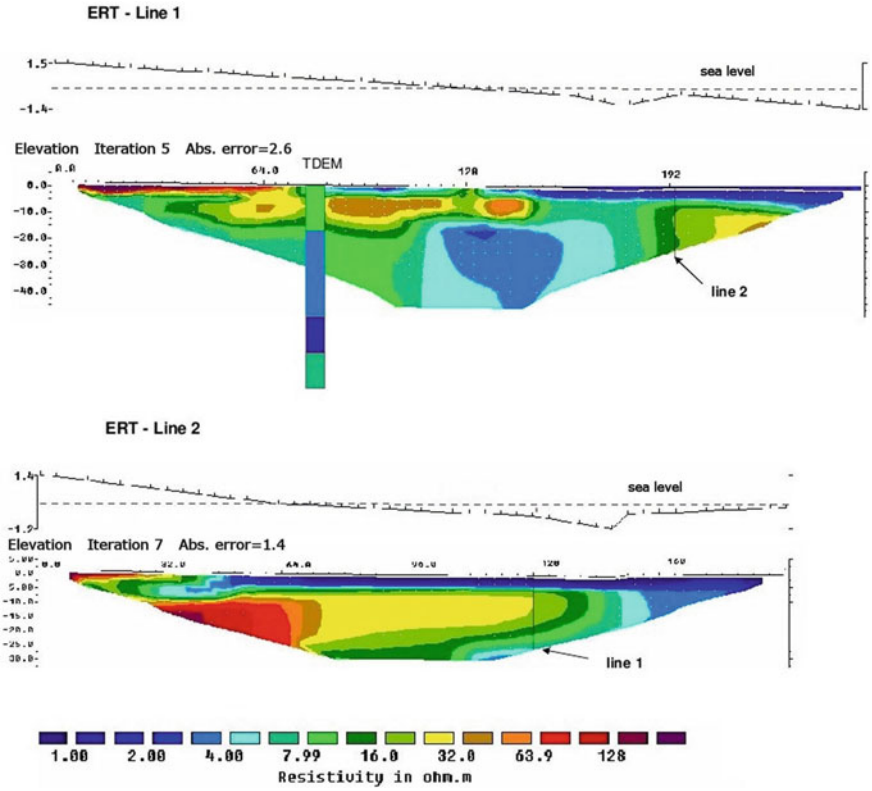


Fig. 30 2-D ERT resistivity cross-sections along lines 1 and 2 showing SGD (high resistivities) at shallow depths below the sea. 1-D TDEM columnar resistivity cross-section shows, in addition, freshwater (high resistivity) below the seawater intrusion at greater depth

Figure 32 shows an example of a quasi-2D resistivity cross-section, based on the interpreted 1-D TDEM measurements along line F-F′. To enable the above interpretation, the following resistivity and appropriate hydrogeological units have been taken into account and introduced:

A discrepancy was found between some of the TDEM resistivities and resultant water salinities and water salinities of adjacent boreholes. Lower bulk resistivity values were obtained than the expected for the adjacent appropriate water salinities. For example, groundwater chlorinities between 3100 and 3400 mg/l, from the depth interval of 100–200 m in the Sabkha 19 and 20 wells correspond to low resistivities below 0.5 ohm-m, typical to concentrated brines rather than to the above described brackish waters (Fig. 32). This discrepancy is explained herein by the existence of fairly thick clay intercalations, which assumingly contain unflushed brines which contribute to the integrated lower bulk resistivity, whereas the borehole water salinities represent solely the aquiferous fresher portions of the aquifer.

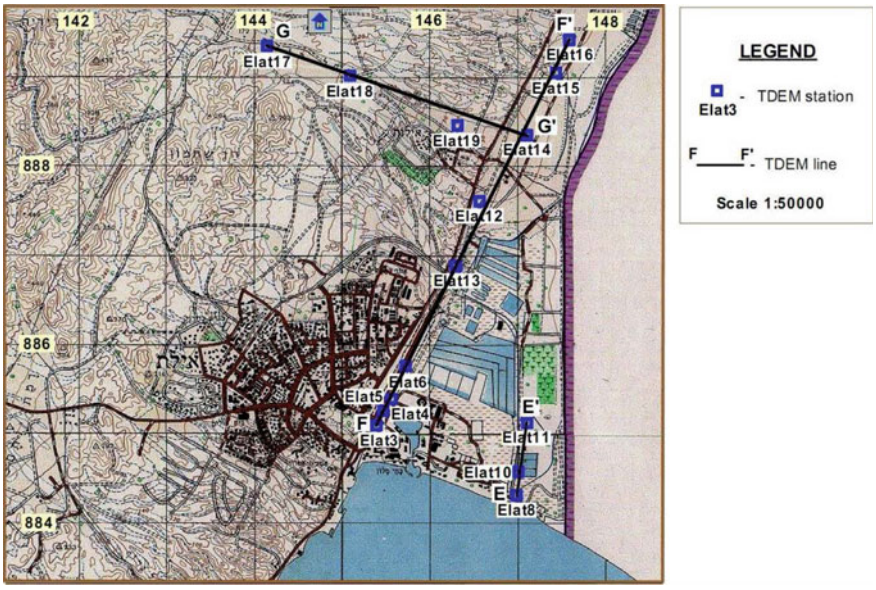


Fig. 31 Location map of the TDEM measurements at the northern coast of the Gulf of Eilat

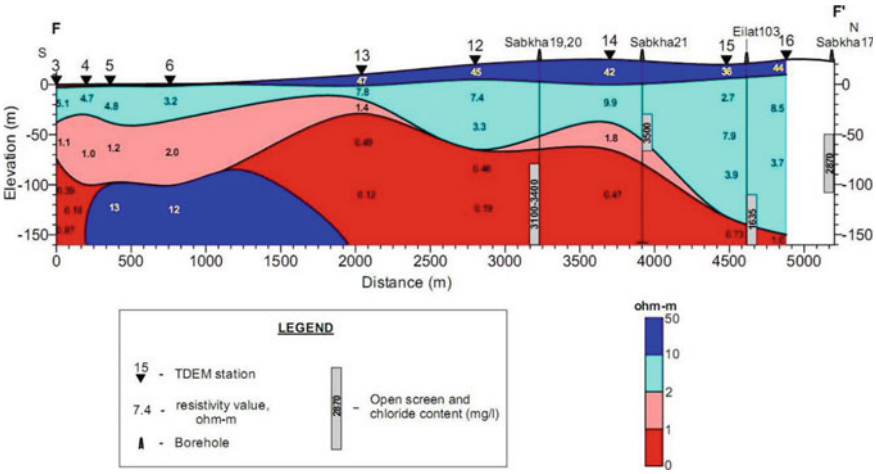


Fig. 32 Quasi-2D resistivity cross-section along line F-F′ accompanied by the chlorinity data from nearby observation wells

The resistivity results of the above-mentioned lines combined with individual TDEM soundings from the western coastal strip enabled to draw a schematic 3D hydrogeological model of the Gulf of Eilat coastal aquifer (Fig. 33). The above findings can be summarized as follows:

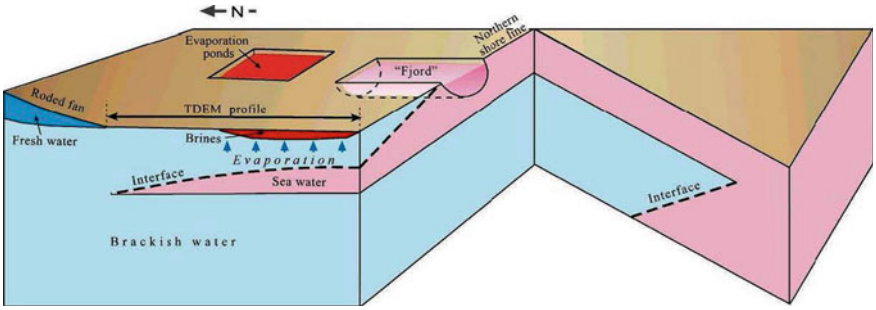


Fig. 33 Schematic hydrogeological model of the Gulf of Eilat coastal aquifer

- TDEM resistivities below 2 ohm-m uniquely characterize the salinity of seawater intrusion. The depth to the brine–freshwater interface, within the upper sub-aquifers is accurately detected by TDEM measurements and fits the one obtained from borehole measurements.
- In most cases, the TDEM method manages to characterize groundwater salinity within the lower sub-aquifers, except for those sub-aquifers which include saline clay horizons where the resolution of TDEM method is sometimes insufficient to detect the fresh/brackish water bodies.

4 Geoelectric Characterization of Groundwater Salinity Along and Close to the Dead Sea Rift (DSR)

4.1 The Sea of Galilee

The Sea of Galilee is located within the DSR and occupies approximately 170 km^2 with a maximum water depth of 46 m (Fig. 34). In fact, it is a freshwater lake, which serves as one of the most important water reservoirs of Israel. The lake is subjected to salinization from different sources that results in the average chlorinity of approximately 220 mg/l.

In order to study possible salinization sources, attempts have been made in the past to quantitatively evaluate the amount of salts entering the lake from the bottom. Since drilling within the lake is prohibited by law, the above study has been limited by extracting shallow cores from the lake bottom. Consequently, nineteen short (50 cm) cores and three long (500 cm) ones were taken at various locations on the lake floor (Fig. 34) in order to examine their chlorine content (Stiller 1994). The water in all cores exhibited a roughly linear increase in chlorine concentration from about 220 mg/l at the top of the core (water-sediment interface) to 350–600 mg/l at the depth of 50 cm, and 2000–3500 mg/l at the depth of 500 cm. A rough estimate based on the linear extrapolation of the observed chlorinity gradients in the cores has revealed a diffusion and advection rate of 5.3 mg Cl cm^{-2} yr^{-1} from the lake's

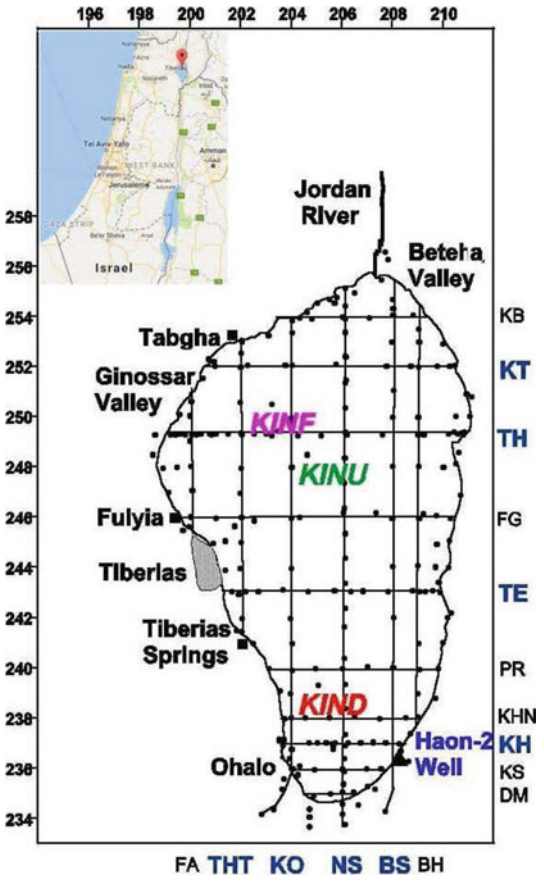


Fig. 34 TDEM location map of the Sea of Galilee and its vicinity. KINF, KINU and KIND designate the locations of the core samples. KB, KT, TH, etc., are the names of the traverses used in the text (modified after Hurwitz et al. 1999)

bottom sediments, resulting in an annual contribution of 9000 tons of Cl to the lake. (Stiller 1994). These results clearly indicate that saline sources are located at shallow sub-bottom depths over large parts of the lake. However, the spatial distribution of the saline groundwater and actual sub-bottom depth was unknown.

In light of the legal forbiddance to drill boreholes within the lake, the only reasonable way to tackle the above problem was by applying geophysical, more precisely GE/GEM, methods. As a result a novel marine TDEM system has been developed and widely applied in the Sea of Galilee (Hurwitz et al 1999). The system consisted of a 25 m diameter circular transmitter loop, which floated on the water surface and was towed by a motorboat. A 1-m-diameter receiver coil was rigidly connected to the center of the transmitter loop and to another rubber non-motorized boat, which carried an operator and a receiver console (Fig. 35).



Fig. 35 Central loop marine TDEM system applied at the Sea of Galilee (modified after Hurwitz et al. 1999)

Since the measurements with the described system did not require assembling and disassembling of the measurement array at each point, the productivity of the survey was considerably greater than that of the conventional onshore TDEM (30–40 offshore soundings per day vs. 6–8 onshore soundings). This, in turn, allowed to obtain a dense grid of soundings covering the entire lake and its close vicinity (Fig. 34).

A total of 269 offshore and 33 supplementary onshore TDEM soundings have been carried out along six N-S and thirteen W-E profiles as well as at selected points both offshore and onshore, along the entire coastline (Fig. 34). The interpreted resistivities were calibrated with the salinity measurements in nearby boreholes available in the vicinity of the lake and from deep cores taken from the lake bottom (Hurwitz et al. 1999). The calibration revealed that resistivities below 1 ohm-m are solely indicative of groundwater chlorinity exceeding 10,000 mg/l. Such low resistivities (high salinities) were detected practically everywhere at depths greater than 15 m below the bottom of the lake. At some parts of the lake, particularly in the south, the saline water was detected at shallower depths, sometimes at only a few meters below the bottom (Fig. 36).

A relatively high resistivity identified with fresh groundwater was detected along the margins of the lake to the depths of at least 100 m, which represents the maximum exploration depth of the marine TDEM system. The detected sharp lateral contrasts between high and low resistivities apparently represent a fault controlled lateral separation between carbonate and clastic units (Figs. 36 and 37), where the uplifted block, consisting of the carbonate aquifer is saturated with fresher groundwater. The geometry of the fresh/saline groundwater interface below the central part of the lake approximately follows the shape of the lake's bathymetry.

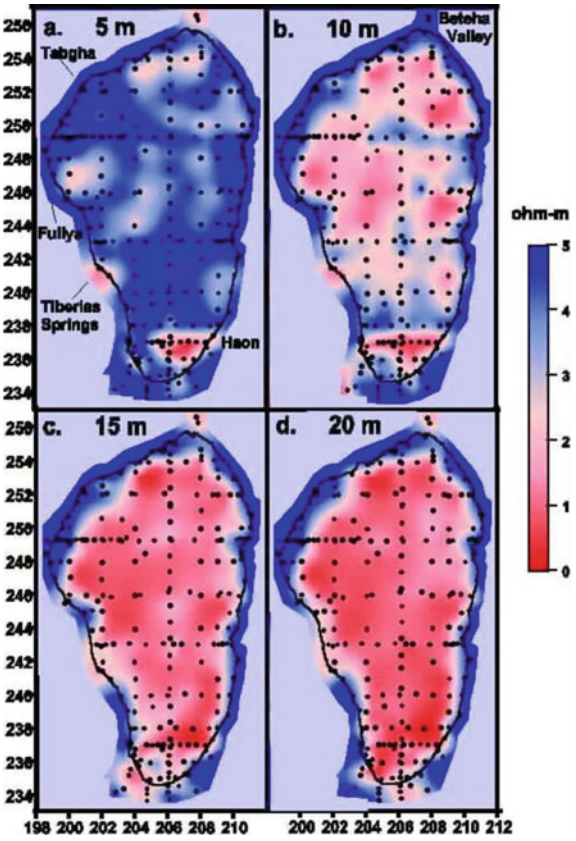


Fig. 36 Distribution of resistivities at different depths below the lake's bottom (modified after Hurwitz et al. 1999)

The described TDEM results indicate that the distribution of the saline groundwater entrapped within the lake's sediments is highly heterogeneous. The greatest variability, both in concentration and depth, is limited to a narrow strip of 1–2 km wide near the shores. A significant change is also observed between the southern and northern parts of the lake. The interpreted TDEM resistivity maps and cross-sections define two zones with different transport mechanisms. These are the diffusion zone under most parts of the lake and the advective flux zone in the margins, where diffusion is negligible.

The similar shape of the lake's bathymetry and the top of the submarine saline groundwater body in the central part of the lake suggest that in the past, a saline lake covered the area and its water was trapped in the sediments. Consequently, when the Sea of Galilee became a freshwater lake, a diffusion of salinity from the sediments into the lake began to take place.

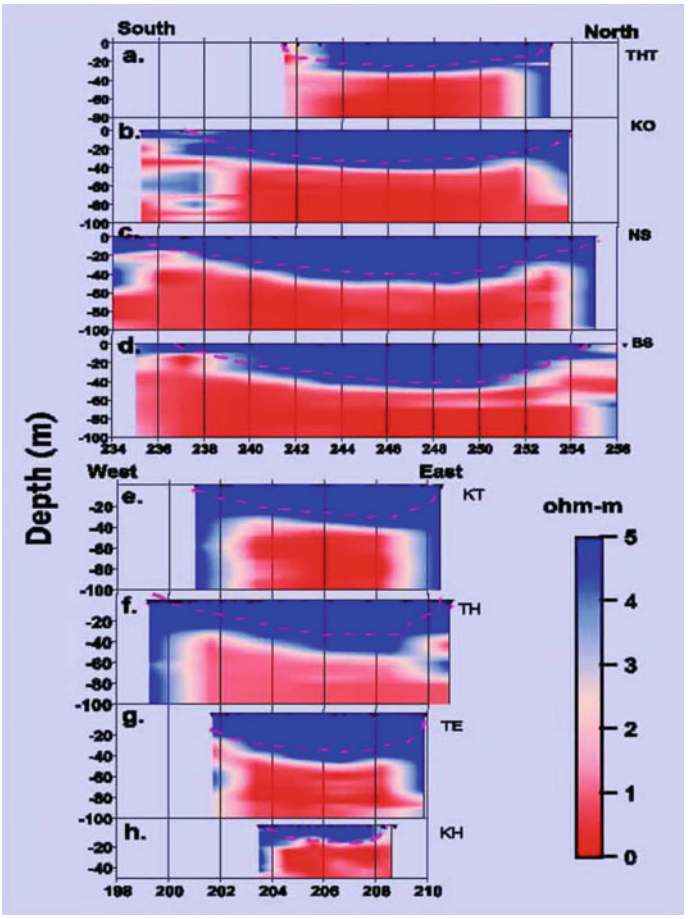


Fig. 37 TDEM quasi-2D resistivity cross-sections along the SN and WE traverses (modified after Hurwitz et al. 1999)

In the southeastern part of the lake, the depth to the top of the saline groundwater body below the bottom is small and sometimes negligible. This implies that this part of the Sea of Galilee turned to be a freshwater lake later than in the northern part.

The Beteha Valley, in the northeast looks as an isolated pocket (Fig. 36). Since it is practically a delta of freshwater rivers, entering the lake, this observation seems surprising. Delta sediments are known there to be coarse grained with a high permeability. Therefore, either the Beteha Valley was also covered by the same lake as the one below the Sea of Galilee or alternatively that it was an isolated separated lake.

4.2 The Dead Sea Coast

The Dead Sea (hereafter DS) is the terminal base level of the DSR system (Fig. 38) (see also Chapter "The Eastern Dead Sea Rift Continental Groundwater Base Level"). It occupies hyper-saline brines typical of a very low electric resistivity. The DS is bounded to the west by north–south directed major faults. The coastal plain between the faults escarpment and the DS consists mainly of Quaternary continental detrital sediments (Fig. 39). Since the area is arid, the main freshwater recharge into the Quaternary aquifer is a lateral flow across the faults scarp from the Judea Group aquifer (Fig. 39), which is replenished in the mountain area some 10–30 km to the west. The DS is hydraulically connected to the adjacent Judea Group and the coastal aquifers and thus the continuous decline of the DS level, affects the parallel decline of the groundwater level in the above-mentioned aquifers (Kafri 1982; Yechieli 1993).

TDEM measurements

In order to understand the hydrogeological setup and its related salinity distribution, TDEM soundings were performed along generally W-E traverses between the DS

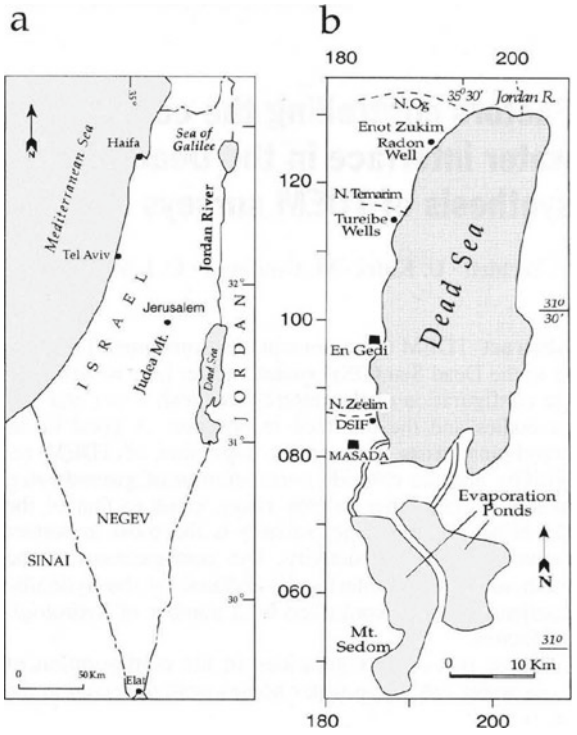


Fig. 38 Location maps: general map (**a**); location map (**b**) of TDEM traverses (dashed lines) (after Yecheli et al. 2001)

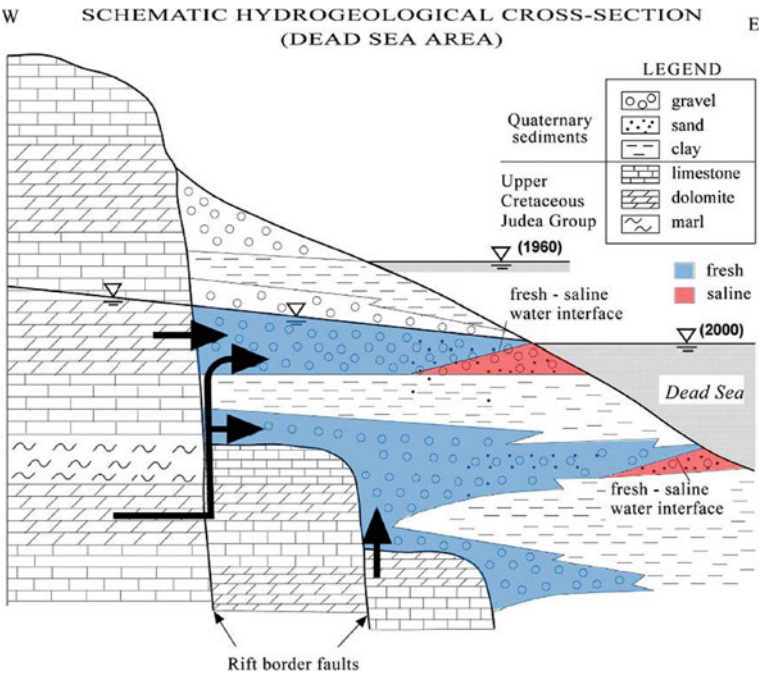


Fig. 39 Schematic hydrogeological cross-section of the Dead Sea aquifer system (after Yecheli et al. 2001)

shore and the fault scarp (Fig. 38b). The Temarim stream traverse serves, for example, (Fig. 40) enables a spatial interpretation of the salinities in the coastal aquifer as follows:

The obtained configuration of the fresh (>5 ohm-m)-saline (<1 ohm-m) water interface exhibits its westward dipping and the transition (mixing) zone (1–5 ohm-m) that exists in between them (Fig. 40). Two features are recognized, common to other traverses in the region:

1. An unflushed brine body (sounding 23/7), below and above the present water table, and
2. A westward steepening of the interface in the west close to the fault scarp.

The suggested hydrogeological interpretation of the both phenomena is as follows:

1. The DS coastal aquifer in most places is subdivided into sub-aquifers, as detected in some of the boreholes (Yechieli 1993). This setup permits the possibility that the lower fresher water body being separated from the upper more saline one.

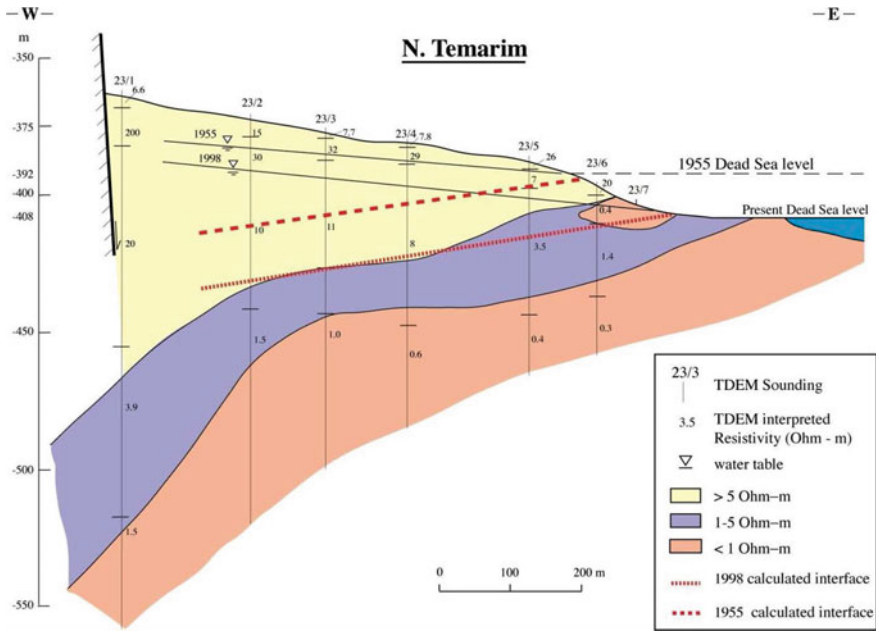


Fig. 40 TDEM traverse at Nahal Temarim. Location of interface was calculated using Ghyben–Herzberg approximation modified for the Dead Sea conditions (after Kafri et al. 1997)

Moreover, the irregularity may also be the result of the fact that the DS system is not in steady-state conditions, due to the continuous drop of the DS and the adjoining groundwater levels. As a result, parts of the sequence have not yet been completely flushed, although they are already located in the vadose zone far above the fresh–saline water interface, as happens in some other locations in the region.

2. The steepening gradient of the water levels across the border faults, caused by anisotropic permeability in the fault zone and/or by different permeability in both sides of the fault (Yechieli et al. 2001), would demand there a parallel steepening of the freshwater/brine interface, as was indeed recognized here.

4.3 Geophysical Delineation of Sinkhole Development in the Dead Sea Coastal Plain

Thousands of sinkholes have been developed and are still developing, since the early 1980s, along the Israeli, as well as the Jordanian coastal plains of the Dead Sea (hereafter DS) following the rapid recession of the DS level since the 1960s. The

phenomenon and the mechanism of sinkholes formation was studied in detail and discussed in several articles as well as in a separate chapter of this book.

Arkin and Gilat (2000) attributed this phenomenon to the resultant removal of fine sediments and to piping within the young detrital sequence of the DS coastal aquifer. The more commonly accepted mechanism of the sinkholes formation, however, explains the phenomenon by dissolution of a salt layer known at a depth of 20–70 m below surface (i.e., Yechieli et al. 2006; Abelson et al. 2006). Until the beginning of the rapid drop of the DS level, this salt layer was saturated with the concentrated DS brine (hereafter DSB), which was unable to dissolve salt. Above the interface, fresher groundwater or diluted DSB (termed herein DSBD) prevailed. Following the continuous drop of the DS level, accompanied by the drop of the interface, the salt layer was "exposed" to the DSBD, which has the capability to initiate salt dissolution, resulting in the appearance of subsurface cavities, collapse of overlaying sediments and sinkholes formation.

The general mechanism of sinkholes formation all over the world is related to dissolution of soluble rocks leading to appearance of subsurface cavities that collapse and form sinkholes. The dissolved rock types exhibit different dissolution rates, which dictate a strategy of geophysical delineation of zones subjected to potential sinkhole development. In the case of carbonates, typical of relatively slow dissolution rates, geophysical methods usually focus on detecting only the cavities. In the case of salts, which exhibit high dissolution rates, detecting only cavities might be inefficient and the methods are required to delineate and forecast preferable zones or paths of salt dissolution prior to the creation of the cavities. Thus, in the case of the DS coastal plain sinkholes, the geophysical methods were primarily used to delineate the underlying salt layer and to detect the zones, where groundwater salinity is less than that of the DSB. Two other possible mechanisms, namely subsidence and erosion ("piping"), which might also contribute to the sinkhole development, are beyond the scope of this chapter.

The most suitable combination of geophysical techniques to delineate salt layers and to detect the DSBD bodies includes the seismic refraction and high-resolution reflection methods, the TDEM and/or the integrated TDEM and SNMR methods. The seismic refraction method was used to delineate the salt layer and to locate a buried "salt edge," which assumingly represents a dissolution front controlling, at least partly, the sinkhole formation (Ezersky et al. 2017). The use of seismic refraction to delineate the "salt edge" seems somewhat problematic though. The high-resolution seismic method allowed to detect faults, which might serve as preferential hydraulic conduits to DSBD flow toward a contact with the salt layer (Abelson et al. 2003). The TDEM method is known to be feasible for characterization of groundwater salinity (Goldman et al. 1991), but might be more efficient in detecting the DSBD when combined with SNMR (Legchenko et al. 2009).

Due to the complexity of the problem, additional geophysical methods were involved in the study of sinkhole development in the DS coastal plain. Among them are micro-gravimetry (Rybakov et al. 2001), electrical resistance tomography (ERT) (Ezersky 2008; Al-Zoubi et al. 2007), ground-penetrating radar (GPR) (Frumkin et al. 2009, 2011), gravimetry and radar differential interferometry (Closson et al. 2003,

2005; Nof et al. 2013), nano-seismicity and microseismic activity (Wust-Bloch and Joswig 2006; Abelson et al. 2013). The comprehensive review of all these methods and the obtained results is discussed by Ezersky et al. (2017).

Since most of the above geophysical methods, besides seismic refraction/reflection, TDEM and SNMR, contributed only partly or indirectly to the understanding the main mechanism of the Dead Sea coast sinkhole development, the further discussion focuses exclusively on these four methods.

Seismic refraction

A typical P-wave velocity versus depth profile of the Nahal Hever South test site is shown in Fig. 41. The cross-section consists of three velocity layers, the lowermost, with a velocity of 2890 m/s has been identified with a salt layer (Wachs et al. 2000). The depth to this layer varies between 20 and 25 m in the western part of the profile, dipping to some 30–35 m in its eastern part. Borehole HS-2, located close to the western segment of profile RP-004, has in fact encountered the salt layer at a depth of 24 m (Wachs et al. 2000). The geophysical and log data of this borehole allowed to correlate the velocity layers with the borehole lithology as follows:

Further seismic refraction surveys all over the DS coastal plain, supported by borehole data, has shown that the P-wave velocity of the salt layer varies between 2900 and 4250 m/s (Ezersky 2006). The good velocity contrast between the target salt layer and the background lithology allowed a reliable delineation of the salt layer along the entire DS coast of Israel (Ezersky et al. 2006). Unfortunately, the second

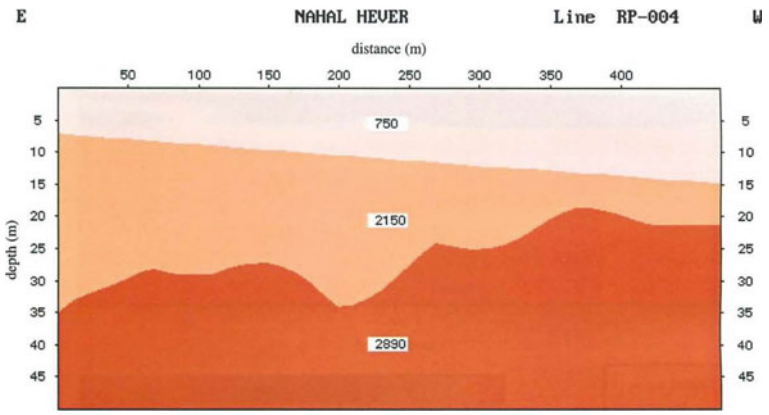


Fig. 41 Typical velocity (in m/s) versus depth profile along line RP-004 (after Wachs et al. 2000)

important element related to the delineation of the DSBD turned out to be more complicated and difficult for geophysical treatment.

TDEM

Countless TDEM surveys, carried out all over the world, showed the great efficiency of the method in characterization of groundwater salinity and in accurate delineation of saline groundwater bodies at different depths, from tens of meters to several kilometers (Fitterman and Stewart 1986; Goldman et al. 1991; Kafri et al. 2008). No wonder that the TDEM method was considered as a principal technique to delineate the DSBD in the DS region. The very first survey in the Nahal Hever South test site has indeed exhibited, as described before, very promising results (Goldman and Ezersky 1999a). Figure 42 shows the quasi-2D TDEM resistivity cross-section obtained for one of the N-S TDEM profiles in the Nahal Hever South test site. The profile crosses the E-W sinkhole lineament between TDEM stations NH5-1 and NH5-2.

Two main resistivity features are recognized in the cross-section, both exhibiting significantly higher resistivities than those of the background ones. The relatively shallow and smaller resistive body is located below station NH5-1, at the depth interval between 30 and 35 m. Since the water table here is at a depth of approximately 20 m, this feature, most likely, testifies to either a lower porosity of the aquifer or to a lower salinity of its hosted groundwater, or a combination of both. Since no information regarding the porosity of the aquifer was available, it was assumed that it represents the DSBD body. According to borehole HS-2, the bottom of the salt layer roughly coincides with the depth to the DSBD, enabling the process of salt dissolution there. The location of this feature just below the sinkhole lineament in the Nahal Hever South test site testifies to the validity of the proposed mechanism (Goldman et al. 2000).

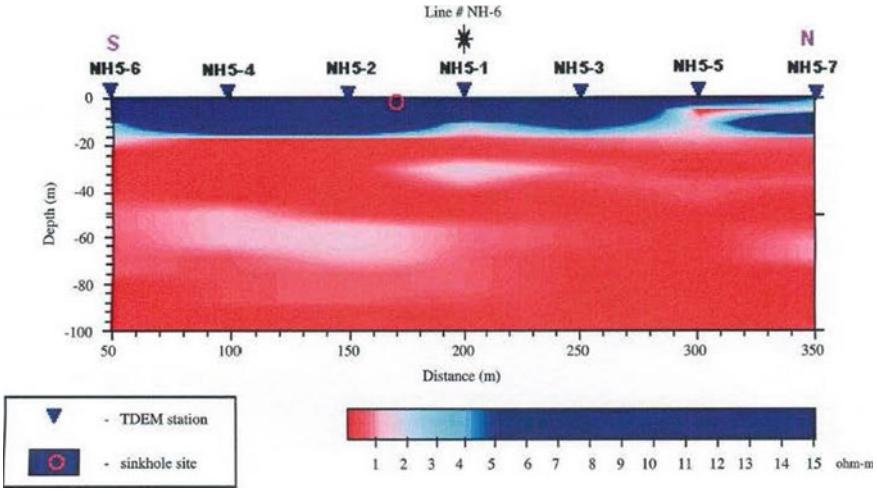


Fig. 42 An example of a quasi-2D TDEM resistivity cross-section in the Nahal Hever South test site (after Wachs et al. 2000)

Concerning the deep feature at the depths between 50 and 60 m (Fig. 42), the occurrence of clay interlayers within the salt layer and the assumed DSDB might have blocked the process. This could have been overcome by an existing fault that cuts across the clay unit forming a hydraulic conduit, which was indeed detected by the high-resolution seismic method (see below).

The apparent success of the described integrated seismic/TDEM method at the Nahal Hever South test site greatly motivated a wide application of the method along the entire DS coastal plain of Israel. However, it was realized that the ability of TDEM method to characterize extremely saline groundwater is rather limited. In the next survey, at the Nahal Hever North site, despite an abundance of sinkholes on the surface, no increased TDEM resistivities have been found underneath them in the entire site (Goldman and Ezersky 2000). All TDEM resistivities, at depths below the water table, varied within a narrow range between 0.4 and 0.6 ohm-m. Such small resistivity variations could not be attributed to the corresponding salinity variations in the absence of the information regarding porosity or any other factor, which might have affected the measured bulk resistivity. As a result, further application of the TDEM method has been temporarily abandoned until empirical quantitative relationships between the TDEM bulk resistivities and chloride concentration were derived for different lithologies in the DS coastal aquifer (Ezersky and Frumkin 2017). These relationships were based on the correlation of TDEM results and borehole data using statistically significant number of measurements adjacent to 28 boreholes throughout the DS coastal plain. An example of the relationships between the measured bulk TDEM resistivities and the hosted groundwater chlorinity for different lithologies that comprise of aquifer are shown in Fig. 43 (Ezersky and Frumkin 2017). It is evident that the chlorinity is in the range between 80 and 200 g/l whereby even small variations in the bulk resistivities reflect considerable chlorinity changes. Thus, the required quantitative estimation of the chlorinity is still ambiguous. However, knowing the lithology underneath the salt layer, one can discriminate DSB and DSBD more accurately based solely on measured TDEM resistivities. For example, if bulk resistivities exceed 0.55 ohm-m, the chlorinity of groundwater is less than 100 g/l for all sedimentary types of the aquifer, and salt dissolution is thus feasible. If resistivities vary between 0.3 and 0.4 ohm-m, then, in the case of a silty aquifer (DS mud), dissolution is possible, whereas in case of the sandy-gravel aquifer, these resistivities are indicative of DSB, which is not capable to dissolve salt. Resistivities below 0.3 ohm-m, most likely, correspond to the DSB for all types of aquifers.

Thus, knowing the lithology of the aquifer, for the above considered range of groundwater salinities, is crucial for the determination of groundwater's capability to dissolve salts. Unfortunately, due to the scarcity of boreholes and high lateral lithological variability, such a crucial information in most cases cannot be obtained from boreholes. Therefore, an attempt was made to tackle this problem using only

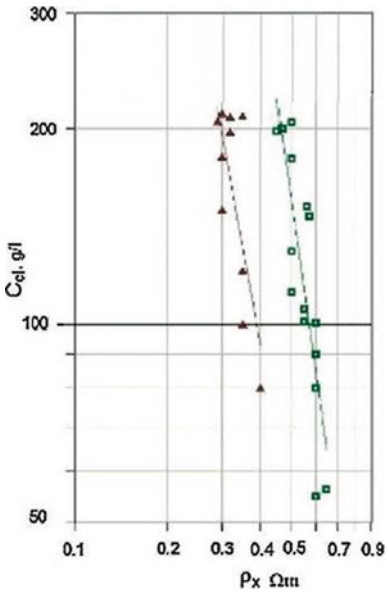


Fig. 43 Chlorinity versus bulk resistivity relationships for different lithologies of the Dead Sea coastal aquifer: triangles—silty lithology, squares—sandy-gravel lithology (modified after Ezersky and Frumkin 2017)

geophysical methods, namely the combined three-dimensional (3D) SNMR and TDEM methods (Legchenko et al. 2009).

Combined 3D SNMR-TDEM

The feasibility study for the combined method was carried out at the Nahal Hever South test site. It included 24 SNMR soundings, each accompanied by a TDEM measurement at exactly the same location (Legchenko et al. 2009). Two important hydrological parameters were extracted from the 3D SNMR interpretation. The first one, which is directly obtained by the SNMR inversion, is water content, and the second is hydraulic conductivity of the aquifer, (K_{SNMR}), calculated using inverted SNMR relaxation time (T_1) and water content (w), using the following empirical relationship (Legchenko 2013): $K_{\mathrm{SNMR}} = C_p w T_1 **2$, with C_p being an empirical constant that can be calibrated to similar hydrogeological parameters obtained from pumping tests.

It should be noted that SNMR is an efficient tool not only for characterizing aquifers but also for locating water-filled voids in the subsurface (Vouillamoz et al. 2003; Boucher et al. 2006).

The results of the 3D SNMR interpretation are shown in Fig. 44. The relaxation time (T_1), presented in Fig. 44a, varies between approximately 50 and 400 ms. Higher T_1 values correspond to a sandy-gravel formation. The map (Fig. 44b) and the 2D cross-section (Fig. 44c), showing the water content, which was derived from the SNMR inversion, enable to locate and delineate the aquifer. Substituting this

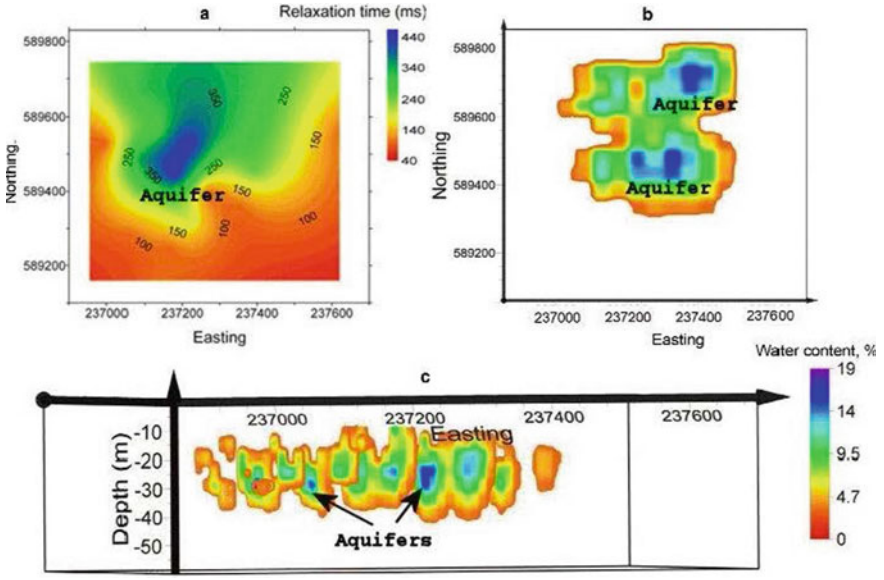


Fig. 44 3D-SNMR results in the Nahal Hever site: **a** Map of the relaxation time T_1 derived from 3D-SNMR inversion; **b** map of the water content distribution at a depth of 26 m, and **c** variations of the water content within the investigated area (modified after Legchenko et al. 2009)

information along with the measured TDEM bulk resistivities into Fig. 43 allows to estimate the chlorinity of groundwater within the aquifer. The latter results, combined with seismic refraction data regarding the location and depth to the salt layer, allow to delineate the potential zones of sinkholes formation.

As mentioned before, the existence of impermeable clay interlayers within the salt layer and the aquifer required the detection of faults that facilitate the emergence of the DSBD toward the salt layer.

Regarding the above, there are two main mechanisms of the sinkhole formation due to the salt dissolution. The first one, based on the so called "structural model," was described by Abelson et al. (2003). It assumes that sinkholes tend to develop along faults, which serve as preferential hydraulic conduits to emerging confined aggressive water upward to reach salt layer. The second mechanism is based on the so called salt edge model that associates sinkholes with karstification of the salt edge by deep and shallow aggressive groundwater (Ezersky and Frumkin 2013). The latter model is handled using the above described seismic refraction method, whereas the former model is better geophysically treated by means of the high resolution seismic reflection method (HRSR).

High-resolution seismic reflection (HRSR)

The results of a high-resolution seismic reflection survey at the Nahal Zeelim site are presented in Fig. 45. The survey included a 340-m-long 2D profile, which was shot in NE-SW direction across the sinkhole line (Ezersky and Frumkin 2013).

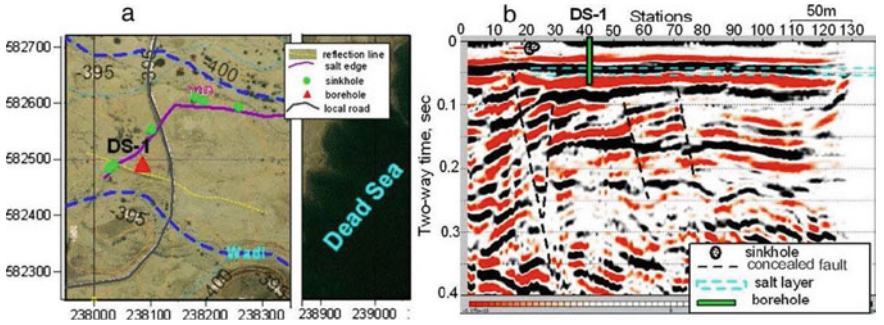


Fig. 45 **a** Location map of the Nahal Zeelim site. Reflection line (yellow) crosses the salt layer edge and a small sinkhole, which is a part of a sinkhole line. **b** 2-D seismic reflection section through the site in the NW-SE direction (modified after Ezersky and Frumkin 2013)

The edge of a 7–10-m-thick salt layer has been previously delineated by seismic refraction study and verified by borehole DS-1 (Fig. 45a, b). A distortion zone is clearly seen in the vicinity of the intersection of the reflection profile and the sinkhole line (Fig. 45b). The results testify to the presence of faults close to the sinkhole zone. The upper reflector is generally smooth but distorted between stations 10 and 30. Two sinkholes and the salt edge indeed coincide with the fault site.

The following procedure for geophysical evaluation of a sinkhole development hazard in a selected site, based on the above, is presented herein:

1. Seismic refraction survey for the detection of a salt layer in the site, and delineation of its edge.
2. A combined 3D SNMR-TDEM survey for evaluating the chlorinity of the groundwater.
3. A HRSR survey, to detect faults that serve as conduits for emerging DSBD.

5 Integrated Geophysical Surveys in the Mediterranean and Dead Sea Coastal Aquifers

Each GE/GEM method has inherent limitations, which reduce its application to specific targets, geoelectric conditions, noise levels and surface conditions. Sometimes, the choice of an appropriate single method might be sufficient to solve the required hydrogeological problem, but in most cases a single method provides only partial information that must be accompanied by independent data. In the past, such data were mainly extracted from nearby boreholes, but the scarcity or absence of boreholes, required to look for an alternative solution. Several decades ago, the idea to jointly use different GE/GM methods for solving specific geological and hydrogeological problems was proposed by Jupp and Vozoff (1975). Initially, the idea was only realized in the so-called joint inversion of different GE/GM data for reducing ambiguity of geophysical interpretation. At present, integrating different

GE/GM and also other geophysical techniques is widely used both for more accurate geophysical interpretation and for solving particularly complicated geological and hydrogeological problems. Two examples of the latter applications are discussed herein.

5.1 SNMR-TDEM

It is widely recognized that among all geophysical techniques, the GE/GEM methods are the most efficient ones in solving various hydrogeological problems, particularly those related to delineating and monitoring groundwater bodies and processes. In the case of saline groundwater, in most cases the methods are feasible irrespective of the environment, since saline groundwater provides excellent resistivity contrast practically with any background lithology, whether dry or saturated with freshwater. Unfortunately, it is not always the case with freshwater-saturated aquifers, which in many cases cannot be even delineated by the GE/GEM methods, not mentioning more delicate problems such as hydraulic properties.

On the other hand, the SNMR method proved to be of high efficiency, not only in detecting groundwater of any salinity, but also in quantitative estimation of porosity and transmissivity of aquifers. However, the method suffers from a rather limited exploration depth, and in most cases is unable to detect a fresh/saline water interface. Therofore,by combining the TDEM and SNMR measurements in coastal aquifers, one might expect not only to delineate the freshwater portion of the aquifer, but also to quantitatively estimate the amount of freshwater within the aquifer and, under favorable conditions, the transmissivity of the latter. The feasibility of this integrated technique has been examined in two coastal aquifers of Israel.

The Mediterranean coastal aquifer

Two perpendicular profiles, each including four SNMR/TDEM stations, were run in close proximity to existing observation wells at the Nitzanim coastal area (Fig. 46). The comparison of the geophysical and borehole information from all three wells available clearly indicated the limits of applicability of each method. As an example, Fig. 47 shows the results of the comparison.

The SNMR method manages to accurately determine the depth to the water table, while the TDEM method misses it. The opposite is true with regard to the fresh/sea water interface. Due to the limited exploration depth, the SNMR data does not manage to detect the interface, whereas the TDEM method does it with high accuracy. Combining both interpreted models, and taking into account the discussed limits of application of each method, the freshwater portion of the aquifer was found in the depth interval of 10–66 m, as compared to a 12–65 m interval in an adjacent borehole. In the absence of pumping tests, carried out in that well, the alternative way is to quantitatively estimate the water content in the borehole by the SNMR method. Based on statistically significant correlation data gathered in granular coastal aquifers

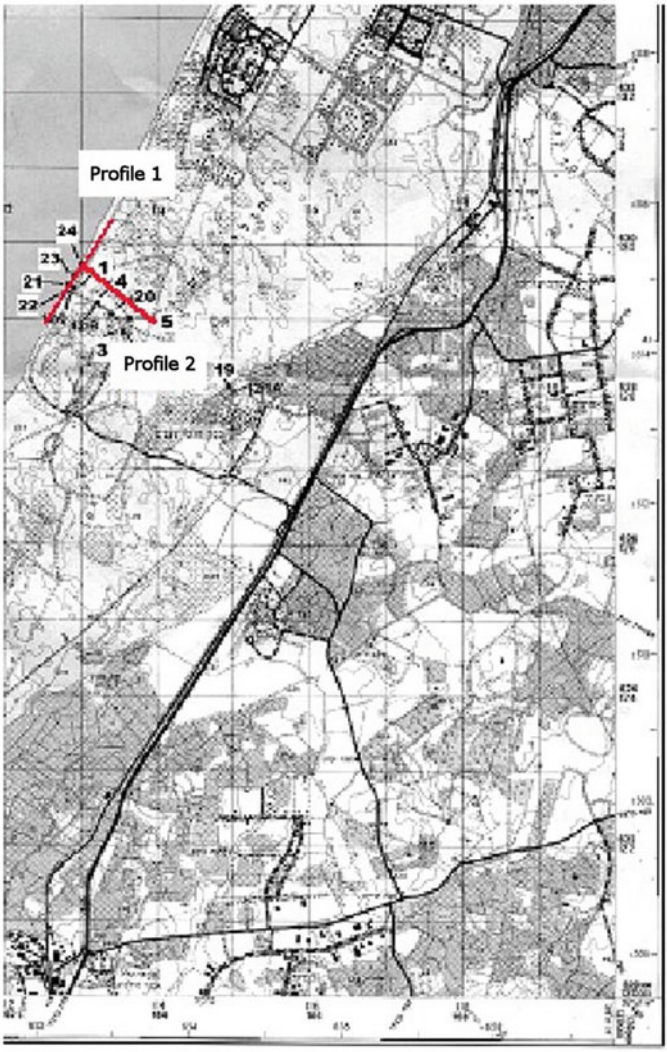


Fig. 46 NMR/TDEM survey location map in the Nitzanim area

all over the world (Yaramanci 2003), one can use the interpreted SNMR water content to calculate the amount of fresh ground water with great confidence.

Figure 48 compiles the results of the integrated SNMR/TDEM measurements along profile 2 perpendicular to the seashore (Fig. 46). The freshwater portion of the aquifer is clearly indicated in the combined quasi-2D resistivity/water content cross-section. The fairly consistent water content data along the entire profile is evident, varying in a narrow range, roughly between 15 and 20%.

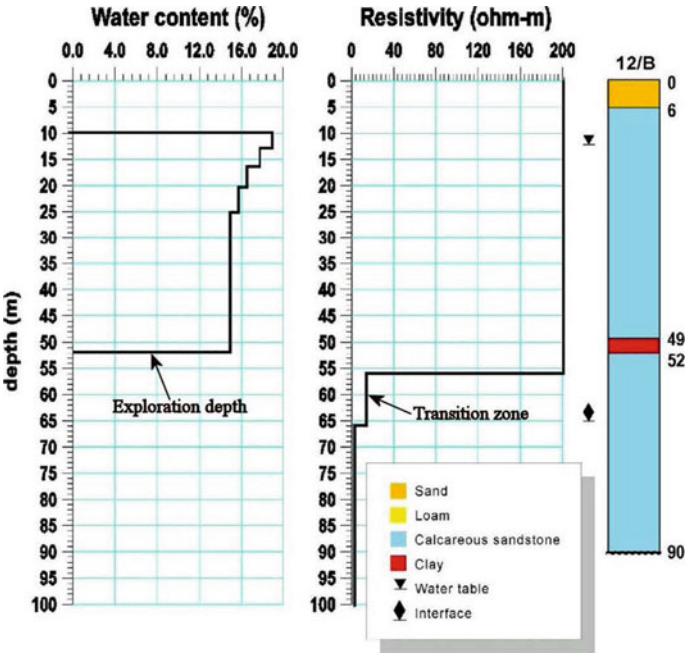


Fig. 47 SNMR-TDEM interpretation results near observation well 12/B (after Legchenko et al. 1998)

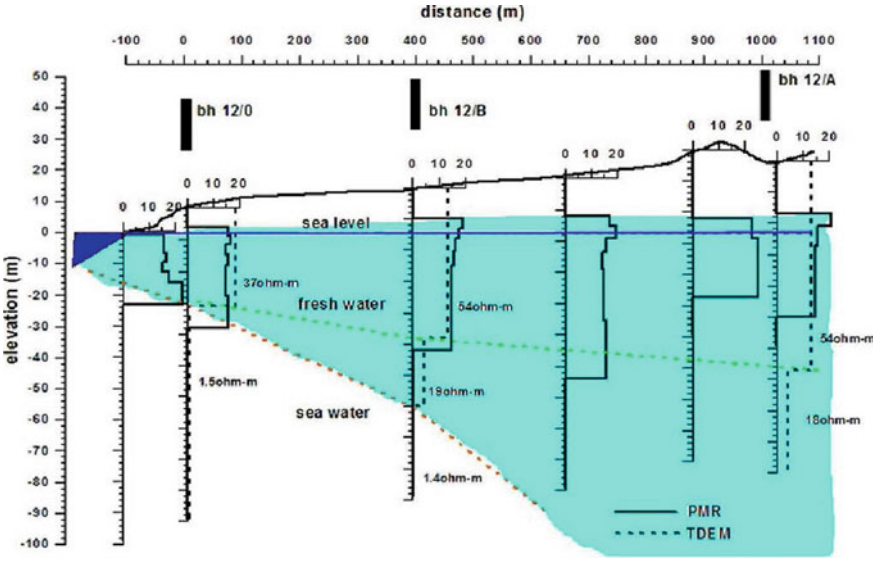


Fig. 48 Combined SNMR-TDEM columnar cross-section along profile 2. Blue color designates fresh groundwater. PMR = SNMR (after Legchenko et al. 1998)

The Dead Sea coastal aquifer

In order to test the proposed integrated SNMR-TDEM technology under a more complicated hydrogeological setup in an arid area, a similar survey has been carried out in the Ein Gedi area (Fig. 49). Due to topographic and technical limitations, it was impossible to align the SNMR-TDEM stations along straight lines. Therefore, the discussed profiles include also projections of the real measurements on the profile lines.

Unfortunately, in the absence of observation wells in the close vicinity of the SNMR-TDEM stations, the calibration measurements were performed under a similar hydrogeological set up of the nearby Tureiba area, adjacent to observation wells T_2 and T_4 (Fig. 49).

Figure 50 shows, as an example, the comparison between the geophysical results and the well log data from observation well T_4. One can see the same phenomenon

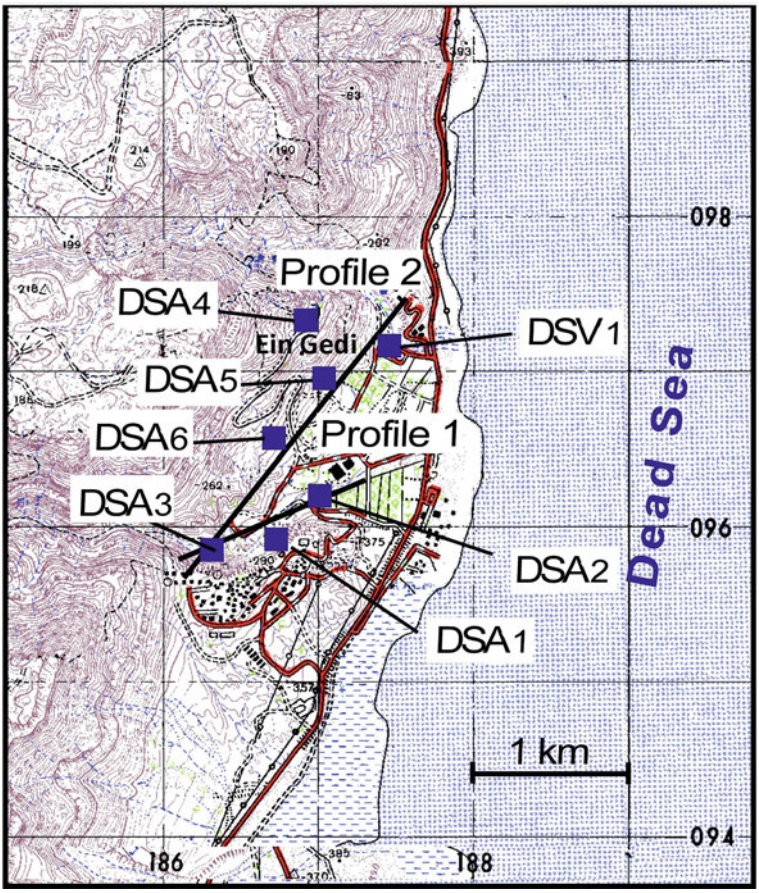


Fig. 49 SNMR-TDEM survey location map in the Ein Gedi area

that was observed in the Mediterranean coastal aquifer. Due to a rather low resistivity of the vadose zone, that does not provide a sufficient resistivity contrast with the saturated part of the aquifer, the TDEM method failed to detect the water table. However, the latter was accurately detected by the SNMR method (Fig. 50). Unfortunately, the Tureiba wells did not reach in depth the intruded DS brine body, and thus, a quantitative comparison with the TDEM results was not possible. Nevertheless, taking into account the very high reliability of the TDEM method in the detection of fresh/saline water interface, it is reasonable to assume that the resistivity drop slightly below 1 ohm-m at a depth of approximately 60 m (Fig. 50) reflects the upper boundary of the intruded, and somewhat diluted DS brine. Thus, similarly to the Mediterranean coastal aquifer, one can quantitatively estimate the amount of fresh groundwater in the Dead Sea coastal aquifer. Figure 51 demonstrates such an estimate along profile 1 as shown on the location map (Fig. 49).

It should be noted that near the Tureiba T_4 observation well, the SNMR decay time, which is closely connected to a transmissivity of the aquifer, accurately fits the location of the pebble layer in the depth interval of 33–40 m (Fig. 50). Unfortunately, such a good fit is not very common and is not statistically supported by the rest of the SNMR measurements.

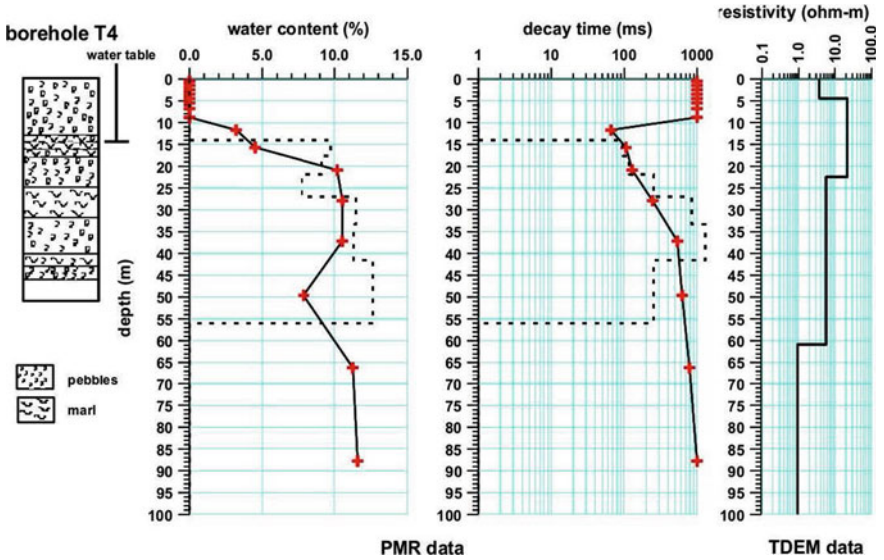


Fig. 50 SNMR-TDEM calibration measurements near the Tureiba T_4 observation well. The dashed and solid lines in the SNMR interpretation represent different inversion algorithms used (after Legchenko et al. 1998)

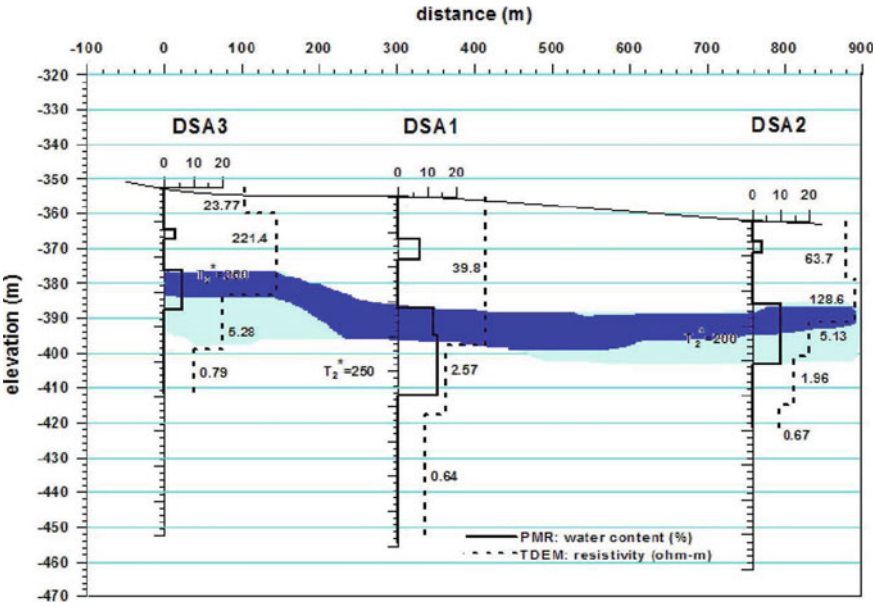


Fig. 51 SNMR-TDEM columnar cross-sections along profile 1. Dark blue color designates fresh groundwater, and light blue color designates brackish water. The amount of water is estimated by using the water content provided by SNMR. PMR = SNMR (after Legchenko et al. 1998)

5.2 Reflection Seismics—TDEM

The so-called hydrological reversals representing freshwater bodies below saline ones exist in all coastal multiple aquifer systems of Israel (Mediterranean, Dead Sea and Gulf of Eilat (Aqaba). They were detected and studied by hydrogeological and geophysical methods (e.g., Kafri et al. 1997; Goldman et al. 2003; Kafri and Goldman 2006). In many cases, the TDEM method alone is able to detect these bodies. However, if the thickness of the electrically conductive impermeable layers dividing the aquifer into sub-aquifers is similar or exceeds the thickness of the freshwater-saturated sub-aquifers, the resolution of the method could be insufficient to detect the latter. This phenomenon was observed at some places in the Mediterranean coastal aquifer and was particularly significant in the Gulf of Eilat (Aqaba) coastal aquifer (Goldman et al. 2003). In order to resolve this problem, additional detailed information, regarding the subsurface geological sequence, is essential. Unfortunately, obtaining such information from boreholes is problematic due to both their scarcity and high lateral lithological variability. Integrating TDEM, which is most sensitive to groundwater salinity with high-resolution reflection seismics, which is able to differentiate between the impermeable (clays and loams) and aquiferous (calcareous sandstones) layers was found as a reasonable choice to resolve the problem.

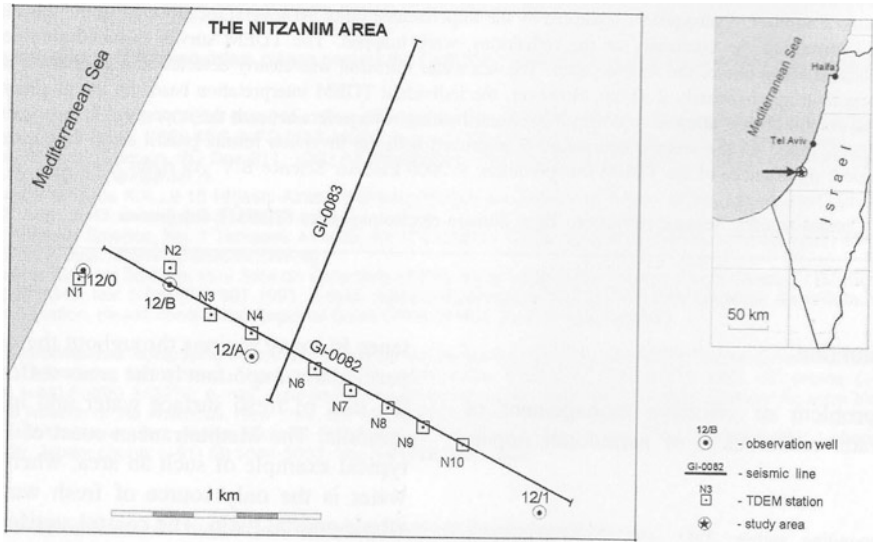


Fig. 52 Location map of the combined reflection seismic-TDEM survey in the Nitzanim area of the Mediterranean coastal plain

In order to test this approach, the TDEM and reflection seismic methods were jointly applied at the Nitzanim area in the Mediterranean coastal plain of Israel (Shtivelman and Goldman 2000) (Fig. 52). This area has been selected, since previous TDEM measurements testified to the existence of the above-mentioned hydrological reversal and lithological logs and salinity measurements from nearby observation wells were available to assist the interpretation of the geophysical data. Figure 53 shows the schematic W-E hydrogeological cross-section, based on the borehole data across the investigated site. Three of the wells (12/0, 12/A and 12/1) penetrate the entire aquifer down to the Saqiye Group, located at a depth of about 160 m in this area. The water table along the W-E profile (Fig. 52) varies from 2 to 3 m above mean sea level (MSL). The seawater intrusion into the upper sub-aquifers extends to approximately 1 km east of the shoreline.

The high-resolution seismic reflection survey included two seismic lines, GI-0082 and GI-0083 (Fig. 52), which were shot using the conventional *P*-wave technique. The TDEM measurements consisted of nine central loop soundings carried out along seismic line GI-0082 (Fig. 52). The transmitter loop size varied in accordance with the distance to the seashore. The measurements, close to the seashore, were performed using 50 by 50 m transmitter loops, while those located in the eastern part of the profile employed 200 by 200 m transmitter antennae. The corresponding penetration depths varied between roughly 100 m and approximately 300 m, respectively.

An example of a seismic time section for the western part of the reflection line GI-0082 is shown in Fig. 54. The horizontal axis on the section shows the station numbers, and the vertical axis exhibits the two-way travel time in milliseconds.

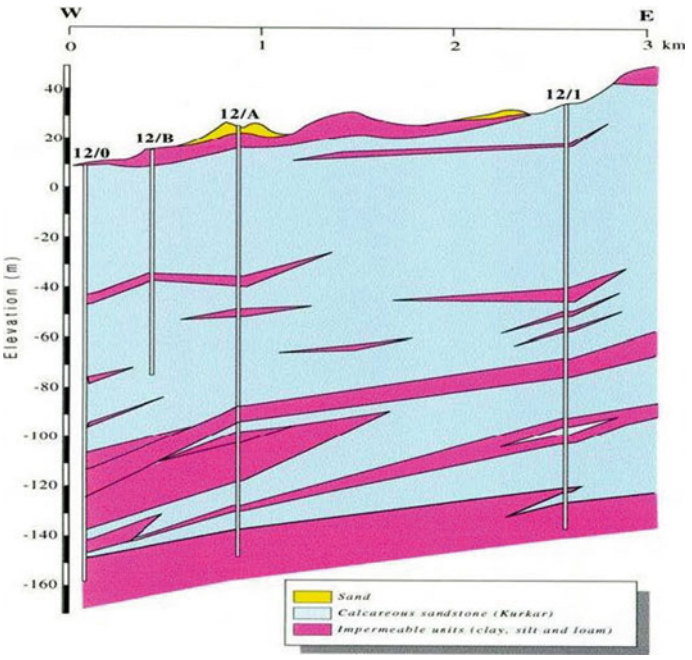


Fig. 53 W-E geological cross-section of the Nitzanim site

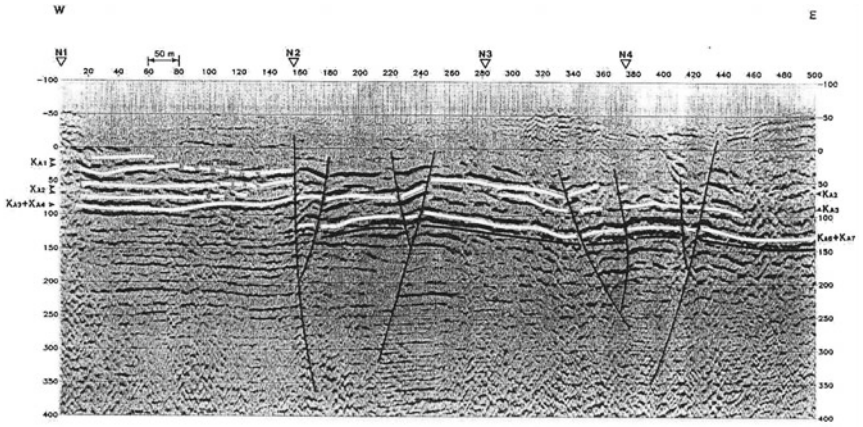


Fig. 54 Seismic time cross-section for the western part of the reflection line GI-0082 shown in Fig. 52 (modified after Shtivelman and Goldman 2000)

The times for the section located above the MSL datum appear as negative. The boreholes and the TDEM stations located in the vicinity of the line are shown above the section. Unfortunately, no velocity information from boreholes was available in the investigated area. Therefore, no depth conversion of the time sections was carried out. However, a *P*-wave velocity of 2000 m/s, measured on the average in boreholes throughout the coastal plain, was assigned to the relevant calcareous sandstone sequence (Fig. 53). This in turn enabled approximate estimates of the elevations of the reflected horizons, appearing on the sections.

A total of nine TDEM measurements were carried out along the entire seismic line GI-0082 (Fig. 52). Four of the measurements executed in the western part of the seismic line GI-0082 (Fig. 54) were used for the integrated interpretation.

The collected TDEM data were processed and interpreted using the Interpex TEMIX-XL 1-D interpretation software (Interpex 1996), while no a priori information has been used at this stage. The initial model for each sharp boundary inversion was obtained from the smooth Occam inversion (Constable et al. 1987) by reducing the number of layers in the recovered smooth model to a possible minimum. According to the above-described hydrogeological setting of the area, the interpreted resistivities were divided into four groups:

- Very low resistivities (less than 3 ohm-m) attributed to saline water-saturated lithologies (hosted in aquifers or trapped in aquicludes).
- Low resistivities (between 3 and 8 ohm-m), characterizing either brackish water-saturated aquifers or aquicludes (mainly clays).
- Moderate resistivities (between 8 and 15 ohm-m) typical for freshwater-saturated aquifers and some aquicludes (loams).
- High resistivities (above 15 ohm-m), typical of either unsaturated or freshwater-saturated aquifers.

The lateral extent of sea water intrusion into the aquifer was estimated from the TDEM measurements alone, by taking into account that the depth to the base of the aquifer (top Saqiye) is consistent at 160 m below sea level (Fig. 53). The fact is that the saline Saqiye beds, in most known cases, are characterized by low resistivities similar to those of seawater-intruded aquifers. Therefore, if the detected very low resistivity unit was located at depths shallower than 160 m, it was supposed to represent a seawater intrusion, otherwise it represented the Saqiye Group clays underlying the aquifer.

Figure 55 shows the pseudo-2D resistivity cross-section of the western part of the profile, compiled from the interpreted 1-D models. Two moderately resistive layers, which can be identified with freshwater-saturated sub-aquifers were detected beneath station N3. However, the lateral extent of the lower layer, hydrogeologically interpreted as a sub-aquifer, is unclear since it was not recognized at stations N2 and N4. It seems to wedge out laterally toward an aquicludal layer, having a resistivity of approximately 5 ohm-m, somewhere between stations N3 and N4, whereas between N3 and N2 the sub-aquifer probably becomes saturated with saline water. Thus, according to the preliminary TDEM interpretation, which is not based on any

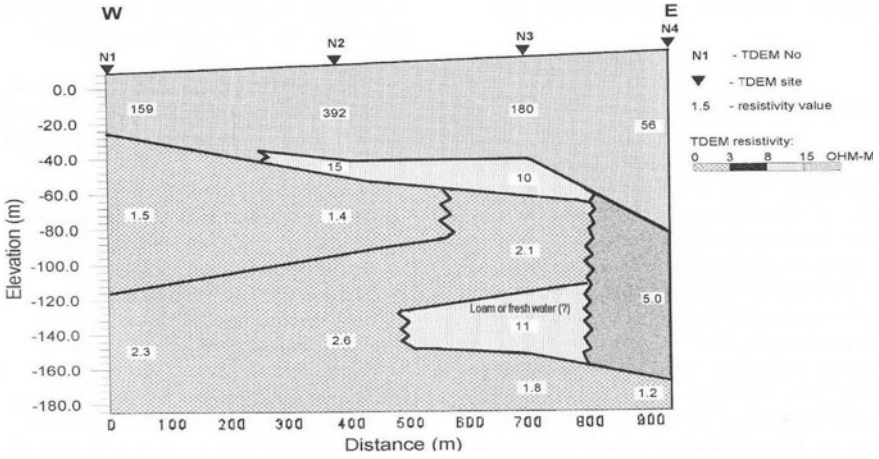


Fig. 55 Pseudo-2D resistivity cross-section of the Nitzanim site (preliminary interpretation) (modified after Shtivelman and Goldman 2000)

additional independent information, the lower freshwater-saturated sub-aquifer is confined to an area just below station N3.

In order to verify the above preliminary inversion results, a combined interpretation of the TDEM and seismic data was performed. According to the seismic interpretation, the lower calcareous sandstone units, designated as K_{A6} and K_{A7} extend from the vicinity of TDEM station N2 continuously through stations N3 and N4 and farther eastward (Fig. 54). The moderately resistive layer, detected beneath station N3 at approximately the same elevation (−120 m), is most likely identified with the detected seismic horizon saturated with freshwater. Provided that this assumption is correct, the data collected at station N4, where the lower sub-aquifer was not detected by the preliminary TDEM inversion, should be re-interpreted. In order to make the TDEM interpretation consistent with the seismic results, an additional moderately resistive layer has been included in the initial model for the follow up TDEM inversion. The results of this inversion are shown in Fig. 56b. For the sake of convenience, the figure also shows the results of the preliminary inversion (Fig. 56a). The uncertainty in the model parameters is approximately estimated by using the so called linear equivalence analysis (Goldman et al. 1994b). The best-fit models are drawn by solid lines in the resistivity versus depth sections in Fig. 56a, b. Dashed lines in the figures represent alternative (equivalent) models, for those where the misfit error is only slightly greater than that for the best-fit model and whereby the misfit error is still bellow an assumed noise level of the measurements.

As expected, both the resistivity and thickness of the lower moderately resistive layer are poorly resolved, and therefore the inclusion of this layer in the interpreted model is only justified, if it is supported by additional independent information, such as in our case the results of seismic interpretation in this case. Figure 56b shows that the inclusion of this layer in the final combined interpretation decreased

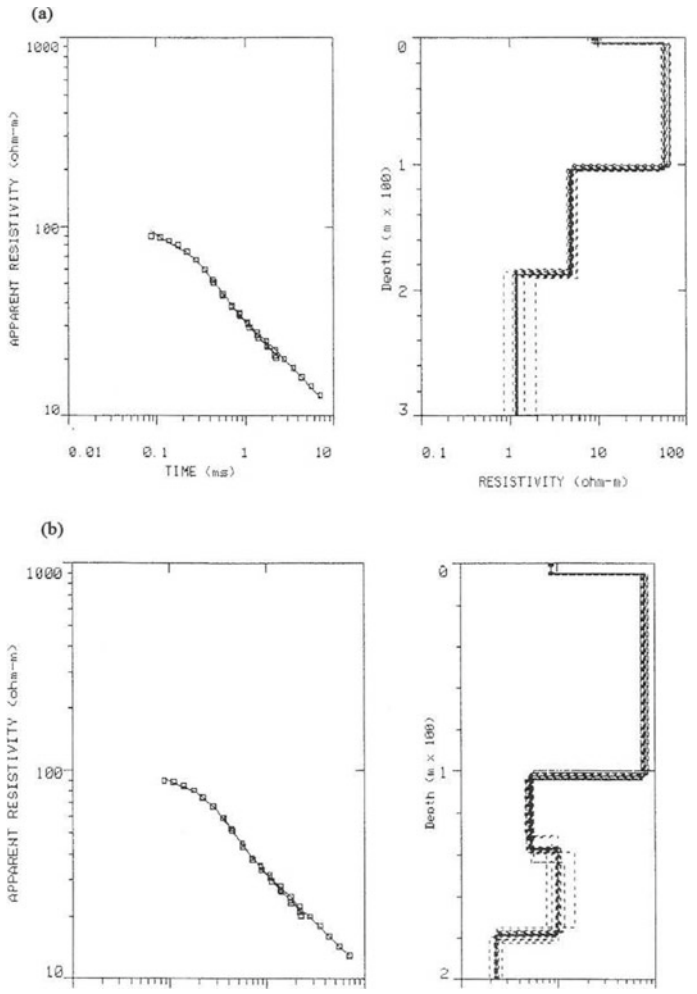


Fig. 56 TDEM results at station Nitzanim N4. Left panels show the acquired data (open squares) and the best-fit synthetic responses (solid lines). Right panels show the best-fit (solid lines) and equivalent (dashed lines) models. **a** Preliminary interpretation based on the minimum possible number of layers. **b** Final interpretation based on the independent seismic results (modified after Shtivelman and Goldman 2000)

the misfit error of the TDEM inversion from approximately 3%, obtained in the preliminary inversion, to slightly more than 2%. From a practical point of view, such an improvement of the inversion by itself is not that crucial because the estimated total noise level is larger than 3% anyway. Therefore, it can be considered just as a bonus to the main result, namely the consistency of the final TDEM inversion with the independent seismic results. Thus, the hydrogeological interpretation, of the

combined TDEM-seismic results, which testifies to the continuation of the relatively resistive freshwater body below station N3 toward station N4, seems to be justified.

The final resistivity cross-section, accompanied by the borehole data, is shown in Fig. 57a. One can see that the location of the lower moderately resistive unit roughly coincides with the lower sub-aquifer in well 12/A, although the coincidence is not perfect, due to the above-mentioned uncertainty in the layer parameters. The resistivity of the aquiclude layer, separating the two sub-aquifers, varies laterally from approximately 5 ohm-m in the east (station N4) to slightly more than 2 ohm-m

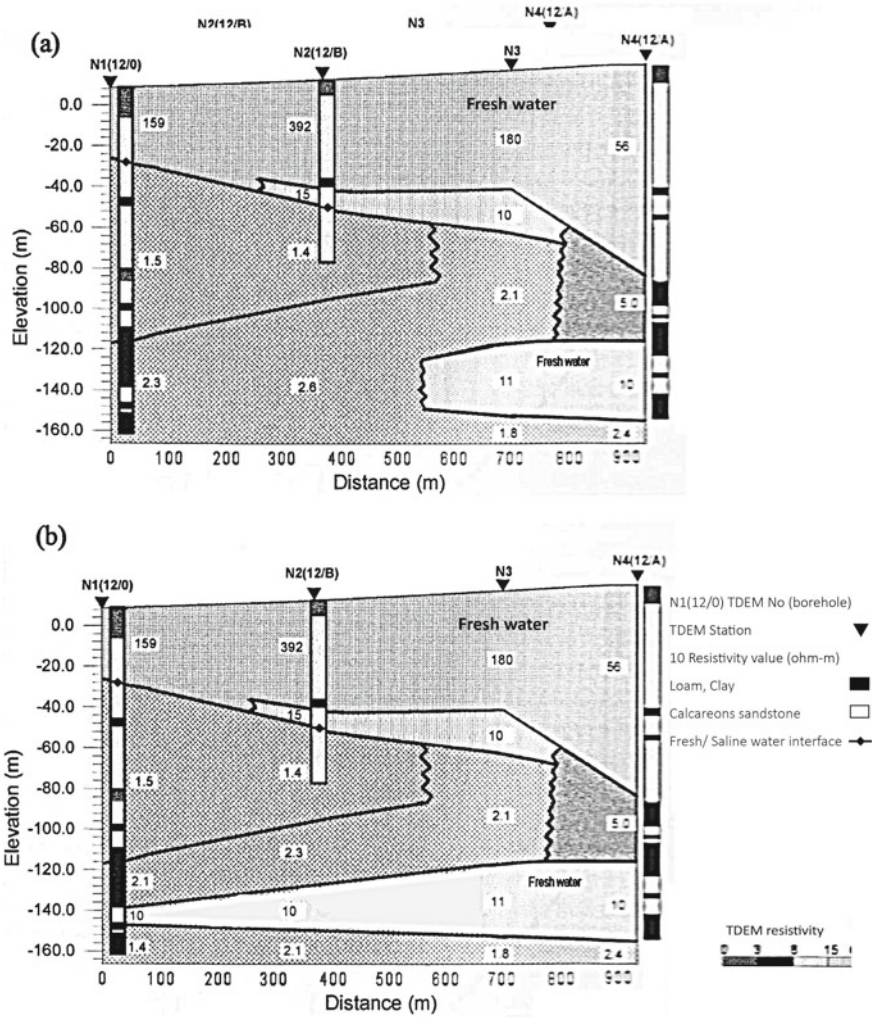


Fig. 57 Pseudo-2D resistivity cross-section of the Nitzanim site (final interpretation) (modified after Shtivelman and Goldman 2000)

in the west (station N3). Such a lateral variation can be reasonably explained by expected higher salinities of water, trapped in the aquiclude in the west, closer to the sea. It also explains, why the lower sub-aquifer was not detected in station N4 during the preliminary interpretation. Most likely, it happened due to the much smaller resistivity contrast between the aquiclude and underlying sub-aquifer at station N4, as compared to station N3.

Another small resistivity contrast within the very low-resistivity unit of approximately 1.5 and 2.3 ohm-m can be most likely identified with a boundary between the sea water-saturated aquifer and the aquiclude. Finally, the boundary between the moderately/highly resistive layers and very low-resistivity unit exactly coincides with the freshwater/seawater interface detected in wells 12/0 and 12/B. The interface was not detected in well 12/A, apparently due to the impermeable clays that exist elsewhere in the vicinity of station N3.

Also, the re-interpretation of TDEM measurements N1 and N2 similarly to the one shown in Fig. 57a, led to the appearance of a thin freshwater-saturated sub-aquifer (Fig. 57b). This sub-aquifer was indeed known at the depth interval of −138 and −154 m in the nearby observation well 12/0. The unit was not detected by the seismic measurements, apparently since it is too thin at this location.

Comparison of the seismic results (Fig. 54) with the geological cross-section (Fig. 53) interpolated using the well log data between boreholes 12/A and 12/B shows some discrepancy typical for regions with sparse borehole data. In this case, seismic sections usually provide a continuous and, therefore, more diagnostic image of the subsurface.

Thus, the consideration of independent seismic results motivated a re-interpretation of the preliminary TDEM data, leading to the resolution of non-uniqueness in the original TDEM inversion. As a result, the combined interpretation of seismic and TDEM data resulted in an improved hydrogeological significance of the geophysical results.

6 TDEM Characterization of Groundwater Salinity in Deep Carbonate Aquifers

Saline groundwater bodies hosted in deep, mostly carbonate, aquifers were detected in deep boreholes from all over Israel. The geochemistry of these aquifers was investigated in several studies (e.g., Starinsky 1974; Kafri and Arad 1979; Nativ 1984). In addition, it was essential to spatially delineate these bodies and assess their salinity (whether seawater or brine concentrations) that could have helped in obtaining their overall salinity, and to shed light as to their origin. For this purpose, the geophysical methods seemed to be the most suitable and, among them, the TDEM method was found to be superior thanks to its high accuracy and sensitivity to electrically conductive targets (Goldman et al. 1988, 1991).

The first TDEM measurements using SHOTEM transmitter/receiver configuration were aimed to check the possibility of subsurface seawater intrusion into the carbonate Judea Group Aquifer (hereafter JGA) close to the seashore. The survey was carried out in the Mount Carmel area and, later on, more to the south in the central coastal plain (Goldman et al. 1988). The obtained promising results allowed to tackle a more complicated geophysical and hydrogeological issue, namely the possibility of subsurface seawater intrusion from the Mediterranean Sea to the DSR base level as was suggested by Kafri and Arad (1979). In order to examine this assumption, deep TDEM traverses connecting the Mediterranean Sea and DSR were carried out to provide the resistivity and thus the salinity of the groundwater bodies along the traverses as well as the configuration of the interfaces between the saline bodies and the overlying fresh groundwater system.

A total of four generally W-E directed deep TDEM profiles were run, between the Mediterranean Sea and the northern DSR and between the Mediterranean Sea and the Dead Sea. Also, a deep TDEM traverse was executed in the south along the Arava Valley between the Dead Sea and the Gulf of Eilat (Aqaba). These measurements were performed in order to delineate and characterize the different saline water bodies in depth and thereby to determine their origin.

The results of all the above-mentioned studies are described herein.

6.1 Mount Carmel-Central Coastal Plain Area

A number of SHOTEM measurements were carried out to detect possible subsurface seawater intrusion into the JGA in the Mount Carmel area in northern Israel close to the Mediterranean Sea. The aquifer there is phreatic and open to the sea. Salinization processes were detected in some of the groundwater exploitation wells in this area. As a result, it was assumed that the aquifer is intruded by seawater (Kafri 1967, 1969; Arad et al. 1975; Kafri and Arad 1979). For calibration purposes, the measurements were performed adjacent to relevant existing water wells (Fig. 58).

The results of the measurements can be summarized, as follows (Goldman et al. 1988):

1. The actual freshwater/seawater interface was not encountered in the discussed wells since it was assumingly below the bottom of the wells. Therefore, in order to be correlated with the relevant SHOTEM results, the depth to the interface was calculated for each well, based on the freshwater head through the Ghyben–Herzberg equation.
2. The K1 TDEM station is adjacent to Nahal Oren 2 well. As shown in Fig. 59, the upper resistivity contrasts coincide with parallel lithological boundaries in the well log. However, the lower TDEM boundary between very high and low resistivities, which is typical to lithological unit saturated with seawater, does not correspond to a lithological boundary. The depth of this boundary is indeed compatible with the depth of the calculated interface at around 200 m.

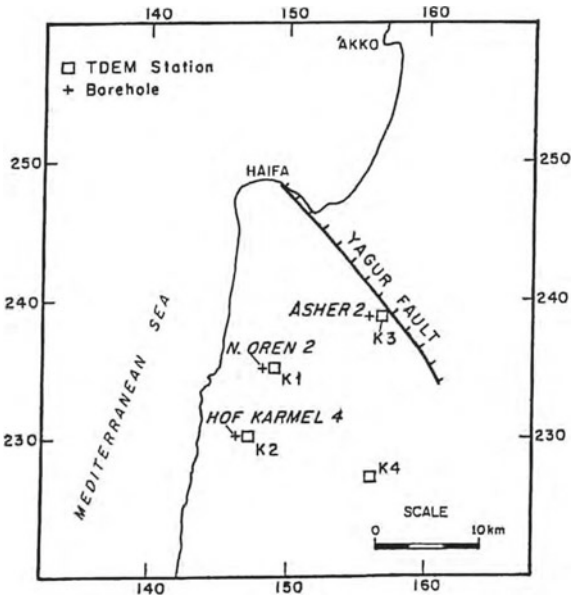


Fig. 58 Location map of the TDEM survey in the Mount Carmel area

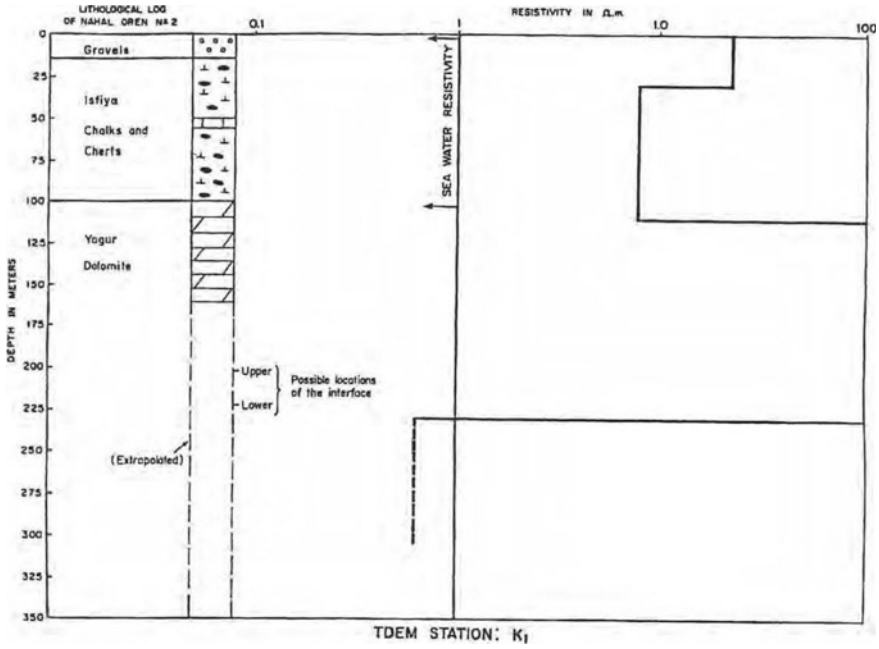


Fig. 59 Correlation of the TDEM results and well log data from the Nahal Oren-2 site (after Goldman et al. 1988)

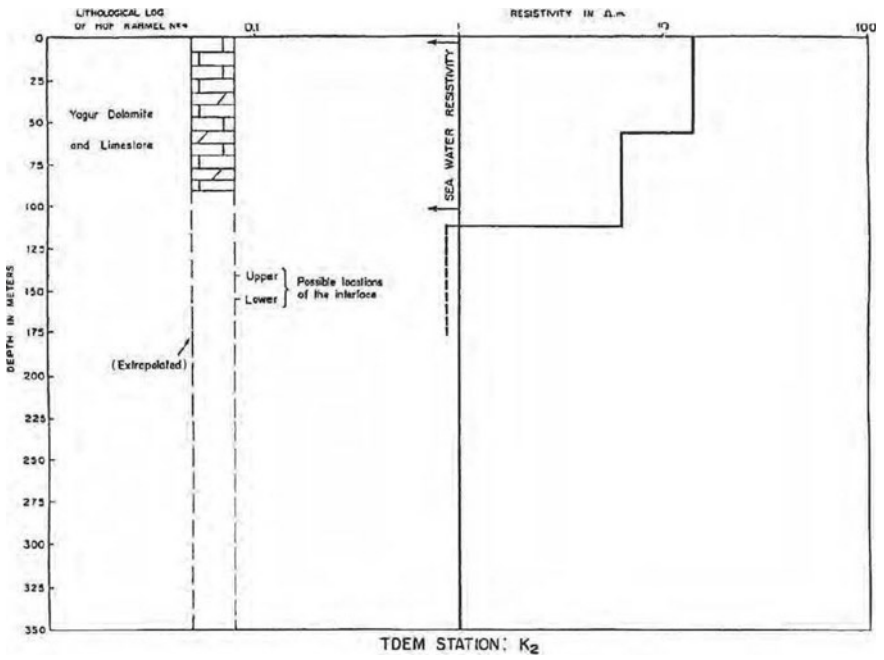


Fig. 60 Correlation of the TDEM results and well log data from the Hof Carmel-4 site (after Goldman et al. 1988)

3. Similarly, the K2 TDEM station is adjacent to Hof Carmel 4 well. In this case (Fig. 60), the upper resistivity contrast within a homogeneous lithology assumingly represents a boundary between the upper freshwater body and a mixing zone with a saline end member underneath. Here, the depth to the lower resistivity boundary is somewhat lower than that of the calculated fresh/seawater interface (roughly at 140 m), apparently, due to the existence of the interpreted mixing zone.

In the cases of TDEM stations K3 and K4, the penetration depth of the measurements was not sufficient enough to detect the calculated depths to the interface.

In the southern fringes of Mount Carmel and farther south in the central coastal plain of Israel, groundwater salinity was encountered in some of the wells exploiting the JGA as part of the northern Yarkon–Taninim aquifer system. It was agreed in the past that the system is blocked to the sea and thus not subjected to seawater intrusion. Contrary to it, Kafri (1967) and Kafri and Arad (1979) claimed that the encountered salinity in those wells is related to current seawater intrusion. In exploration boreholes, drilled later on in the northern Yarkon–Taninim aquifer system, indeed, groundwater of approximately seawater salinity was encountered at their bottom at a considerable depth (Goldman et al. 2006). Unfortunately, due to the "masking" effect of the overlying rather thick conductive Saqiye Group shale sequence, this target

was beyond the penetration depth of the SHOTEM method. The LOTEM method, however, provided sufficient penetration depth and, in some cases, succeeded to detect low resistivities corresponding to the high salinities in the boreholes. Unfortunately, in most cases, the LOTEM method also failed to accurately detect the encountered deep fresh/seawater interface (Goldman et al. 2006).

Fortunately, in the case of the Taninim deep well, where the Saqiye Group is missing, both methods managed to accurately detect the fresh/seawater interface (Figs. 61 and 62). The interface within the JGA is encountered at the depth of 600 m in accordance with the salinity in the Taninim deep well.

6.2 Mediterranean Sea-Northern DSR Traverse

Both the JGA and the Sea of Galilee are subjected to salinization processes through deep-seated saline water bodies, which occupy the DSR base level and the northern valleys (Fig. 63). Both sources and their mechanism of salinization have been debatable for a long time (see also Chapter "The Eastern Dead Sea Rift Continental Groundwater Base Level"). In this regard, Kafri and Arad (1979) suggested that part of the salinization is connected to current subsurface seawater intrusion from the Mediterranean to the DSR base levels due to the elevation difference between them.

In order to detect those deep-seated saline water bodies, the geoelectromagnetic (GEM) methods were employed. The objective of the initiated survey was to obtain the spatial distribution of the deep-seated saline water bodies as well as the configuration of the interface between them and the overlying fresher ones. This in turn could help to clarify the source and mechanism of groundwater salinization in the region. The results of the study were detailed by Kafri et al. (2007) as follows:

A total of 28 central loop SHOTEM soundings were executed between the Mediterranean Sea and the Sea of Galilee along two traverses (Fig. 63) using mostly 500 by 500-m transmitter loops. The maximum penetration depth of the SHOTEM soundings varied between 500 and 800 m. LOTEM measurements, to a depth interval between 300 and 3000 m in the same places, were conducted using 16 transmitters (Tx) and 29 receiver (Rx)-stations with Tx-Rx separations between 3 and 4 km. Calibration measurements were also carried out adjacent to existing deep water wells.

Based on the above-mentioned working hypothesis (Kafri and Arad 1979), TEM, both LOTEM and SHOTEM measurements were conducted between the Mediterranean and the DSR, primarily at those locations, where deep-seated saline water bodies were expected to exist. The alignment of the measurement sites and the resultant traverses (designated A-A′ and B-B′ in Fig. 63) were supposed to follow expected intruded sea water flow paths from the Mediterranean to the DSR. The target of the survey was the relevant deep electric conductor that reflects the saline water bodies, occupied within the JGA which could have been obtained mainly by the LOTEM measurements. The bulk resistivities of the shallower, mostly younger graben fill

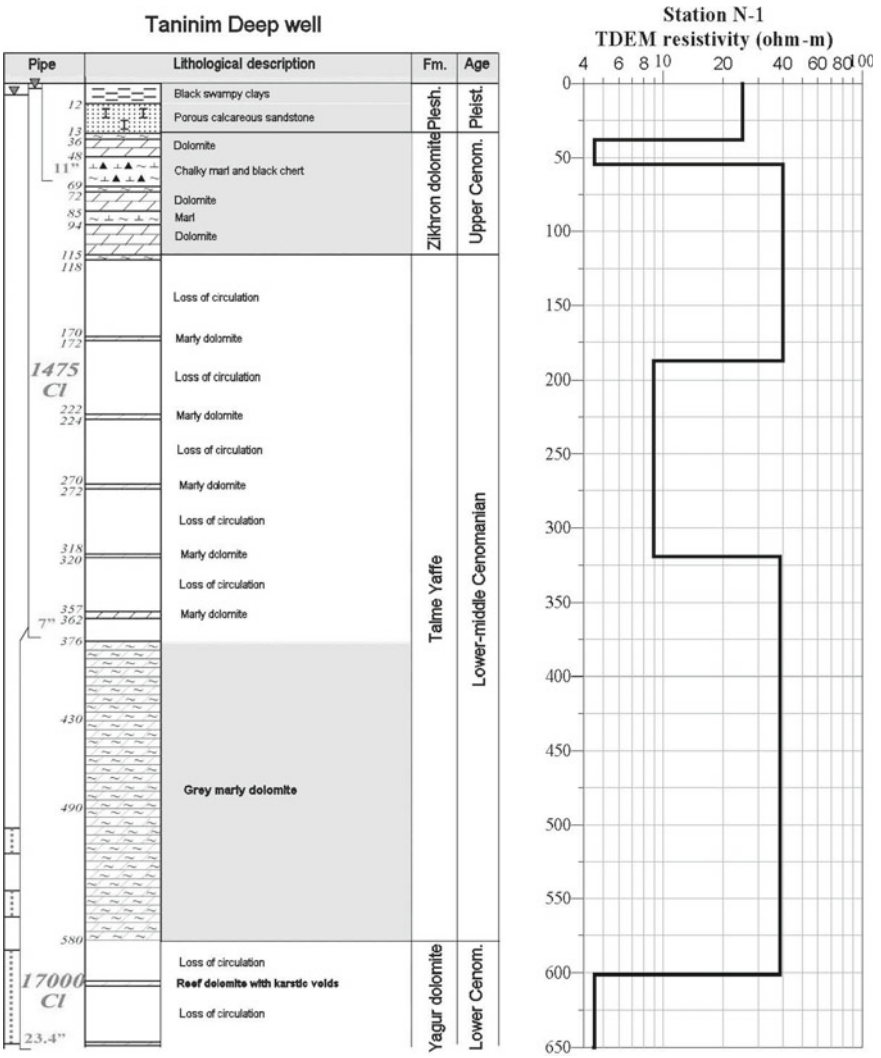


Fig. 61 Comparison of the deep SHOTEM (Cycle-5 M) and borehole results at the Taninim deep well site (after Goldman et al. 2006)

sediments, that overly the deep-seated JGA, were obtained by the SHOTEM measurements as also described by Kafri et al. (2007). However, in order to remain focused only on the deep target, the shallow data appear on the TDEM cross-sections in blank (Figs. 65 and 66).

Figure 64 exhibits the columnar resistivity cross-sections obtained by the mainly LOTEM measurements along traverses A-A′ and B-B′. Both exhibit deep-seated conductors with resistivities below 10 ohm-m, which were interpreted as groundwater

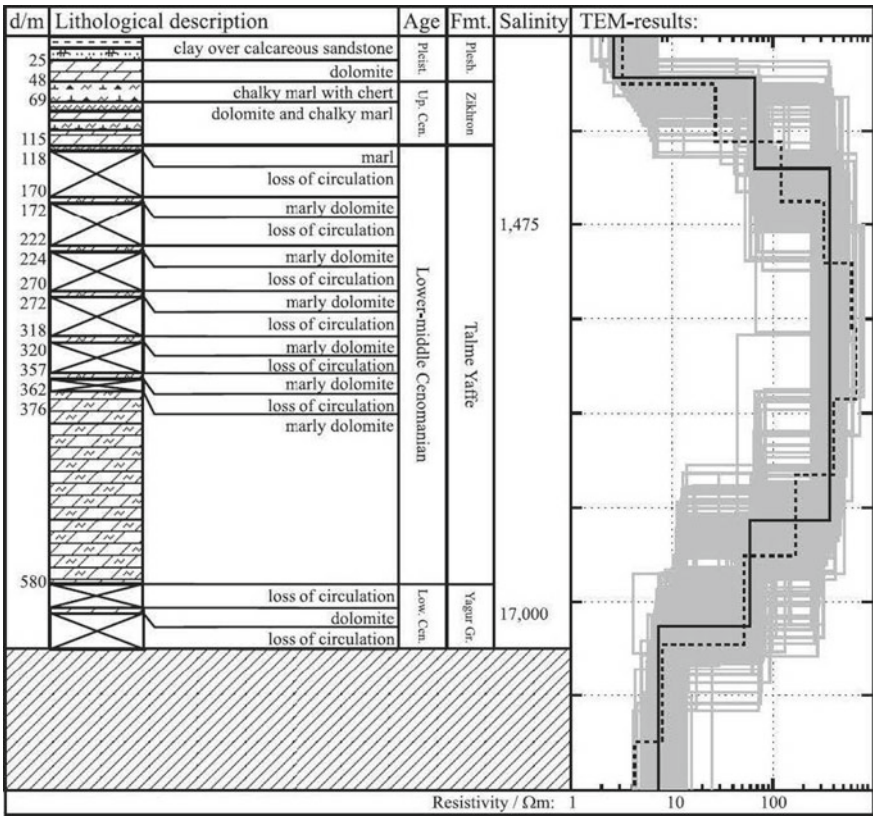


Fig. 62 Comparison of the LOTEM and borehole results at the Taninim deep well site. The black solid line represents the best-fit layered model, the dashed line shows the smooth model. The family of gray lines exhibits all equivalent inversion results (after Goldman et al. 2006)

of brackish to seawater salinity, in accordance with the calibration measurements in adjacent deep water wells (Scholl 2005).

Figures 65 and 66 exhibits a simplified geological cross-sections along the same traverses, coupled with the superimposed water table and the assumed fresh/saline groundwater interface representing the top of the detected deep conductor.

The Afeq-Bet Shean traverse A-A′ (Fig. 65)

The fresh-saline water interface along this traverse continuously declines eastward within, above and below the JGA, crossing different geological structures all the way from the Mediterranean Sea to the DSR. The decline of the interface follows that of the freshwater table, both below sea level along most of the eastward flow system. Close to the DSR margin, near the Bet Shean site, the deep-seated saline waters emerge to shallower levels or even to the surface through the fault system by the driving head of the freshwater mechanism, as described by Goldshmidt et al.

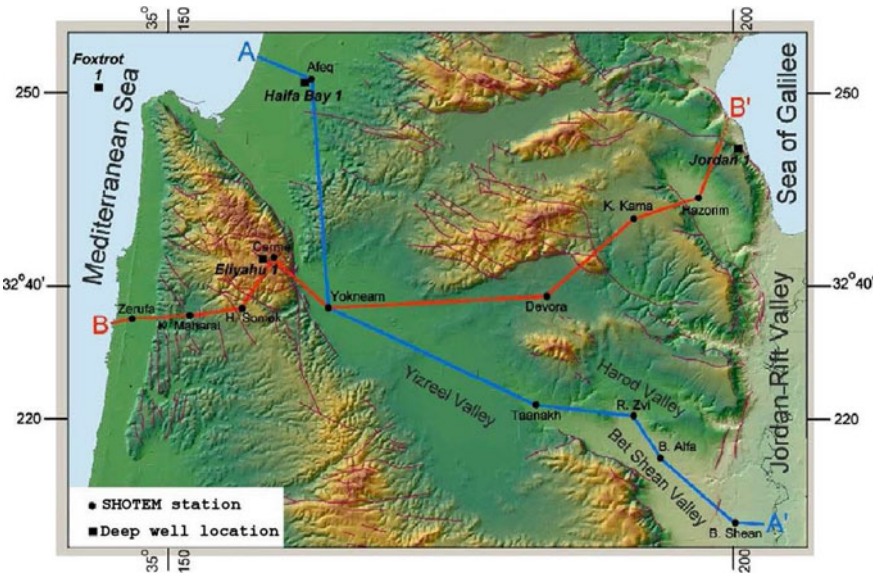


Fig. 63 Location map of the deep TDEM survey in northern Israel. Lines A-A′ and B-B′ designate the geophysical traverses (modified after Kafri et al. 2007)

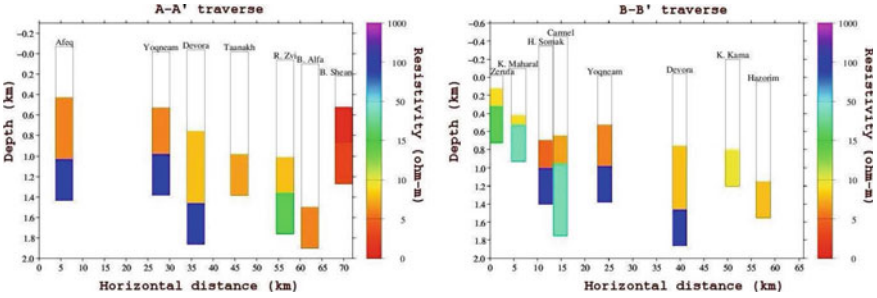


Fig. 64 Columnar resistivity cross-sections obtained by the TEM (mainly by LOTEM) measurements along traverses A-A′ and B-B′. The shallower resistivity sequence obtained by SHOTEM appears in blank (modified after Kafri et al. 2007)

(1967) and Kafri and Arad (1979). The steepness of the interface between the Bet Alfa and Bet Shean sites (Fig. 65) is controlled, as expected, by the very steep fresh groundwater flow gradient between those sites.

The Zerufa–Hazorim traverse (B-B′) (Fig. 66)

This traverse is running from the Mount Carmel coastal plain across the uplifted Mount Carmel block, along the northern valleys, crossing a structurally more elevated area close to the Sea of Galilee. The obtained configuration of the interface is as follows:

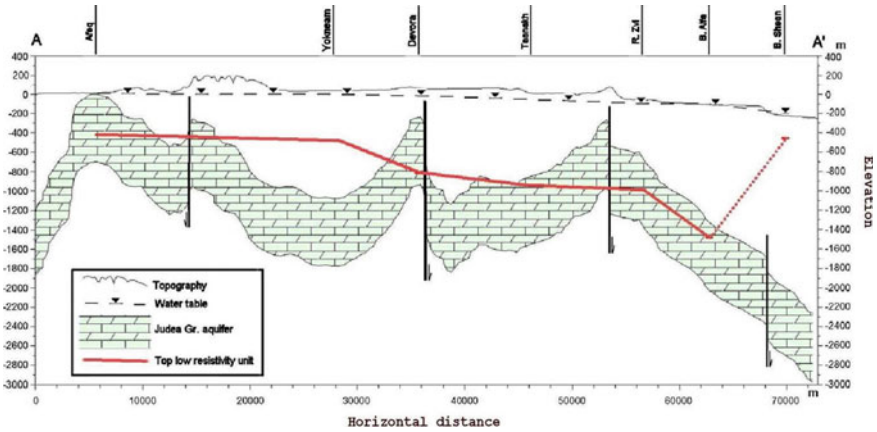


Fig. 65 A simplified hydrogeological cross-section along traverse A-A′ with superimposed water table and top of the deep conductor detected by TEM and identified with fresh/saline groundwater interface (modified after Kafri et al. 2007)

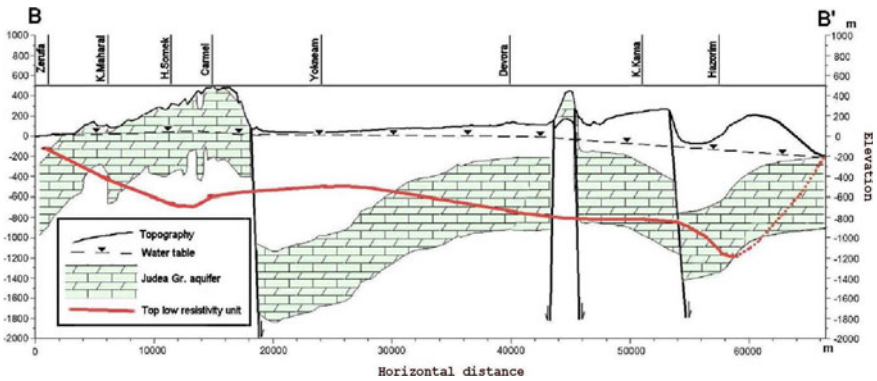


Fig. 66 A simplified hydrogeological cross-section along traverse B-B′ with superimposed water table and top of the deep conductor detected by TEM and identified with fresh/saline groundwater interface (modified after Kafri et al. 2007)

Groundwater level in the Mount Carmel block declines from the divide to the west and as a result, the interface declines eastward in accordance with the Ghyben–Herzberg approximation. From the divide eastward and farther to the Yoqneam site, the interface starts, as expected, to rise due to the drop of the groundwater level east of the divide. Farther east, groundwater levels mildly decline to slightly below sea level, accompanied by a mild decline of the interface. Close to the DSR, where the groundwater level gradient steepens, a steeper decline of the interface takes place accordingly. Due to the absence of TDEM measurement and hydrological information east of the Hazorim site, the extrapolated interface is shown as a dashed

line in Fig. 66. The obtained configuration provides a picture similar to the one obtained in the Bet Shean site in traverse A-A′.

All the above apparently support the proposed idea of subsurface seawater intrusion from the Mediterranean to the DSR. In addition, a hydrological model, carried out is also in accordance with the above described results (Kafri et al. 2007).

6.3 Mediterranean—Dead Sea Traverse

The subsurface of the Be'er Sheva valley between the Mediterranean Sea and the DSR occupies a relatively shallow and rather fresh groundwater system, which is underlain by deeper saline water bodies The different deep-seated saline sources were discussed and interpreted by Starinsky (1974), Nativ (1984) and Fleischer et al. (1997). Two basic types of brines were proposed as follows:

1. The C-type brine with rNa/rCl ratio similar to the marine one (0.82–0.86), found in the western part of the study area and
2. The R-rift type of Ca-chloride brine with rNa/rCl ratios below the marine ones found mostly in the DSR.

Stanislavsky and Gvirtzman (1999), demonstrated both the distribution of the above brine underneath the fresh groundwater system. They showed that the R-type brine is being density driven from the DSR westward to underneath the C-type brine. According to their model, the fresh groundwater system close to the DSR directly overlies the westward intruding Dead Sea brine (hereafter DSB) of the R-type.

Regarding geoelectric studies, shallow TDEM measurements managed to detect the interface between a fresher water body and the DSB body in the Dead Sea coastal plain aquifer (Kafri et al. 1997).

A deeper SHOTEM survey, aimed to the JGA, adjacent to the Dead Sea coastal plain (Levi et al. 2008), exhibited a different sequential salinity configuration from the one described by Stanislavsky and Gvirtzman (1999). The upper, high-resistivity freshwater body was found to be underlain below the theoretical depth of the freshwater-DSB interface, by a water body with resistivities higher than the expected ones for the latter. This could hint to the existence of an additional intermediate normal intruded seawater body from the west, between the upper freshwater body and the lower DSB one, as proposed for northern Israel (see Sect. 6.2).

In order to check this hypothesis, a deep TDEM survey from the Mediterranean Sea to the DS was executed, combined with geochemical considerations. The results of this study (Kafri et al. 2014) suggest that the hydrogeological setup is basically similar to the one discussed one in the northern valleys. It is assumed, contrary to previous assumptions, that the JGA is not blocked to the Mediterranean Sea, enabling seawater intrusion into it. However, unlike the case in the north, the DSR contains here a concentrated brine.

In order to understand the type and genesis of the different groundwater bodies in the study area and to delineate their spatial distribution, two roughly WNW-ESE

directed TDEM traverses were executed between the Dead Sea and the Mediterranean coastal plain (Figs. 67 and 68).

The traverses consist of 30 deep TDEM measurements located as close as possible to the existing wells for calibration (Fig. 68). Unfortunately, the TDEM traverses could not be reaching the west to the proximity of the Mediterranean shoreline since the shallow subsurface of the western coastal plain consists of a thick Neogene shale sequence of a very low resistivity which electrically masks the underlying

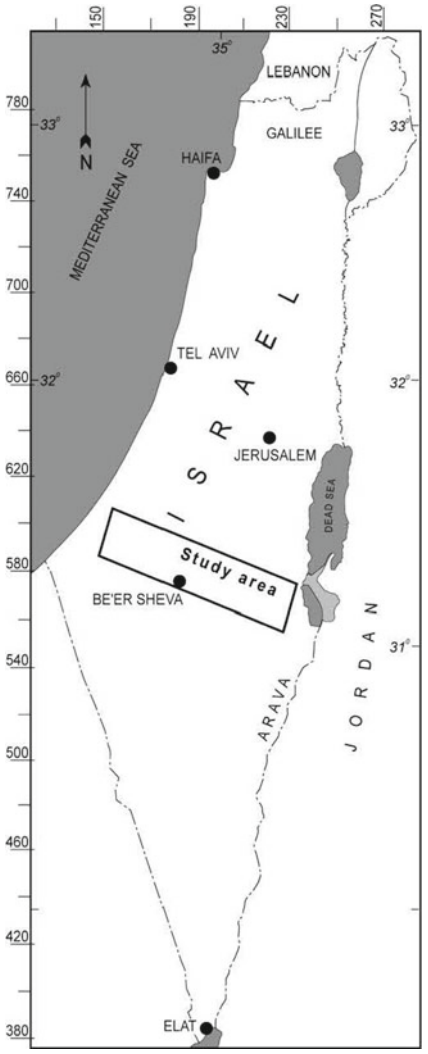


Fig. 67 Location map of the study area

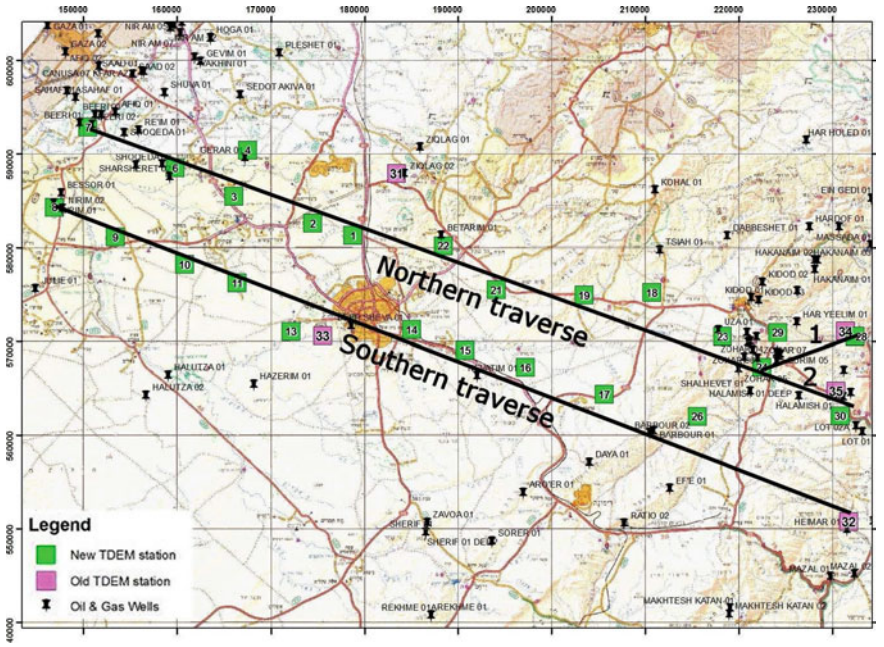


Fig. 68 Location map of the TDEM traverses (after Kafri et al. 2014)

depositional sequence. Hence, measurements were carried out only at sites, where the thickness of the shale sequence does not exceed some 250 m.

To enable a reasonable interpretation a known resistivity–salinity relationship was employed based on the following:

1. In the Mediterranean coastal aquifer with porosities between 0.2 and 0.4, the bulk resistivity of the seawater-intruded portion is known to be between 1 and 3 ohm-m with the mean value of approximately 1.7 ohm-m (Goldman et al. 1991, 2003: Kafri and Goldman 2006).
2. The Dead Sea coastal aquifer that is intruded by the concentrated DSB exhibits resistivities well below 1 ohm-m (Kafri et al. 1997).
3. Seawater-intruded carbonate aquifers in Israel (see also Sect. 7) are characterized by higher resistivities as compared to the granular aquifers (approximately, 5 ohm-m) due to their lower porosities (Goldman and Kafri 2004; Kafri and Goldman 2005).

The results of the TDEM campaign are shown on three columnar resistivity cross-sections along the measured traverses (Figs. 69, 70 and 71). These include the TDEM resistivity models, the main geological units, the water table and the groundwater salinities in g/l TDS (in blue when rNa/rCl ratio matches the marine one and in red when it is considerably low). Also, the minimum calculated possible depth to the assumed deep-seated brines, which are located beyond the maximum exploration depth of the TDEM measurements, is designated by a dashed line. Bulk resistivities

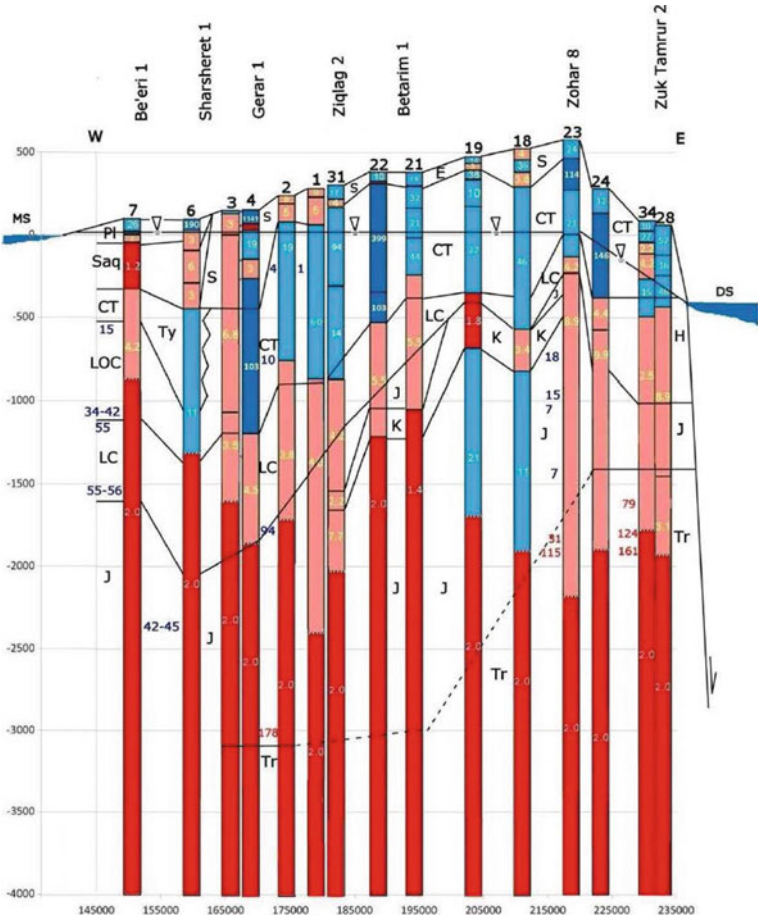


Fig. 69 Northern-1 TDEM columnar cross-section (after Kafri et al. 2014)

exceeding 10 ohm-m, represent fresh to slightly brackish water and appear on the cross-sections in blue color. Those between 3 and 10 ohm-m, representing roughly seawater concentration, appear on the cross-sections in pink color. A diluted deep DSB, evidenced by hydrochemical information from below a depth of 2 km, appears in red color on the cross-sections. For the assumed 0.1 porosity, a roughly calculated 2 ohm-m bulk resistivity was obtained for such brines and at some sites, as was indeed revealed by the TDEM method in depth. At all other sites, where these brines were assumed to be found below the maximum exploration depth of the TDEM

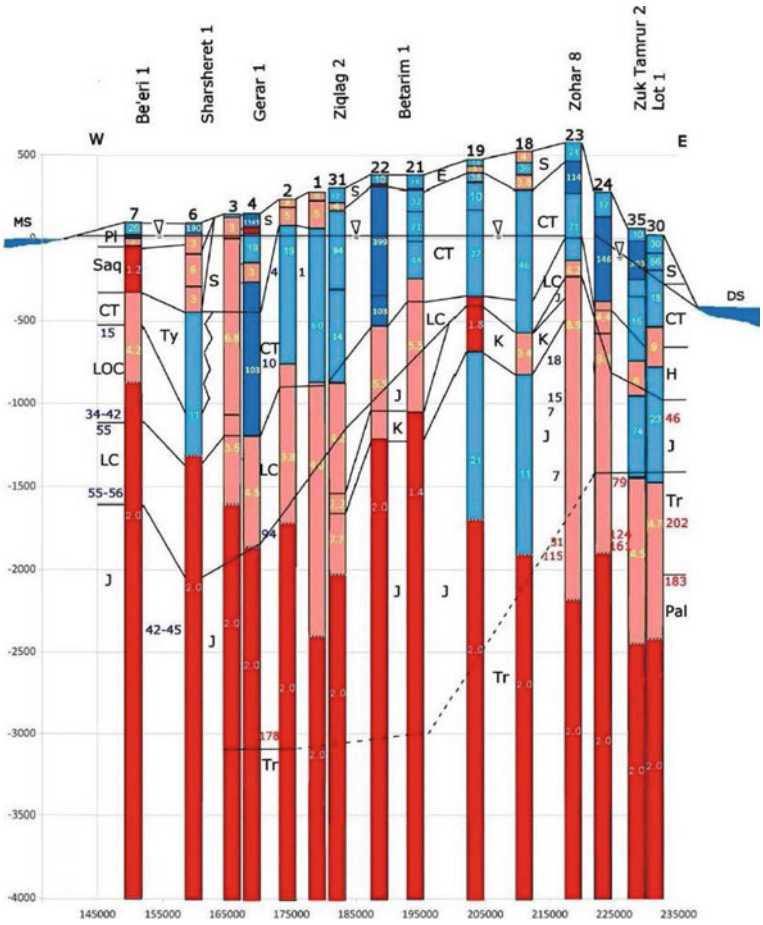


Fig. 70 Northern-2 TDEM columnar cross-section (after Kafri et al. 2014)

measurements, the 2 ohm-m resistivity was designated to a fictitious deep layer located at a minimum theoretically possible depth.

Northern cross-sections 1–2 (Figs. 69 and 70)

These two cross-sections are basically the same except that they end up at different locations in the easternmost part, close to the Dead Sea. Their proposed interpretation if as follows:

The uppermost high-resistivity fresh to somewhat brackish water body extends along the entire cross-sections from west to east. The depth of its base is assumingly controlled by the equipotential groundwater table at 10–20 m asl along most of the cross-section, except that it drops steeply to the DSR in the east. A relatively conductive unit (1.8–3.4 ohm-m) is recognized in measurements 18 and 19, "interbedded" within the uppermost high-resistivity layer most likely correlates well with the known

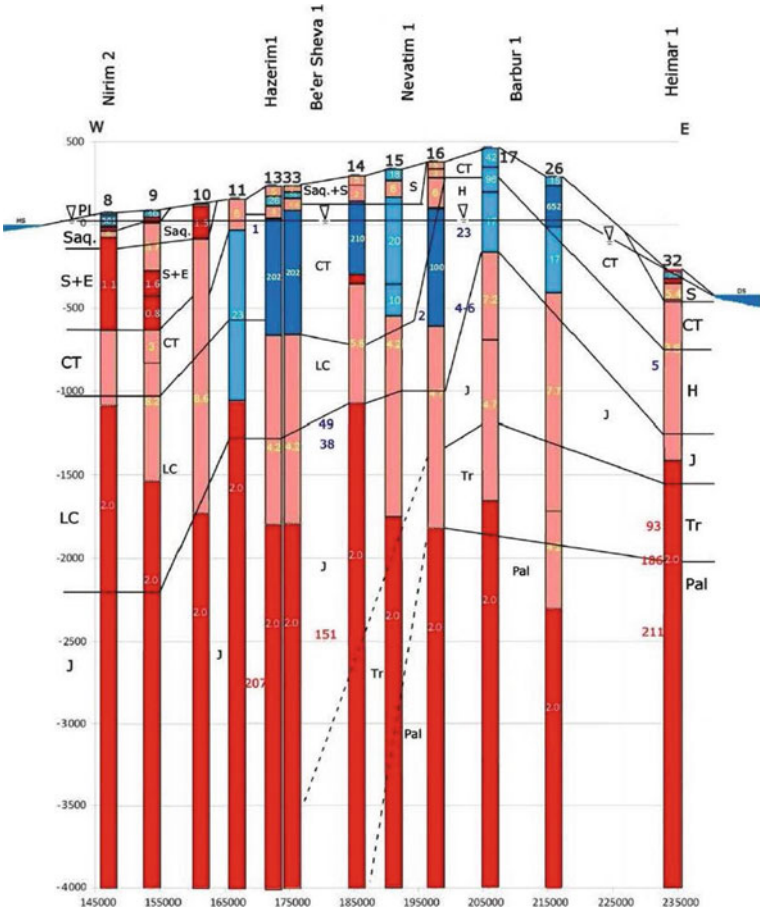


Fig. 71 Southern TDEM columnar cross-section (after Kafri et al. 2014)

conductive Jurassic Kidod shale Formation (designated K on the cross-sections) and known at the same depth interval. The electrical conductivity of this unit stems from its considerable amount of pyrite. Toward the Dead Sea in the east and the Mediterranean Sea in the west, the thickness of the uppermost high-resistivity body is being reduced considerably as expected toward both base levels on top of the declining interfaces underneath. In the west, its thickness is also reduced due to the existence of the overlying low-resistivity Neogene shale sequence.

The intermediate moderate resistivity layer, beneath the high-resistivity one, extends along almost the entire northern cross-section except for the more structurally elevated area in the north (measurements 18 and 19), which is outside the path of the assumed intruding seawater. Based on the hydrochemical interpretation, this layer represents a triple component mixture of deep-seated brines, intruding seawater and freshwater from above.

The deep and very low-resistivity brine body (≈2 ohm-m) was not detected in this cross-section except for two measurements (21 and 22). Its upper boundary matches the low-resistive Kidod Formation, whereby it assumingly masks the underlying sequence and, as a result, as an artifact, the brine body is artificially detected at a rather shallow level. Thus, the existence at depth of the brines in the west and east is detected, whereas in the middle of the cross-section it is mainly deduced from boreholes hydrochemical information.

Figure 72 represents a simplified northern cross-section that exhibits the main resistivity units.

Southern cross-section (Fig. 71)

The southern cross-section is located closer to the center of the Be'er Sheva Valley and thereby is less affected by the elevated higher groundwater levels to the north of it. It is thus most likely located entirely along the expected path of the intruding seawater.

Here too, the uppermost high-resistivity freshwater body is hosted in the Judea Group (CT) and the lower Cretaceous (LC, H) aquifers, extending along the entire

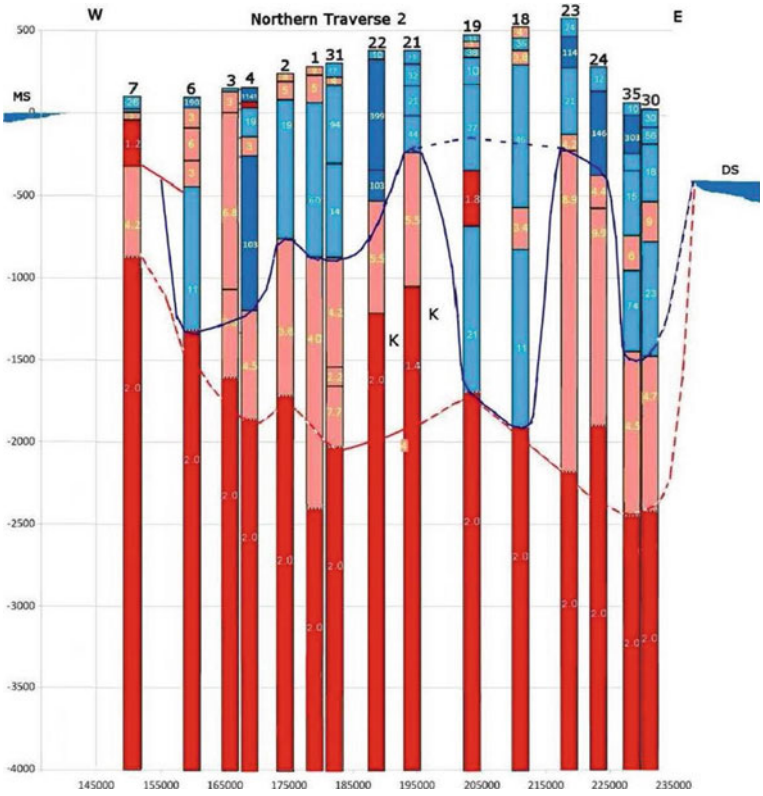


Fig. 72 Simplified northern-2 TDEM columnar cross-section (after Kafri et al. 2014)

cross-section. Here too the depth of its lower boundary is controlled by the equipotential water table, as noticed in the northern cross-section. Similarly, its thickness is being reduced toward the Mediterranean and the DSR base levels as expected.

The underlying intermediate moderate-resistivity body also extends here along the entire cross-section.

The very low-resistivity unit (2 ohm-m) underneath the above, which assumingly represents the deep-seated brines, was detected here in TDEM measurements 11, 5, 17 and 32 whereby its uppermost theoretical possible depth was calculated for the rest of the measurements. Similarly to the northern cross-section, the upper boundary of the unit roughly exhibits the presumed inclined interfaces the eastern and western base levels (Fig. 73) except of measurements 11 and 14, where the anomalously shallow appearance of this boundary seems to be affected by the rather shallow Kidod Formation at this site and thus, the top of the very low-resistivity brine unit is expected to be, in fact, at a deeper level in accordance with the rest of the cross-section.

Figure 73 represents a simplified southern cross-section that exhibits the main resistivity units.

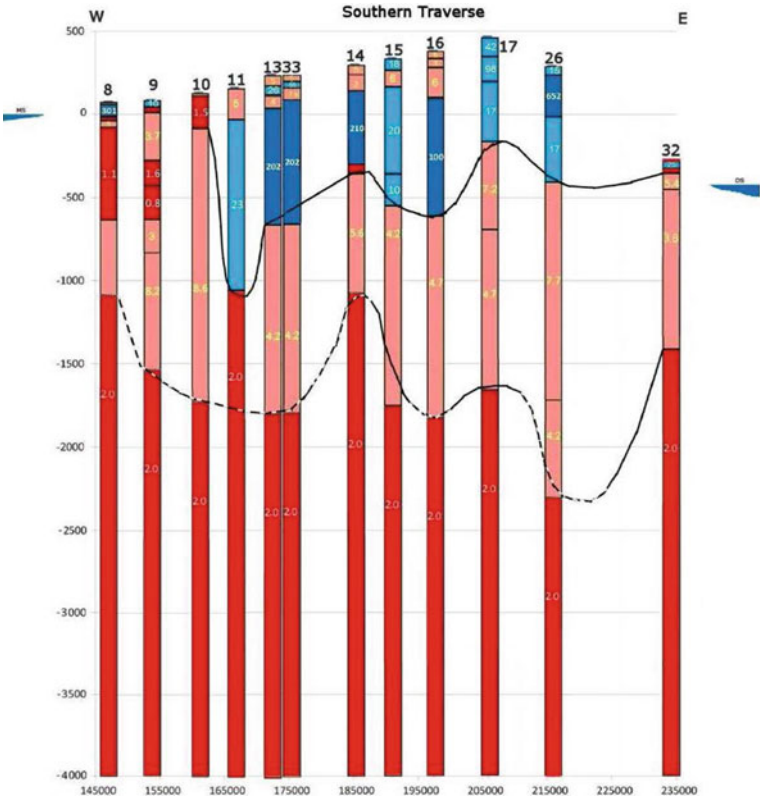


Fig. 73 Simplified southern TDEM columnar cross-section (after Kafri et al. 2014)

Based on the TDEM results, the combined geochemical considerations and the employed hydrological FeFlow model, the discussed groundwater system includes the following resistivity/salinity units from top to bottom:

1. An upper high-resistivity fresh to brackish water unit, which flows to the western Mediterranean and eastern DSR base levels, occupied by the Judea Group and the Lower Cretaceous aquifers.
2. An intermediate moderate resistivity unit, interpreted also geochemically as a mixing zone of three components, namely deep-seated brines, seawater intrusion from the west and freshwater.
3. A lowermost, very low-resistivity body representing the C- and R-type brines, whereby the latter are density driven from the DSR westward.

6.4 Dead Sea—Gulf of Eilat Traverse

The Arava Rift Valley extends between the Gulf of Eilat (Aqaba) in the south and the Dead Sea in the north (Fig. 74). The main problem there is related to the salinization processes of the groundwater systems caused through different mechanisms as well as from deep-seated saline sources *r* as proposed (see below) in numerous studies. In most cases, the existing boreholes or geophysical soundings were not deep enough to penetrate or detect those deep saline sources. A deep TDEM survey was thus carried out along the entire Arava Valley, targeted to detect the deep saline sources, and to spatially delineate their extension and to reveal the genetic relationship between them. It also aimed to delineate the different base levels and paleo-base levels in the south and north. The results of the survey were described by Kafri et al. (2008).

The hydrogeology and groundwater chemistry of the Arava Rift Valley and its western margins were discussed in numerous studies (i.e., Bartov and Bein 1994; Bein et al. 1995, 1998; Fleisher et al. 1996, 1997; Guttman et al. 1999; Issar et al. 1972; Kroitoru et al. 1981; Kronfeld et al. 1992; Magal and Yechieli 2002; Rosenthal et al. 1992). A concise summary based on previous studies as well as on some of the present observations is given below:

The regional mountain aquifers include the Paleozoic to early Cretaceous Kurnub Group (or Nubian sandstone) aquifer (see also Chapter "The Nubian Sandstone Aquifer in the Sinai Peninsula and the Negev Desert") and the late Cretaceous Judea Group aquifer. The graben fill aquifer, mainly within the Rift, includes the Neogene to Recent Hazeva Formation, the Arava conglomerate and the younger graben fill.

The general groundwater flow is from both western and eastern mountain aquifers into the graben fill aquifer. A hydrological divide within the Rift Valley (see below) dictates the separation and diversion of the flows to the Gulf of Eilat base level in the south and to the Dead Sea base level in the north.

Regarding—salinity, the Kurnub and the JG aquifers contain only fresh to brackish waters along the entire Arava Valley and its margins. Within the graben fill aquifer, freshwater is found close to the divide and to the outlets of the large streams that drain from the west the northern part of the valley. Elsewhere, fresh to brackish waters

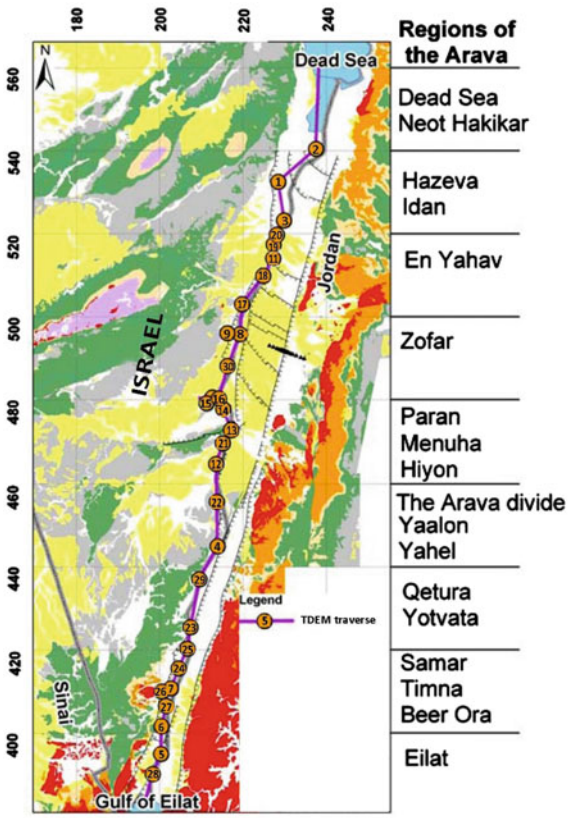


Fig. 74 TDEM locations superimposed on the geologic map of the study area (after Kafri et al. 2008)

are abundant. Groundwater with seawater salinity is encountered in the immediate vicinity of the Gulf of Eilat, whereas concentrated brines, with up to 202 gr/l Cl, were detected in the Arava 1 deep well close to the Dead Sea. Superficial to shallow brines were encountered in the so-called Sabkha sediments at Yotvata (up to 100 gr/l Cl) and Evrona. The latter were formed by emergence of brackish waters from depth and their evaporation close to the surface (Michaeli and Galai 1963; Michaeli 1971; Kafri 1977).

Several shallow TDEM surveys, up to a depth of a few hundred meters, were executed in the past along the Arava Valley, the results of which are described by Bartov et al. (1998), Goldman and Ezersky (1999b), and Kafri et al. (2001). Geoelectromagnetic surveys conducted in the framework of the Dead-Sea-Rift-Transec (DESERT) project provided additional information regarding both shallow (TDEM) and deep (magnetotelluric) structures in the eastern margins of the central Arava valley (Ritter et al. 2003; Koch et al. 2004; Maercklin et al. 2005). In general, the shallow measurements revealed in both the southern and northern Arava, rather

shallow, low-resistivity body interpreted as saline groundwater bodies of limited extension. In the north, in the vicinity of the Dead Sea those measurements detected a very low-resistivity unit (below 1 ohm-m) at a depth of several hundred meters. The magnetotelluric measurements in the eastern Arava valley revealed a considerable drop of resistivity to roughly 1–2 ohm-m at a depth of approximately 1.2 km.

The working hypothesis of Kafri et al. (2008) assumed that the regional base levels in the north and south served as sources of salinization in the past and at present. It was thus most important to reveal the salinization as well as flushing mechanisms of the system during the different time intervals of the system. Those mechanisms were suggested below for the conceptual model of the Arava Valley (Fig. 75).

The present base levels are at the elevations of 0 m and at 430 m bsl in the Gulf of Eilat and the Dead Sea, respectively. Based on their respective dated deposits, the Pliocene marine base level in the northern and southern Arava valley is assumed to be at 60–70 m asl, and the Quaternary Lisan lake base level, which existed only in the northern Arava valley at 180 m bsl. The marine base level, at the peak of the last glacial, all over the world and thus also in the Gulf of Eilat is at 120 m bsl. Current water table elevations are monitored and known. In the case of paleo-water tables, these were reconstructed, taking in account the location of the paleo base levels and the accepted assumed groundwater flow gradients, similar to the present day ones. The known saline water densities that occupy or were occupied in the different base levels are: 1.029 for normal seawater, 1.115 for the Lisan lake brine and 1.23 for the present Dead Sea brine. The configurations of the interfaces for each paleo-stage

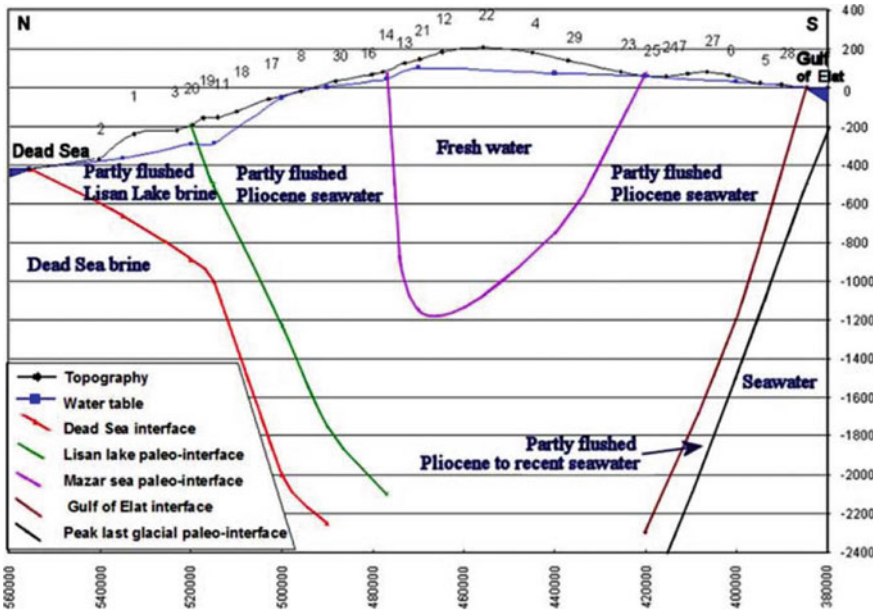


Fig. 75 A N-S cross-section of the conceptual model (after Kafri et al. 2008)

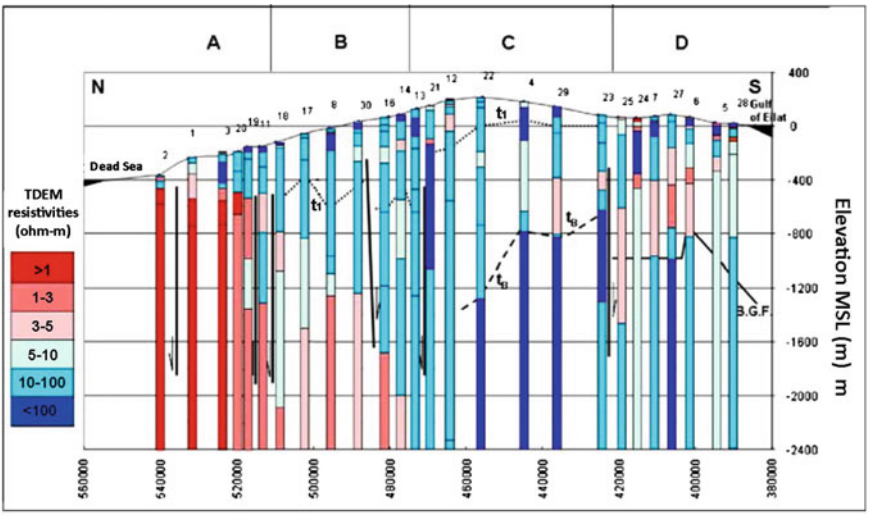


Fig. 76 A transversal columnar TDEM resistivity cross-section between the Gulf of Eilat and the Dead Sea. tB—top basement, tJ—top Judea Group, B.F.G—base graben fill (after Kafri et al. 2008)

were calculated using the Ghyben–Herzberg approximation. Based on the results of the above mentioned deep TDEM survey along the Arava valley, a cross-section between the Gulf of Eilat and the Dead Sea was constructed (Fig. 76). The cross-section exhibits, in addition to the resistivity intervals, the/also the top Basement, top Judea Group and the base graben fill aquifers. This enabled to attribute a specific resistivity interval to either a granular or to a carbonate aquifer, based on porosity considerations as discussed below (Sect. 7). The correlation was also insisted on borehole data.

Based on the structural, hydrological and the resistivity characteristics, the longitudinal cross-section was divided into four segments or basins (A, B, C and D in Fig. 76) from north to south and the resistivities, discussed below, are already translated herein to salinities:

Segment A (The Hazeva-Dead Sea basin)

The entire segment consists of the granular graben fill aquifer bottomed at depths between 2500 and 4000 m bsl. The interface between fresh and brackish waters and the underlying, brine of resistivity below 1 ohm-m, declines from north to south as encountered at measurement points (mp) 2, 1, 3, 20. The top of the very low-resistivity unit reaches a depth of at least 1400 m bsl. In mp 3, the brine is overlain by a 2 ohm-m resistive body, interpreted as a mixing zone with fresher water or as unflushed trapped seawater. Moreover, the brine body exhibits a gradual increase of resistivity southward, which indicates an increasing dilution with distance from the base level. Farther south (mp 11, 19), a southward declining interface between fresh to brackish waters and underlying water of approximately seawater concentration, is encountered.

Segment B (The En Yahav-Zofar basin)
This segment is aligned at the margins of the Rift where by the soundings penetrated only the mountain aquifers. The Judea Group carbonate aquifer contains fresh to brackish waters in its upper part, underlain in places (mp 17 and 8) by water of seawater concentration. The upper Kurnub Group granular aquifer below it contains waters of seawater concentration.

Segment C (The Menuha-Qetura basin)
This segment partly coincides with the central structural divide of the Arava Valley, where the Rift narrows and the graben fill aquifer, overlying the Judea Group aquifer, is very thin. The top of the crystalline basement here (mp 22, 4, 9 and 23) is relatively shallow, between 600 and 1200 m bsl, as estimated by downward stratigraphic extrapolation and through geoelectromagnetic measurements (Kafri et al. 2001; Ritter et al. 2003).

In its northern part, the segment is characterized by the absence of low-resistivity saline water along the entire penetrated sequence, supported by the occurrence of only freshwater encountered in the Yaalon 7 borehole adjacent to mp 4. Farther south (mp 23, 29), high (freshwaters) resistivities are encountered, also in complete agreement with adjacent boreholes both within the Judea and the Kurnub Group aquifers. However, contrary to the northern part of the segment, here those high resistivities are underlain by lower resistivities (brackish water) within the Kurnub Group aquifer.

Segment D (The Yotvata-Eilat basin)
This segment is aligned in the graben fill aquifer that thickens southward. Thin and low-resistivity bodies are encountered up to the depth of 100 m. These are interpreted herein as saline water bodies, related to existing or ancient sabkhas or to alternatively to unflushed clay layers (Kafri et al. 2001).

The upper high-resistivity (freshwater) body is not thick, and is bottomed by an interface declining from 100 to 600 m bsl in a south–north direction, and the underlying water body exhibits brackish to seawater concentrations. Moreover, the current seawater intrusion here from the south is limited to the close vicinity of the Gulf of Eilat, and it only occupies the uppermost graben fill aquifer (Bein et al. 1998). The underlying high resistivities are interpreted herein as the crystalline basement since no freshwater is expected at such depths below the saline ones.

No deep, very low-resistive brines have been encountered in the entire segment despite the fact that saline water in the nearby Timna site were interpreted as originating from deep-seated brines. Based on the above, an alternative interpretation suggests herein that the latter are density-driven brines downward, from the nearby existing or ancient surficial hallow sabkhas.

It is, thus, concluded that the overall salinity distribution configuration along the Arava Valley, as deduced from the longitudinal TDEM cross-section (Fig. 76) is basically in agreement with the one shown in the conceptual model (Fig. 75), Kafri et al. (2008).

7 Evaluating Porosity/Storativity of Saline Ground Water-Saturated Aquifers Using TDEM Method

7.1 General

Contrary to groundwater salinity and geometry of hydrogeological targets, the determination of hydraulic properties of porous media, using conventional geophysical methods, is highly inaccurate and ambiguous. This is caused by an inherent non-uniqueness of the geophysical interpretation accompanied by numerous uncertainties in mostly empirical relationships between hydraulic and geophysical parameters (Archie 1942; Winsauer et al. 1952; Glover 2016). Therefore, as mentioned before, during the last two decades major efforts were made in the development of novel geophysical methods (e.g., borehole tomography, SNMR, seismoelectric, electroseismic methods, etc.) as well as in the adaptation of some conventional geophysical techniques (e.g., induced polarization (IP) and self-potential (SP)), which are based on studying parameters more closely related to porosity, hydraulic conductivity and transmissivity of the aquifers than traditional electric resistivities or seismic velocities (e.g., Garambois and Dietrich 2001; Rubin and Hubbard 2005; Linde et al. 2006; Hördt et al. 2007). Unfortunately, most of these studies are still experimental and some are only performed on samples in labs or using synthetic modeling (Binley et al. 2005). In fact, among surface geophysical methods, only SNMR has an extensive record of the successful field applications for direct detection of the water content and, to some lesser extent, the transmissivity of aquifers (Yaramanci 2003). However, taking into account the very limited penetration depth and high sensitivity to ambient EM noise, the application of SNMR in Israel seems to be questionable.

An alternative approach to obtain the aquifer's porosity or storativity was to employ the TDEM method to aquifers that occupy saline water of a known salinity (Kafri and Goldman 2005). The approach is based on the use of conventional GEM methods and particularly recommends to use the TDEM method, which allows to overcome the above mentioned limitations of the novel techniques.

Similar to previous conventional studies, the theoretical basis of the described approach is being constituted by the empirical Archie's equation (Archie 1942):

$$\rho_w/\rho = \alpha\Phi^m \tag{1}$$

where ρ_w is the resistivity of water within the pore space, ρ is the bulk resistivity of the rock, Φ is porosity of the rock that approximately represents the volume of water filling the pore space, α and m are material-dependent empirical parameters, which are introduced to force the equation to fit the behavior of a rock in question.

Since m increases with cementation, Archie (1942) named it *cementation index*. Its value varies from 1.3 for unconsolidated sands to 1.8–2 for consolidated sandstones.

The most widely used simplified form of the Archie's equation for sandstone aquifers is obtained from (1) by letting $\alpha = 1$ and $m = 2$. Thus,

$$\rho_w/\rho = \Phi^2 = 1/F, \tag{2}$$

where F is the so-called formation factor, related to the porosity for $m = 2$ and $F = 1/\Phi^2$.

Even in such an oversimplified case, the required porosity cannot be determined from Eq. (2) without a priori knowledge of the water resistivity, ρ_w. Moreover, the bulk resistivity of rocks, ρ, which is the only parameter provided by surface GE and GEM measurements, in many cases is ambiguously determined due to the so-called equivalence problem (e.g., Kafri and Goldman 2005).

The resistivity of the pore water, ρ_w is customarily estimated using borehole salinity measurements. Also, an ambiguity in geophysical interpretation is mainly based on using independent borehole information. Unfortunately, in most cases, the number of existing boreholes is insufficient to provide accurate estimates of the both parameters Therefore, most of the attempts to apply surface GE and GEM methods to determine hydrological parameters in general and porosity in particular suffer from one common drawback: They are apparently "successful" in close proximity to boreholes, but could be rather unreliable far away from them. Moreover, even in the proximity of boreholes, the required a priori information provided by well logs is inevitably scale dependent and, thus, does not necessarily coincide with the required bulk geoelectric parameters, as will be discussed below. The scale dependency phenomenon, which is less pronounced in the case of granular clastic aquifers, becomes crucial for fractured, cavernous, karstic or crystalline aquifers.

In order to significantly reduce ambiguities of both hydrogeological and geoelectrical interpretations in case of coastal aquifers, particularly, in view of scarce amount or even a total absence of boreholes, Goldman and Kafri (2004) suggested to use properly selected GE and/or GEM techniques for detecting the fresh/saline groundwater interface, followed by application of the Archie's equation only to the seawater-intruded portions of the aquifer.

By using this approach, one can exactly determine one of the parameters in the Archie's equation and significantly improve the accuracy of the other. Indeed, the pore water resistivity, ρ_w is exactly known for the seawater salinity (i.e., Repsold 1989) and the use of the TDEM method allows to significantly reduce the ambiguity in determining the bulk resistivity, ρ, of the seawater-intruded portions of the aquifer (discussed below). Then, by applying Eqs. (1) or (2), one can more accurately calculate the porosity of the saline water-saturated portion of the aquifer. Since hydraulic properties of the aquifer should not depend on the location of the fresh/saline groundwater interface, one might expect that the porosity determined below the interface adequately represents the entire aquifer or sub-aquifer, in which the interface was detected.

It is important to emphasize that the approach in question is particularly feasible in complicated groundwater systems including partial saturation or clay conductivity effect. Such systems are described by more general equations, connecting fluid and electrical properties, such as the empirical Schlumberger equation or the theoretical Clausius–Mossotti equation, the application of which require additional a priori information (Wempe 2000). In brine-saturated aquifers, however, where the clay

conductivity effect becomes insignificant (Hoyer and Spann 1975), the amount of the required independent information can be essentially reduced.

The proposed methodology is particularly feasible in coastal aquifers, where ρ_w of the intruding seawater is well known (Repsold 1989) and where the normally wide extend of the intrusion allows the gathering sufficient amount of GE/GEM data to be statistically representative of the entire aquifer.

The main requirement for a GE/GEM method is the ability to accurately detect the fresh/saline interface and to determine the bulk resistivity of the aquifer below the interface. Experience shows that, in most cases, the TDEM method could be a right choice for this purpose (Goldman et al. 1991). However, the application of TDEM by itself is sometimes limited, due to technical reasons, such as a high level of ambient EM noise (in urban areas) or laying down transmitter loops, in a rough topography. In such cases, the 2D resistivity imaging method seems also to be feasible for this purpose. In addition, the latter can be used either instead of, or along with TDEM method in areas with expected sharp lateral variations of the subsurface resistivity.

The most important feature, which can significantly affect the ability of a geophysical method to quantitatively estimate the hydraulic properties of the aquifer, is an inherent non-uniqueness of the geophysical interpretation. The following example demonstrates how crucial is this feature for the accuracy in estimating porosity of the coastal aquifer. Figure 77 shows TDEM data collected by the Geonics EM-67 system (open squares at the left-hand side) along with a family of the equivalent interpreted models (at the right-hand side of the figure). The model, drawn by a solid line, represents the so-called best-fit model, whereas the rest of the models, shown by dashed lines, exhibit the equivalent models. At the left-hand side, the solid line represents a family of synthetic responses for both the best-fit and the equivalent

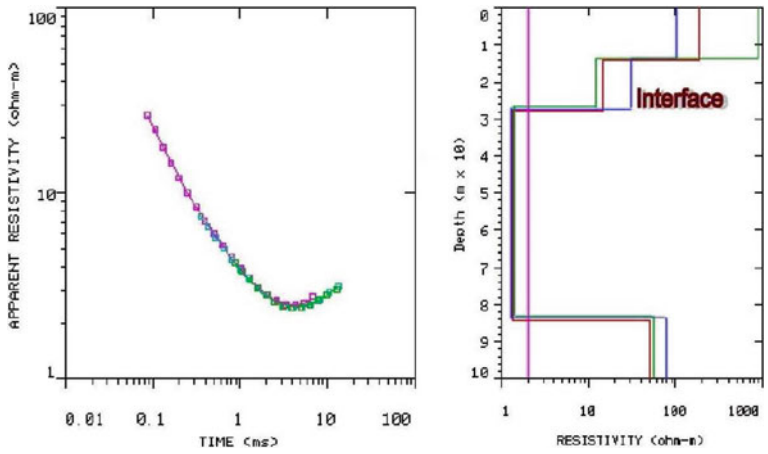


Fig. 77 TDEM data collected at station HS557 in the coastal aquifer, and the interpretation, including linear equivalence analysis by means of TEMIX-XL interpretation software (modified after Kafri and Goldman 2005)

models. The difference between the responses is so small that they all appear as a single solid line. In this specific example, the best-fit model provides a misfit error of approximately 1.9%, while misfits for all the equivalent models do not exceed 2%. Thus, the maximum difference between all the synthetic responses does not exceed 0.1%, which is considerably less than the overall noise level of the measurement. Thus, in practical terms all the interpreted models are equally consistent with the data and cannot be distinguished without a priori information.

In this example, both the data interpretation and the equivalence analysis have been carried out using TEMIX-XL interpretation software (Interpex 1996). It should be noted that the software does not provide a full (nonlinear) equivalence analysis, but only the one, which is based on the use of a parameter resolution matrix (Goldman et al. 1994b). Nevertheless, for illustration purposes, it is enough to consider the partial equivalence analysis since in reality the variations of some model parameters could be even wider than those shown in Fig. 1. Some further details regarding the difference between linear and nonlinear equivalence problems can be found in Goldman et al. (1994b).

The equivalence analysis of the TDEM measurement (Fig. 77) exhibits the following resistivity sequences:

It should be noted that since the estimated depth equivalence range (Fig. 77) is considerably narrower than that of the resistivities, the depth values in the above table are taken for the best-fit model only.

7.2 Porosity Calculations

The above example (Fig. 77) is used to describe the methodology for calculating porosity of saline water-intruded aquifers.

Based on the fact that bulk resistivity values below 2 ohm-m are solely indicative of seawater intrusion into the Mediterranean coastal aquifer (Goldman et al. 1991), the location of freshwater/seawater interface was determined at the depth of approximately 26 m. The high-resistivity sequence from the surface to the depth of 14 m represents both the unsaturated zone and the underlying freshwater-saturated aquifer. The thickness of the unsaturated zone at this site is only 2 m, and therefore, it is not taken in the further considerations. In fact, no resistivity drop is detected at the depth of the water table, apparently due to the limited resolution of the TDEM method. The resistivity values within the intermediate zone between 14 and 26 m are typical for either an aquiclude, an aquitard or an aquifer saturated with brackish waters, with a variable salinity.

The very low-resistivity geoelectric layer below 26 m depth is undoubtedly identified with seawater intrusion. By applying Eq. (2) to this layer, one can accurately calculate the porosity of the seawater-intruded portion of the aquifer by using solely the TDEM data without any additional borehole information. Indeed, the salinity of the Mediterranean water is known, and the calculated fluid resistivity (ρ_w), according to Repsold (1989) is 0.18 ohm-m. Similar values were also obtained from measured

boreholes conductivity logs (Friedman 2003, personal communication). Thus, the porosity of the seawater-intruded part of the aquifer, as calculated using Eq. (2), is $\Phi = 0.36$.

Similar porosity calculations for the shallow-resistivity sequence led to a great uncertainty due to the wide equivalence range of resistivities (ρ). Even when fixing $\rho_w = 10$ ohm-m, which is typical for freshwater-saturated aquifers in the area, the calculated porosities are found to vary in a wide range between 0.13 and 0.35, due to the geophysical non-uniqueness only.

The uncertainties could become even greater in the intermediate resistivity sequence, where water salinity varies in a wide range and thus, accordingly leading to a wide range of possible ρ_w values. Since the resistivity, in this sequence, also r varies due to equivalence (Fig. 77), the resulting uncertainty in the calculated porosity might become significant.

Since it is reasonable to assume that the porosity of the aquifer does not significantly depend on the location of the fresh/saline water interface, the suggested approach can be also used to resolve non-uniqueness in the upper resistivity sequence. Indeed, by applying Eq. (2) for $\Phi = 0.36$ and $\rho_w = 10$ ohm-m, one obtains $\rho =$ 77 ohm-m, which is fairly similar to the lowest resistivity value in the equivalence range of this sequence.

Some additional semi-quantitative information, regarding the groundwater salinity, can be also obtained for the intermediate resistivity zone above the interface. Indeed, in order to obtain a porosity value of 0.36, ρ_w must vary in a certain range, corresponding to the equivalence range of ρ in this zone. Applying the Archie's Eq. (2), and considering the possible resistivity range for the intermediate layer (Fig. 77 and the above Table), one can obtain ρ_w values varying between 1.4 and 3.5 ohm-m. This resistivity range roughly corresponds to salinities varying between 300 and 3000 mg Cl/l (Repsold 1989).

The applicability of the proposed methodology has been tested in different coastal aquifers of Israel by comparing TDEM results with borehole data. The study focused mainly on the Mediterranean coastal aquifer, where the greatest amount of both geophysical and borehole data is available. In addition, the analysis also included coastal aquifers of the Gulf of Eilat and the Dead Sea, as well as the regional carbonate Judea Group aquifer.

The porosity of different types of aquifers

The total porosity of the aquifer includes both, the connected pores (named also as the aquifer's storativity) and the unconnected pores. The total porosity is determined usually in a laboratory from core samples, whereas storativity is obtained from hydrological pumping interference tests, which are rarely available in most cases.

Two main types of existing aquifers are granular (Fig. 78a) and hard rock aquifers, where the porosity is mostly gained by solution (Fig. 78b) or fracturing (Fig. 78c).

The porosity of granular aquifers is mostly inter-granular and the aquifers are usually moderately heterogeneous. Therefore, a porosity value obtained from a core sample approximately represents larger volumes of the aquifer. Contrary to the above, in carbonate aquifers, most of the hydraulic conductivity, as well as storativity, is

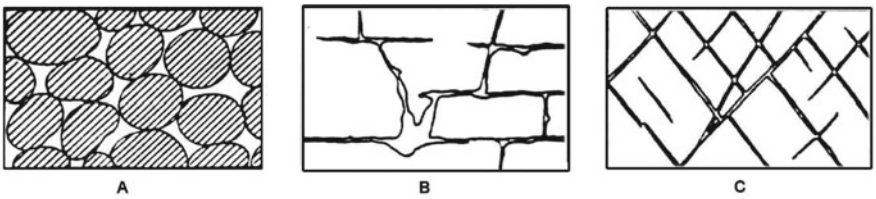


Fig. 78 Diagram showing different types of rock interstices and the relation of rock texture to porosity. **a** Well-sorted sedimentary deposit having high porosity; **b** rock rendered porous by solution; **c** rock rendered porous by fracturing (after Meinzer 1942)

related to open fissures and solution channels, and the aquifers are often extremely heterogeneous. In many cases, an aquifer might have a double or even triple porosity system, which includes both the initial matrix porosity and the secondary fissured and solution porosity. As a result, the hydraulic parameters obtained in carbonate aquifers are usually scale dependent and often show preferable spatial parameters with increasing test radii, from a core sample to a pumping test and toward a regional study (Rover and Cherkauer 1995; Zuber and Motyka 1998; White 2002).

It is thus essential to know, particularly for carbonate aquifers, whether a given porosity value represents a small or a large aquifer volume. Furthermore, when a resistivity value is used for calculating porosity, it should be considered whether it is taken from a borehole log representing a small radius and thus lower volumes, or from TDEM measurements that integrate larger aquifer volumes and thus representing assumingly higher porosities.

Porosity and storativity, both measured and calculated values for different granular and carbonate aquifers of Israel are discussed herein, and are summarized in Tables 1–2.

Storativities in coastal aquifers, based on pumping tests or models, vary between 0.25 and 0.4 representing only the aquiferous portion of the sequence. It is assumed that the storativities in this portion of the aquifer are not quite similar to those of the total porosity of the sequence. The situation, however, is different within the clayey aquicludal interlayers, which might possess a high non-effective porosity that contains trapped and unflushed fluids.

The storativities of carbonate aquifers mostly vary between 0.03 and 0.08, since the initial sedimentary (matrix) porosity is small, and most of the storativity is related to secondary porosity gained by fracturing and solution. In places, like Mount Carmel, where a reef facies is known in both surface and subsurface, the initial porosity and as a result the storativity are high, attaining a value of 0.15. In the case of the discussed carbonate aquifers one should consider, in addition, the large aquifer volume with a small initial non-effective porosity.

Calculated versus actual resistivities in Israeli aquifers

Using the simplified Archie's Eq. (2), one can calculate the expected bulk resistivities of the different aquifers, and based on their related porosities and hosted saline waters compare them with the actually measured resistivities by the TDEM method.

The ρ_w values, for the different saline potential end members, can be obtained from a resistivity/salinity chart (Repsold 1989), and they are 0.18 and 0.04 ohm-m for Mediterranean seawater and Dead Sea concentrated brine, respectively. The ranges of porosities (Φ) used are those described above for different aquifers. The resultant calculated ρ values for different porosities are given in Table 2. It can be seen that in the Mediterranean seawater-intruded coastal aquifer, the TDEM measurements yielded resistivities between 1 and 2 ohm-m, which are similar to the ones shown in Table 2 for the porosity values greater than 0.3. One can note the difference between the lower and the upper parts of the Mediterranean coastal aquifer with regard to the expected and actually measured bulk resistivities. The reason is that in the lower sub-aquifers, the ratio between the cumulative thickness of the clayey interlayers and that of the aquiferous ones is higher, as compared to the situation in the upper sub-aquifers. Therefore, in some cases, where freshwaters were sampled in the aquiferous parts there, TDEM bulk resistivities were below the expected values. It is reasonable to assume that in these cases, the TDEM measurements integrate both aquiferous and clayey layers, which apparently host trapped and unflushed saline water. This

Table 1 Storativity and porosity of granular and karstic aquifers

Aquifer	Location/basin	Storativity	References
Coastal (granular)	Shiqma	0.3–0.4	Sellinger and Kapuler (1972)
	Nizzanim	0.3–0.4	Sellinger and Kapuler (1973)
	Western Galilee	0.25–0.35	Sellinger and Levin (1977)
	General	0.25	Mercado (2000)
	Dead Sea	0.012	Hirsch (1975)
Judea Group (carbonate)	Naaman	0.03–0.08	Michaeli et al. (1973)
	Naaman	0.06	Mercado and Ben Zvi (1983)
	Kabri	0.04	Shachnai and Goldshtoff (1978)
	Yarkon–Taninim	0.05	Baida et al. (1978)
	Bet Shean	0.03–0.07	Goldshtoff and Shaliv (1979)
	Mt. Carmel Na'aman	0.03	Bar Yosef et al. (1997)
	Atlit 1 (Carmel)	0.013	Mercado and Ben Zvi (1983)
	N. Oren Carmel)	0.15	Mercado and Ben Zvi (1983)
		Porosity	
	Mt. Carmel reefs	Up to 0.15	Bein (1971)
	Ashqelon 3	0.15–0.20	Bein (1971)

Table 2 Bulk resistivities calculated using Archie's equation for different porosities and water resistivities and actual resistivities measured by TDEM in different aquifers

Aquifer and saline water body	Φ	ρ_w (ohm-m)	ρ (ohm-m)	
			Calculated	TDEM
Coastal, granular aquifer saturated with normal seawater	0.2	0.18	3.6	Between 1 and 2
	0.3	0.18	1.6	
	0.4	0.18	0.91	
Coastal, granular aquifer saturated with Dead Sea concentrated brine	0.2	0.04	0.81	Between 0.2 and 0.8
	0.3	0.04	0.36	
	0.4	0.04	0.20	
Judea Group carbonate aquifer saturated with normal seawater	0.03	0.18	200	Between 4 and 8
	0.05	0.18	72	
	0.07	0.18	37	
	0.08	0.18	28	
	0.15	0.18	8	
	0.20	0.18	4.5	
Judea Group carbonate aquifer saturated with Dead Sea concentrated brine	0.05	0.04	16	Between 5 and 10
	0.15	0.04	1.8	

In the calculations for granular aquifers, parameter α was chosen to provide a well-known empirical relation $F = 0.81/\Phi^2$ normally used for sands

results in the lowering of resistivities of the entire sequence. As described before, quite a similar phenomenon was encountered in the Gulf of Eilat coastal aquifer.

In the Dead Sea coastal aquifer, where the TDEM resistivities, well below 1 ohm-m, were measured for the density driven intrusion of the Dead Sea brine into the aquifer, similar values were calculated using Archie's equation (Table 2).

The situation is more complicated for the seawater-intruded Judea Group carbonate aquifer. The calculated resistivities for relatively low storativities (0.03–0.08) are significantly higher (between 28 and 200 ohm-m) than those measured by the TDEM method. Only for higher storativities (0.15–0.20), the calculated resistivities become closer to actual TDEM, or well log resistivities.

Deep TDEM measurements near Nahal Oren in the Mount Carmel area, where high storativities were previously obtained from pumping tests in boreholes, yielded bulk resistivities around 1 ohm-m, while elsewhere, these values for the saline water-saturated parts of the Judea Group aquifer varied approximately between 5 and 10 ohm-m (Goldman and Kafri 2002), assumingly due to lower storativities. It should be noted that in a few wildcats (Gaash 2, Netania 1 and Cesarea 3), carbonate sequences saturated with waters of normal seawater salinity (18,000–20,000 mg Cl/l) exhibited well log resistivities between 4 and 6 ohm-m, similar to those obtained by the deep TDEM survey (Kafri et al. 2007).

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